# Class 7
# Physics

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# 1. Units and Measurements

## Solutions

### LEVEL-I

**1. Correct option: (A)**

To know and understand a physical situation completely, one is likely to measure the quantities such as distance, speed, time, mass, acceleration, force etc. which are called the physical quantities. Physical quantity is that in terms of which laws of physics can be expressed and which can be measured directly or indirectly.

**2. Correct option: (B)**

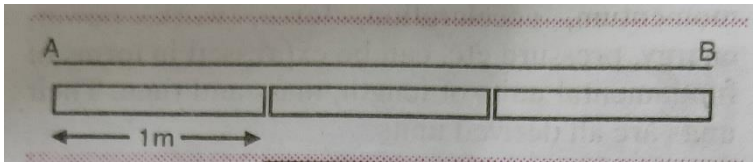
Mass is a physical quantity as it is used to measure a physical situation. Rest all are the emotional situations

**3. Correct option: (B)**

Measurement is basically a comparison process. It involves the selection of a unit of measurement and comparing the quantity with standard unit. The standard chosen should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod, we have to take length as the standard quantity.

**4. Correct option: (A)**

In comparing the quantity with the standard unit, we have to find the number of times this unit is contained in the quantity. For example, suppose we have to measure the length of a wire AB and we select meter as the unit




of measurement. We place the meter rod along the wire AB and find that it is contained 3 times in AB. Thus 3 is the numerical value of the length AB, when m is the unit of measurement. We write AB= 3 meter

**5. Correct option: (B)**

The SI unit of temperature as per the International System of Units is Kelvin which is represented by the symbol K.

**6. Correct option: (B)**

In general magnitude of a physical quantity (P)= numerical value × size of its unit.

For example. If length of a rod = 3 m

Then 3 = numerical value

1m = size of the unit

**7. Correct option:(C)**

Derived quantity is a quantity whose measurement is done with the help of other fundamental physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. For example, velocity is a derived physical quantity which depends on fundamental quantities distance(length) and time.

**8. Correct option: (B)**

The SI unit of temperature as per the International System of Units is Kelvin which is represented by the symbol (K).

**9.Correct option: (A)**
Strength of Electric current is measured in Amperes, Hence SI unit of strength of electric current is Ampere.
Its symbol is (A).

**10. Correct option:(D)**
None of these unit is written by putting a dot between the individual units.

- Units like seconds, meters are fundamental units/independent units.
- units like meter /second are an example of a derived unit/dependent unit.
- Compound units are those composed of more than one unit. For example, miles/hour

**11. Correct option: (A)**
The normal force (thrust)(F) exerted by an object per unit area(A) of the surface in contact with it is called pressure (P).

$$P = \frac{F}{A}$$

SI unit of pressure is Pascal (Pa)
1 Pascal = $1Nm^{-2}$

**12.Correct option:(C)**
Measurement is basically a comparison process. It involves the selection of a unit of measurement and comparing the quantity with standard quantity/unit. The standard chosen should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod we have to take (m) as standard unit.

**13. Correct option:(B)**
The meter is the base unit of length in the International System of Units (SI). Its unit symbol is (m).

**14. Correct option:(C)**
C.G.S is centimeter(cm) –gram(g)- second(s) system of units
Hence,
C.G.S unit of mass is gram.

**15. Correct option:(D)**
M.K.S is the system of units based on measuring lengths in meters, mass in kilograms, time in seconds.
The M.K.S unit of length is meter.

**16. Correct option:(B)**
M.K.S is the system of units based on measuring lengths in meters, mass in kilograms, time in seconds.
The M.K.S unit of time is second having symbol(s).

**17. Correct option:(B)**
**F.P.S** system is the British Engineering system of units, which uses foot as the unit of length, **pound as the unit of mass** and second as the unit of time.

**18. Correct option:(A)**
For a unit named after a scientist, the symbol is a capital letter. For example, symbol for SI unit of force (newton)is (N).

**19. Correct option:(C)**
**SI** unit of work is joule having symbol(J).

**20. Correct option:(A)**
**SI** system is the internationally accepted system of units for measurements. The SI, with standard scheme of symbols, units and abbreviations, was developed and recommended by

General conference on weights and measures in 1971 for international usage in scientific, technical, industrial and commercial work.
There are seven fundamental quantities in SI system: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter and two Supplementary physical quantities.

## LEVEL-II

**1. Correct option:(C)**
Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. Volume depends on other physical quantities for its measurement.

**2. Correct option:(A)**
SI unit of luminous intensity is Candela.

**3. Correct option:(A)**
Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. Units of derived quantities are called derived units.

**4. Answer: C) i**
The symbol used to represent electric current in equations is $i$(or $I$). A (ampere) is the SI unit of electric current, not its symbol as a physical quantity.

**5. Answer: C) $m^3$**
Volume measures the space occupied by an object. Its SI unit is the cubic metre ($m^3$).

**6. Correct Answers: A, B**
Explanation:

- A) Length is independent of other physical quantities, so it is a fundamental quantity.
- B) Length is one of the seven SI base (fundamental) quantities.

**7. Correct options: (A), (D)**
Mass of a body is the quantity of matter it contains.
SI unit of mass is Kilogram.
1 kilogram = 1000000 milligrams
Hence, kilogram and milligram are units used to measure mass. Kilometer and millimeter are used to measure distance.

**8. Correct options: (A), (B), (D)**
Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter/amount of substance. These quantities can be measured independently.

**9. Correct options: (A, B, C)**
Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities.

- Newton is a derived unit for derived physical quantity (force).
- Pascal is a derived unit for derived physical quantity (pressure).
- Joule is a derived unit for derived physical quantity (work/energy).

**10. Correct options: (A, D)**

velocity $= \frac{\text{displacement}}{\text{time taken}}$

SI unit of displacement is m.

SI unit of time is s.

SI unit of velocity is m/s or $ms^{-1}$

**11. Correct option:(B, D)**

Density $= \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kilogram(kg).

SI unit of volume is $m^3$

**So, SI** unit of density is $\frac{kg}{m^3}$ or $kgm^{-3}$

**12. Correct options: (A)**

SI unit of length is meter.

As distance between two points is the path length between two points so its unit is same as the length.

**13. Correct options: (C)**

Unit of time is **second** in all the system of units (F.P.S, C.G.S, M.K.S, SI).

**14.Correct option:(B)**

**Steradian** is the SI unit of solid angle (angle in space)

**15. Correct option: (A)**

As we know

physical quantity (P)= numerical value(N) × size of its unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of its measurement. Hence, we may write

$P = N_1U_1 = N_2U_2$

Where

$U_1, U_2$ are the two unts of measurement of the same quantity and $N_1$ and $N_2$ are their respective numerical values .

If $U_1 > U_2$ , then $N_1 < N_2$ which means larger is the unit , lesser is the numerical value and vice – versa. For example,

Length of rod = 3 m = 300 cm

Here, unit(m) is larger and numerical value 3 is lesser

Unit (cm) is lesser and numerical value 300 is larger

**16. Correct option: (A)**

As we know

physical quantity (P)= numerical value(N) × size of its unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of its measurement. Hence, we may write

$P = N_1U_1 = N_2U_2$

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$U_1, U_2$ are the two unts of measurement of the same quantity and $N_1$ and $N_2$ are their respective numerical values .

If $U_1 > U_2$ , then $N_1 < N_2$ which means larger is the unit , lesser is the numerical value and vice – versa. For example,

Length of rod = 3 m = 300 cm

Here, unit(m) is larger and numerical value 3 is lesser

Unit (cm) is lesser and numerical value 300 is larger

Hence numerical value and size of unit are inversely proportional to each other.

**17. Correct option: (B)**

1 kilogram = 1000 grams or $10^3$ grams

now , 1 gram = 1000 mg

so , 1 kilogram = 1000000 mg or $10^6$mg

Hence 1kilogram can't be expressed as 10000 grams.

- **18. Correct Matching is:** a) Derived unit → Newton (R): Newton is the SI derived unit of force (1 N = 1 kg·m/s²).
- b) Fundamental quantity → Time (S): Time is one of the seven SI fundamental (base) quantities.
- c) Derived quantity → Energy (Q): Energy is a derived physical quantity, obtained from base quantities, and its SI unit is joule (J).
- Candela (P): Candela is the SI base unit of luminous intensity, not a derived unit or quantity in this matching.

Correct Matching: a – R, b – S, c – Q

**19. Correct Matching is: a-T; b-R; c-Q; d-P**

Unit of time is **second** in all the system of units (F.P.S, C.G.S, M.K.S, SI).

**SI** unit of length is meter.

**F.P.S:** It uses foot as the unit of length, pound as the unit of mass and second as the unit of time.

**C.G.S**: It is based on centimeter, **gram** and second as the fundamental units of length, **mass** and time respectively.

## Multiple And Sub Multiples

## LEVEL-I

**1. Correct option: (C)**

1 millisecond = $10^{-3}$seconds

1microsecond = $10^{-6}$seconds

1 nanosecond = $10^{-9}$seconds

1shake = $10^{-8}$seconds

**2. Correct option: (A)**

1Decimeter = 10 centimeter

**3.Correct option:(D)**

The average of the varying solar days, when earth completes one revolution around the sun is called the mean solar day.

**4. Correct option: (C)**

Light year is an important unit of long distances. One light year is the distance travelled by light in vacuum in one year.

as velocity of light in vacuum = $3 \times 10^8$ $ms^{-1}$

1year = $365 \times 24 \times 60 \times 60$ seconds

so, 1 light year = $3 \times 10^8 \times 365 \times 24 \times 60 \times 60$ meter

1 light year = $9.46 \times 10^{15}$m

**5. Correct option:(C)**

1 kg = $10^6$mg

500kg = $500 \times 10^6$ mg

500kg = $5 \times 10^8$ mg

**6. Correct option:(B)**

1g= 1000mg

1 mg=1/1000 g

1mg= 0.001 g

**7. Correct option: (B)**

1 quintal = 100 kg

**8. Correct option: (C)**

The standard unit chosen for measurement should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod, we have to take length as the standard quantity and meter as the standard unit.

**9. Correct option: (C)**

1 hectometer (hm) = 100 m

Meter (m) is a smaller unit of length as compared to hectometer

**10. Correct option: (D)**

1 meter = 100 cm

$1 \text{ cm} = \frac{1}{100} \text{m}$

$= 0.01 \text{ m}$

Hence, 1 centimeter = 0.01 m

**11. Correct option: (D)**

$1 \text{ picometer} = 10^{-12} \text{m}$

$1\text{m} = 10^{2} \text{cm}$

$\text{so, } 1 \text{ picometer} = 10^{-12} \times 10^{2} \text{ cm}$

$1 \text{ picometer} = 10^{-10} \text{ cm}$

**12.Correct option:(C)**

As the gap between two points is measured in units of metres.so, it can measure as length because SI unit of length is meter.

**13. Correct option: (A)**

$1\text{cm} = \frac{1}{100} \text{m}$

$300 \text{ cm} = \frac{300}{100} \text{m}$

$300 \text{ cm} = 3 \text{ m}$

**14. Correct option: (A)**

$1 \text{ gram} = 10^{6} \text{ micrograms}$

**Or**

$1 \text{ microgram} = 10^{-6} \text{grams}$

**15. Correct option: (B)**

(nm) is the symbol for nanometer

$1 \text{ cm} = 10^{7} \text{nm}$

$3\text{cm} = 3 \times 10^{7} \text{ nm}$

$3\text{cm} = 0.3 \times 10^{8} \text{ nm}$

**16. Correct option: (A)**

$1\text{J} = 10^{7} \text{ergs}$

Joule is the SI(MKS) unit of energy and ergs is the cgs unit of energy.

**17. Correct option: (B)**

1 min = 60 s

40 min = 40 ×60 s

= 2400 s

**18. Correct option: (A)**

The diameter of the sun = $1.39 \square 10^{9}$ m

**19. Correct option: (B)**

1 quintal = 100 kg
1quintal = $10^2$kg

**20. Correct option: (D)**

Diameter of hydrogen atom is approximately 1.06□10□10 m. It is defined by the diameter of the spherical
electron cloud around the nucleus.

**21. Correct option: (B)**

A century is a period of 100 years
So, 1 century = 100 years

**22. Correct option: (D)**

1 day = 24 hours
1 hour = 60 minutes
1minute = 60 seconds
So, 1 day =24×60 minutes
=1440 min
Hence, 1 day = 24×60×60 seconds
1 day = 86,400 seconds

**23.Correct option:(A)**

The exponential part of a particular measurement is called the order of magnitude of a quantity. An order of magnitude is an exponential change of plus or minus 1 in the value of a quantity or unit.

**24. Correct option: (D)**

Both option(A) and (B) are the advantages of using scale. It can be easily taken from one place to another and it does not change during measurements.

**25. Correct option: (D)**

1km=100000cm
1cm = 1/100000km
1centimeter=1/100000 kilometer

**26. Correct option: (D)**

1 centimeter = 10 millimeters
Or
1 millimeter = 1/10 centimeter

**27. Correct option: (D)**

The most appropriate unit to measure the length of a book is centimeter. meter would be too large and millimeter or micrometer would be too small to measure the length of a book.

**28. Correct option: (C)**

1 meter = 1000 millimeters
Or
1 millimeter = 1/1000 meter

**29. Correct option: (A)**

1 decade = 10 years

**30. Correct option: (A)**

1 centimeter = 10 millimeters

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## LEVEL-II

**1. Correct option: (C)**

(Mm) is the symbol of Mega meter,
(pm) is the symbol for Pico meter.
1Mm=$10^6$m

$1m=10^{12}pm$
So, $1Mm=10^{6} \times 10^{12}pm$
$1Mm=10^{18}pm$

**2. Correct option: (D)**
$1nm=10^{-9}m$
$1m=10^{10}A°$
So, $1nm=10^{-9} \times 10^{10}A°$
1 nm =10 A°

**3. Correct option: (A)**
**1 fermi or femtometer (FM) =** $10^{-15}m$
These units of length are commonly used for measuring very small distances in scientific work.

**4. Correct option: (B)**
1 ton = 1000 kg

**5. Correct option: (A)**
$1\ m = 10^{9}$ nm
Or
$1\ nm = 10^{-9}m$
(nm) is the symbol for nanometer. It is a smaller unit than meter.

**6. Correct option: (B)**
1 km = 1000 m
Meter (m) is smaller unit of length as compared to km

**7.Correct option:(B)**
(ng) is the symbol for nanogram.
1 metric ton is also known as 1 ton
1 metric ton = 1000 kg
**1**Kg = 1000g
And
$1g= 10^{9}$ ng
So,
1 metric ton = 1000 x 1000 x $10^{9}$ ng
$=10^{15}$ ng

**8. Correct option: (D)**
$1\mu m=10^{-6}m$
$1m=10^{10}A°$
$1\mu m=10^{-6} \times 10^{10}A°$
$1\mu m=10^{4}$ A°

**9. Correct options:(C, D)**
$1mm=10^{-3}m$
$1m=10^{2}cm$
So, $1mm=10^{-3} \times 10^{2}cm$
$1mm=10^{-1}cm$
$=1/10$ cm
1mm =0.1cm

**10.Correct options:(A), (B), (D)**
$1mg=10^{-3}g$--------(1)
$1g=10^{6}\mu g$

Putting in equation 1
$1mg=10^{-3} \times 10^{6}\mu g$
$1mg=10^{3}\ \mu g$
Also
$1g= 10^{9}$ ng
Putting in equation 1
$1mg=10^{-3}$ x $10^{9}$ ng
$1mg =10^{6}$ ng

**11. Correct options:(A, B)**

1 metric ton is also known as 1 ton
1 metric ton= 10 quintal
And we know
1 quintal = 100 kg
So,
1 metric ton =10×100 kg
= 1000 kg

**12.Correct options:(B, C)**

1 decagram= 10 g -------(1)
$1g = 10^{-3}$kg
Putting in equation 1
1 decagram= 10 x $10^{-3}$kg
1 decagram =$10^{-2}$kg

**13. Correct options: (A), (B), (C), (D)**

$1\mu$ g =$10^{-6}$ g
$5\mu$ g =$5\times10^{-6}$ g----(1)
as, $1g = 10^{-3}$kg
So, from equation 1
$5\mu$ g =$5\times10^{-6} \times10^{-3}$kg
=$5\times10^{-9}$ kg - - - - (2)
Also,
$1g = 10^{3}$mg
So, from equation 1
$5\mu$ g =$5\times10^{-6} \times10^{3}$mg
=$5\times10^{-3}$ mg
Also, $1kg = 10^{-3}$tonne
So, from equation 2
$5\mu$ g = $5\times10^{-9} \times10^{-3}$tonne
= $5\times10^{-12}$tonne

**14.Correct option:(A)**

Shake is an informal metric unit of time equal to $10^{-8}$ seconds.

**15. Correct options: (A), (B), (D)**

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.
1 day = 24 hours
1 hour = 60 minutes
1minute = 60 seconds
So,

1 day =24×60 minutes
=1440 min
Hence, 1 day = 24×60×60 seconds
= 86,400 seconds

**16.Correct option:(D)**

**1 fermi or femtometer (FM) =** $10^{-15}$m

These units of length are commonly used for measuring very small distances in scientific work.

**17. Assertion–Reason**

**Correct Answer: A) Both A and R are true, and R is the correct explanation of A.**

**Explanation:**

Fundamental (base) physical quantities are independent because they cannot be derived or expressed in terms of other physical quantities. Examples include length, mass, time, electric current, temperature, amount of substance, and luminous intensity. Since they are not derived from other quantities, they are considered independent.

**18. Assertion–Reason**

**Explanation:**

Plane angle and solid angle are supplementary quantities in the traditional SI classification. They are not included among the seven fundamental (base) physical quantities, which is why they are classified separately as supplementary quantities.

Note: In the modern SI (since 1995), plane angle and solid angle are treated as dimensionless derived quantities. However, if your textbook uses the term supplementary quantities, then Option A is the expected answer**.**

**19. Matrix Matching**

| **Column I** | **Column II** |
|---|---|
| **a) kilo** | **S (1000) or T ($10^3$)** |
| **b) femto** | **P ($10^{-15}$)** |
| **c) nano** | **Q ($10^{-9}$)** |
| **d) pico** | **R ($10^{-12}$)** |

Correct Matching:

- a → S, T
- b → P
- c → Q
- d → R

Explanation:

- kilo = 1000 = $10^3$, so it matches both S and T.
- femto = $10^{-15}$
- nano = $10^{-9}$
- pico = $10^{-12}$

**20. Integer Type**

**Answer: 7**

**Explanation:**

The SI system has seven fundamental (base) physical quantities:

1. Length
2. Mass
3. Time
4. Electric current

5. Thermodynamic temperature
6. Amount of substance
7. Luminous intensity

**21. Integer Type**

**Answer: 9**

**Explanation:**

All the listed quantities are derived quantities because they are obtained from combinations of fundamental quantities.

- Area
- Volume
- Density
- Velocity
- Acceleration
- Force
- Pressure
- Work
- Power

Total = 9 derived quantities.

---

## Length and Time

## LEVEL-I

**1.Correct option:(B)**

Mass of a body is the quantity of matter it contains.

**2.Correct option:(A)**

The kilogram is the base unit of mass in the International System of Units (SI). Its unit symbol is (kg).

**3. Correct option:(A)**

**Scalar quantities:** These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km. similarly mass is a scalar quantity as it has magnitude only and it does not need any other information to specify it.

**4. Correct option:(A)**

The **C.G.S. (Centimetre–Gram–Second)** system uses:

- **Length:** centimetre (cm)
- **Mass: gram (g)**
- **Time:** second (s)

Therefore, the unit of **mass** in the C.G.S. system is **gram**.

**5. Correct option:(A)**

**Scalar quantities:** These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.

Similarly, mass is a scalar quantity as it has magnitude only and it does not need any other information to specify it.

**Vector quantities** are those quantities which have both magnitude(size) and direction.

Velocity, displacement and force are **vector quantities** because they have both magnitude and they act in a particular direction.

**6. Correct option:(B)**

As the gap between two points is measured in units of metres.so, it can measure as length because SI unit of length is meter.

**7. Correct option:(B)**

The meter is the base unit of length in the International System of Units (SI). Its unit symbol is (m).

**8. Correct option:(A)**

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter. These quantities can be measured independently.

**9. Correct option:(D)**

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence, C.G.S unit of length is centimeter.

**10.Correct option:(C)**

1 km = 1000 m

Meter (m) is smaller unit of length as compared to km

1m=100cm

Centimeter (cm) is smaller unit of length as compared to km and m

1cm=10 mm

millimeter(mm) is smaller unit of length as compared to km, m and cm

Meter (m) is smaller unit of length as compared to km

**11. Correct option:(D)**

1 km = 1000 m

Meter (m) is smaller unit of length as compared to km

**12. Correct option:(C)**

Average distance of the center of the sun from the center of the earth is

$1.496 \times 10^{11}$m(1 Asronomical unit)

Which is approximately = 149 million km

**13. Correct option:(B)**

- Mass is the quantity of matter contained by a body. It is a constant quantity and hence cannot be zero for any material body.
- Weight is defined as the force exerted on a body by gravity.

  Weight is often expressed as

  **W= mg**

  Where, w= weight of the body

  m = mass of the body

  g= acceleration due to gravity

  **weight can be zero where there is no gravity(g=0) like in space.**

**14.Correct option:(B)**

The mole is SI unit of amount of substance/quantity of matter.

It has the symbol(mol).

Amount of substance (n) = $\frac{\text{Given weight (g)}}{\text{molecular weight (gmol}^{-1})}$

**15. Correct option:(C), (D)**

Time can be measured in seconds, minutes, hours, days, weeks, months, and years with clocks and calendars.

**16. Correct option:(A)**

**Scalar quantities:** These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.

Similarly, time is a scalar quantity as it has magnitude only and it does not need any other information to specify it.

**17. Correct option:(B)**

1shake = $10^{-8}$seconds

**18. Correct option:(D)**

$$\text{Time} = \frac{\text{distance}}{\text{speed}}$$

Average distance of the center of the sun from the center of the earth is $1.496 \times 10^{11}$m(1 Asronomical unit) or approximately $1.5 \times 10^{11}$m

Speed of light = $3 \times 10^{8}$m/s

$$\text{Time} = \frac{1.5 \times 10^{11}\text{m}}{3 \times 10^{8}\text{m/s}}$$

$$\text{Time} = \frac{15 \times 10^{11}}{3x\ 10 \times 10^{8}}\text{s}$$

Time = 500s

So, the time taken by the light rays to reach the ground from the sun= 500 sec

**19. Correct option:(B)**

1milligram=$10^{-3}$gram

1 centigram = $10^{-2}$gram

1Decagram = 10 gram

1 hectogram = 100 gram

**20. Correct option:(A)**

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.

1 mean solar day = 24 hours= 86400 seconds

$\frac{1}{86400}$th part of the mean solar day = 1 second

**21. Correct option:(A)**

1g= $10^{3}$mg

1 mg =$10^{-3}$ g

**22. Correct option:(D)**

Earth revolves around the sun and completes 1 revolution in 365 days, 5 hours, 59 minutes and 16 seconds.

This time is taken as equal to 1 year.

**23. Correct option: (B)**

1 millisecond = $10^{-3}$seconds

1microsecond = $10^{-6}$seconds

1 nanosecond = $10^{-9}$seconds

1shake = $10^{-8}$seconds

**24. Correct option:(C)**

There are **7** months in a year having 31 days. These are January, March, May, July, August, October and December.

**25. Correct option:(B)**

There are 28 days in the month of February (in normal year).

**26. Correct option:(A)**

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.

1 mean solar day = 24 hours= 24x60 minutes= 1440 minutes

$\frac{1}{1440}$th part of the mean solar day = 1 minute

## LEVEL-II

**1. Correct option:(A)**

The wall clock has a least count (minimum measurable value) of 1 second. It cannot measure a time below one second.

**2. Correct option:(D)**

length of meter scale = 1 m

as the scale is divided into 500 equal parts

so,

500 parts = 1m

1 part = $\frac{1}{500}$m

1 part = 0.002 m

1m = 1000 mm

So, 1part = 0.002 x 1000mm

= $\frac{2}{1000}$x 1000 mm

1 part = 2mm

**Hence,** smallest measurement that can be measured by scale is 2 mm.

**3. Correct option:(B)**

Given:

On atlas

10cm= 15000km

Or

1 cm = $\frac{15000}{10}$km

1cm = 1500km----------(1)

Actual distance between Bhopal and Kochi = 1500km

From equation (1)

Distance between Bhopal and Kochi on atlas = 1cm

**4. Correct option:(A)**

A lunar month is the duration between successive new moons, it has a mean period of 29.53059 days (29 days 12 hours and 44 minutes).

**5.Correct options:(A, B, C)**

Mass can be measured in kilograms, grams, milligrams etc.

**6. Correct options:(C, D)**

Length can be measured in meter, kilometer etc.

Acre and hectare are units of area.

**7. Correct option:(A, B, C, D)**

Time can be measured in seconds, minutes, hours, days, weeks, months, and years with clocks and calendars.

**8. Correct option:(A, B, C)**

**Scalar quantities:** These have a size or magnitude only and need no other information to specify them.

For example, distance/Length between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.

Similarly, time and mass are also scalar quantities as they have magnitude only and they do not need any other information to specify them.

**9.Correct option:(C)**

Mass of earth = $5.972 \times 10^{24}$ kg

**10. Correct option:(A)**

Distance between mars to earth is approximately 54.6 million km

**11. Correct option:(A)**

1 calendar year=365 days
1 day = 24 hours
1 hour = 60 minutes
1 minute=60 seconds

**So,**

1 calendar year = 365 x 24 x 60 x 60 seconds
1 calendar year = 365 x 24 x 60 x 60 seconds
1 calendar year =31,536,000 seconds

**12. Correct option is: a-RT; b-ST; c-QT; d-P**

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. For example, quantities Length, time and mass are measured independently and hence they are fundamental physical quantities.

- **SI** unit of mass is kilogram.
- **SI** unit of length is meter.
- **SI** unit of time is second.
- Weight is defined as the force exerted on a body by gravity.

  Weight is often expressed as:

  W= mg

  Where, w= weight of the body
  m = mass of the body
  g= acceleration due to gravity

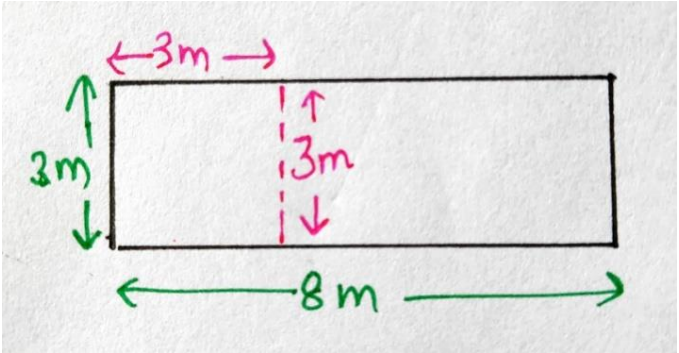
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## Area

## LEVEL-I

**1. Correct option:(A)**

The Maximum length of the side of a square sheet that can be cut off from a rectangular sheet of size 8m x 3m
is 3m because square has all its four sides equal and for that the maximum possible side length which can be cut is 3m.




**2.Correct option:(B)**

Given: edge/side of cube =5m
Total surface area of cube = 6 x $(\text{side})^2$

$= 6 \times (5m)^2$

$= 6 \times 25\ m^2$

Total surface area of cube = $150m^2$

**3. Correct option:(B)**

Area is the amount of surface covered by a two-dimensional shape, measured in square units.

SI unit of area is $m^2$(square meter)

**4. Correct option:(C)**

1 km = 10 hectometer

1km x 1km = 10 hectometer x 10 hectometer

$1km^2 = 100$ hectare

1 sq.km= 100 hectare

**5. Correct option:(C)**

Area is the amount of surface covered by a two-dimensional shape, measured in square units.

SI unit of area is $m^2$(square meter).

**6.Correct option:(D)**

**Given:** area of the square = $256m^2$

To find: perimeter of square

We know

Area of square = $(side)^2$

$256m^2=(side)^2$

$\sqrt{256}$ m = side

16m= side

Or side of square = 16m

Now,

Perimeter of square = 4 x side

= 4 x 16 m

Perimeter of square = 64m

**7. Correct option:(D)**

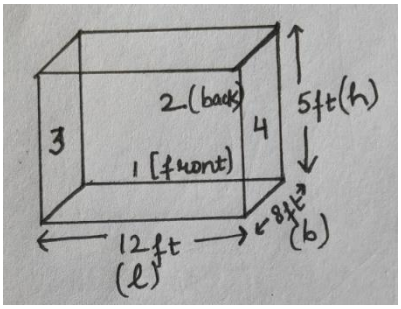
1 km = 1000 m

1km x 1km = 1000 m x 1000 m

$1km^2 = 10^6\ m^2$

1 sq.km= $10^6\ m^2$

**8.Correct option:(A)**




**Given:**

Length of the shop(l) = 12ft

Breadth of the shop(b) = 8ft

Height of the shop(h) = 5ft

Area of the four walls= curved surface area of the cuboid

=area of (wall 1 + 2 +3 +4)

$=lh + lh + bh + bh$

$= 2x\ (lh + bh)$

$= 2x\ (12 \times 5 + 8 \times 5)\ ft^2$

$=2x\ (60 + 40) ft^2$

$= 2 \times 100\ ft^2$

$=200\ ft^2$

**Hence,** area of the four walls is $200\ ft^2$

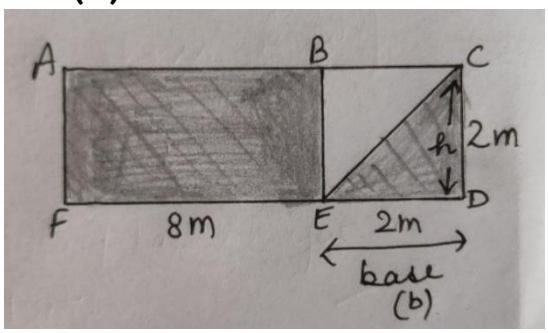
**9.Correct option:(C)**

In order to find the area of a rose leaf the best option is a **millimeter graph paper.**

A millimeter graph paper has ten squares per centimeter.

It would be suitable to find the area enclosed by the smaller portions of the rose leaf.

**10. Correct option:(B)**




Area of rectangle ABEF = Length × breadth

$= 8m \times 2m$

$= 16m^2$

Area of triangle CDE = $\frac{1}{2}$ base × height

$= \frac{1}{2} \times 2 \times 2$

$= 2\ m^2$

Area of shaded region = area of (rectangle ABEF + Triangle CDE)

$= (16 + 2)\ m^2$

$= 18\ m^2$

**11. Correct option: (B)**

Given: radius = 21 cm

And π= 22/7

Area of circle = $\pi r^2$

$=\frac{22}{7} \times 21 \times 21$

$=1386\ cm^2$

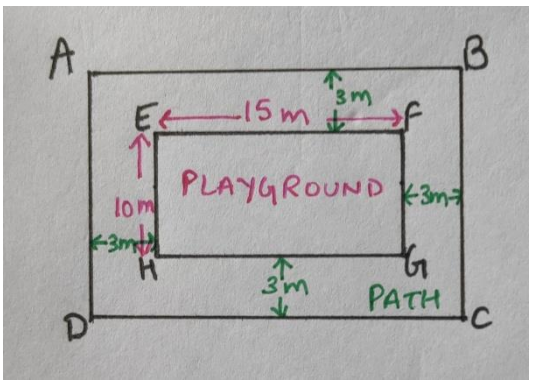
**12. Correct option:(A)**

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Area is the amount of surface covered by a two-dimensional shape, measured in square units.

C.G.S unit of area is $cm^2$(square centimeter).

**13.Correct option:(D)**

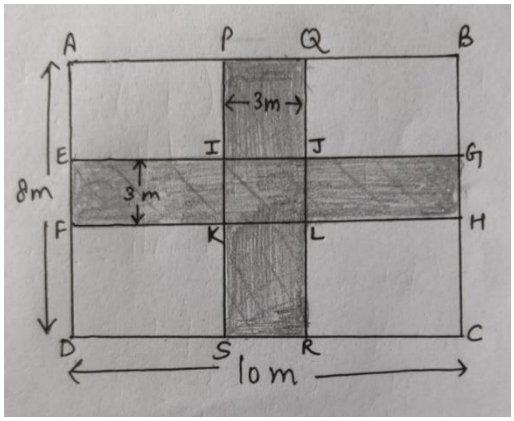



Let EFGH is the rectangular ground.
ABCD is the boundary of the path around the rectangular field.
Length of rectangular ground =15m
Breadth of the rectangular ground=10m
Area of rectangular ground (EFGH)= length x breadth
= 15m x 10m
= $150m^2$
Width of the path = 3m
Length of rectangle ABCD = 15m+3m+3m
= 21m
Breadth of rectangle ABCD = 10m+3m+3m
= 16m
Area of rectangle ABCD = length x breadth
= 21m x 16m
= $336m^2$
Area of the path= Area of rectangle ABCD- Area of rectangular ground (EFGH)
=$336m^2 - 150m^2$
=$186m^2$
Hence, area of the path = $186m^2$

**14.Correct option:(C)**

Hectare is a bigger unit of area.
1 hectare = $10000m^2$
The area of a circular cricket ground is approximately 24.64 hectares.
So, hectare is an appropriate unit to measure the area of a cricket ground.

**15. Correct option: (D)**




Given: length of rectangular field = 10m
Breadth = 8 m
Width of the paths (PQRS and EGHF) = 3 m

Length of Path PQRS =8m
Breadth = 3m
Area of PQRS= length x breadth
= 8m x 3m
= 24 $m^2$
Length of Path EGHF =10m
Breadth = 3m
Area of EGFH= length x breadth
= 10m x 3m
= 30$m^2$
Side of square IJLK= 3m
Area of square IJLK= side x side
= 3m x 3m
= 9 $m^2$
Total area occupied by the paths = area of (rectangle PQRS + rectangle EGHF – square IJLK)
= (24+30-9) $m^2$
=45$m^2$

**16.Correct option:(A)**
The appropriate unit to measure the area of a textbook is $cm^2$,because the dimensions of a normal textbook are in cm.
$km^2$ and $m^2$ would be too large to measure the area of a textbook.
$mm^2$ would be too small to measure the area of a textbook.

**17.Correct option:(B)**
Area of rectangle FEBA = Length × breadth
= 12m × 4m
= 48 $m^2$
Area of triangle EBC = $\frac{1}{2}$ base × height
= $\frac{1}{2} \times 4 \times 4$
= 8 $m^2$
Area of shaded region = area of (rectangle FEBA + Triangle EBC)
= (48 + 8) $m^2$
= 56$m^2$

**18.Correct option:(A)**
1 barn=$10^{-28}$ square meters

**19.Correct option:(B)**
**Given:**
Radius of the wire = 3.5m
Ratio of length to breadth =6:5
Let the length = 6x
Breadth = 5x
**To find:** area of the rectangle
**As the circular wire is bent to form a rectangle, perimeter remains the same**
**Perimeter of rectangle = perimeter of circle**
2× (length + breadth) = 2πr
2× (6x + 5x) = 2 ×$\frac{22}{7} \times 3.5$m
2×(11x) = 22m
22x = 22m

$x = \frac{22m}{22}$

x = 1m

length = 6x = 6 ×1m=6m

Breadth = 5x =5×1m=5m

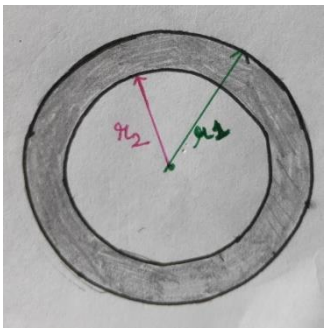
Area of rectangle= length ×breadth

=6m ×5m

$=30m^2$

Hence, area of the rectangle is $30m^2$

**20.Correct option:(A)**



**Given:** outer radius of the ring ($r_1$) =4cm

Inner radius of the ring ($r_2$)=3cm

Area of ring = $\pi(r_1^2 - r_2^2)$

$=\frac{22}{7}(4^2 - 3^2)cm^2$

$=\frac{22}{7}(16 - 9)cm^2$

$= \frac{22}{7} x\ 7cm^2$

Area of the ring $=22cm^2$

**21.Correct option:(A)**

**Given:** ratio of volume of two cubes = 729:1331=$\frac{729}{1331}$----------(1)

To find: ratio of total surface area of two cubes

Let the side of cube 1 = x

The side of cube 2 = y

Volume of cube =$(side)^3$

Volume of 1st cube =$(x)^3$

Volume of 2nd cube =$(y)^3$

ratio of volume of 1st cube to 2nd cube = $\frac{(x)^3}{(y)^3}$ ----------(2)

Equating (1) and (2)

$\frac{(x)^3}{(y)^3} = \frac{729}{1331}$

$\frac{(x)^3}{(y)^3} = \frac{(9)^3}{(11)^3}$

**Cancelling the cubes both side**

$\frac{x}{y} = \frac{9}{11}$ ----------(3)

Total surface area of cube =6 × $(side)^2$

Total surface area of 1st cube =6 × $(x)^2$

Total surface area of 2nd cube =6 × $(y)^2$

ratio of total surface area of 1st cube to 2nd cube = $\frac{6(x)^2}{6(y)^2}$

$=\left(\frac{x}{y}\right)^2$ ----------(4)

Putting value of $\frac{x}{y}$ from equation (3) in equation (4)

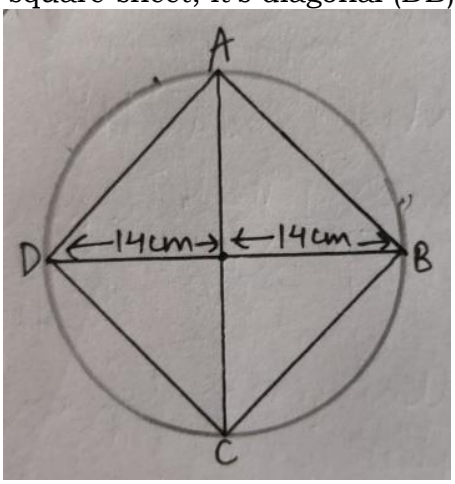
ratio of total surface area of 1st cube to 2nd cube $=\left(\frac{9}{11}\right)^2$

ratio of total surface area of 1st cube to 2nd cube $=\frac{81}{121}$

ratio of total surface area of 1st cube to 2nd cube =81:121

**22. Correct option: (A)**

To get maximum area of the square sheet, it's diagonal (DB) is equal to the diameter of the




circle

Given: radius of the circle= 14cm

Diameter (DB) = 2r

= 2 x 14 cm

= 28 cm

Diagonal of square (DB)= diameter of the circle

$\sqrt{2}$ x side = 28 cm

Side = $\frac{28}{\sqrt{2}}$ cm

Area of the square ABCD = side x side

$= \frac{28}{\sqrt{2}}$ cm x $\frac{28}{\sqrt{2}}$ cm

$= \frac{28\times28}{2}$ $cm^2$

= 392 $cm^2$

Area of the circle = $\pi r^2$

$=\frac{22}{7}\times 14\times 14$

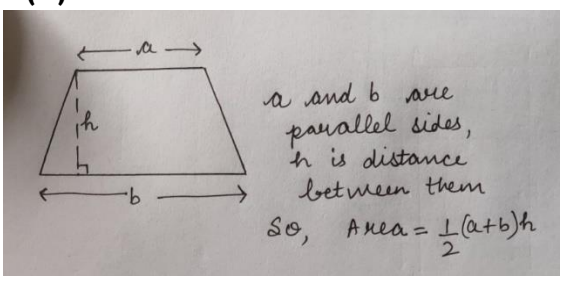
$=616\ cm^2$

Area of remaining cardboard = Area of circle – area of square ABCD

= 616-392 $cm^2$

= 224 $cm^2$

**23. Correct option: (A)**



Area of trapezium $=\frac{1}{2}\times$ (sum of parallel sides) $\times$ (distance between them)

$=\frac{1}{2}\times(a+b)\times(h)$

$=\frac{1}{2}\times(24+20)\times(7)$

$=\frac{1}{2}\times(44)\times(7)$

$=$**154** $cm^2$

Area of trapezium =**154** $cm^2$

According to question

Area of circle = area of trapezium

$\pi r^2 = 154cm^2$

$\frac{22}{7}r^2 = 154\ cm^2$

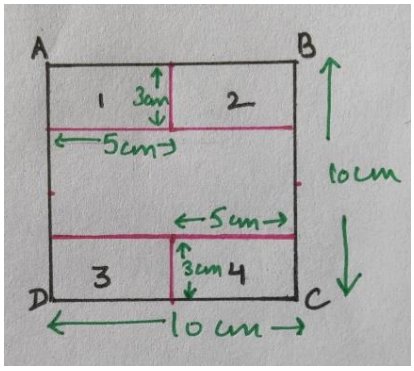
$r^2 = \frac{7}{22}x\ 154cm^2$

$r^2 = 49cm^2$

$r = \sqrt{49}cm$

r = 7 cm

**radius of the circle = 7cm**




**24. Correct option: (B)**

Given: side of the square sheet =10cm

Length of rectangular sheet =5cm

Breadth of rectangular sheet= 3cm

Area of the square sheet (ABCD)= side x side

= 10cm x 10cm

$=100cm^2$

Area of 1 rectangular piece =length x breadth

=5cm x3cm

$=15cm^2$

Area of 4 rectangular pieces =4 x$15cm^2$

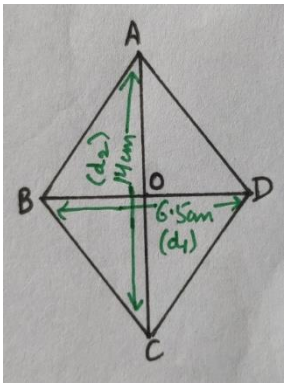
$=60\ cm^2$

Area of aluminum sheet leftover = Area of the square sheet (ABCD)- Area of 4 rectangular pieces

= 100 $cm^2$-60 $cm^2$

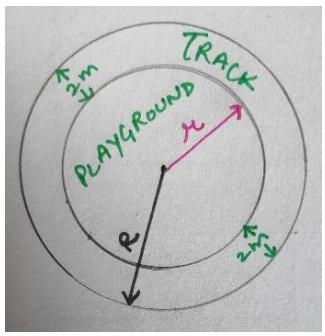
=40 $cm^2$

**25.Correct option:(A)**




Given: diagonal ($d_1$) = 6.5cm
diagonal ($d_2$) = 14cm
Area of rhombus = $\frac{1}{2}$ x $d_1$ x $d_2$
Area of rhombus = $\frac{1}{2}$ x 6.5cm x 14cm
Area of rhombus = $\frac{1}{2}$ x91 $cm^2$
Area of rhombus = 45.5 $cm^2$
As rhombus is divided into 4 equal triangles
So,
Area of one the 4 triangles (AOD) = $\frac{1}{4}$x Area of rhombus
Area of one the 4 triangles (AOD) = $\frac{1}{4}$x 45.5 $cm^2$
Area of one the 4 triangles (AOD) = 11.375$cm^2$

**26.Correct option:(C)**




Given: circumference of the playground = 132m
Width of the circular track = 2m
To find: area of the circular track

- Calculation for Radius and area of the playground

We know
Circumference of circular track = 2πr

$$132m=2\pi r \ (\pi = \frac{22}{7})$$

$$132m = 2x\frac{22}{7} x\ r$$

$$132 = \frac{44}{7} r$$

$$\frac{132x7}{44} = r$$

$$21m = r$$

So, radius(r) of the playground = 21m
Area of the playground = $\pi r^2$

$$=\pi \times 21 \times 21\ m^2$$

Area of the playground =441π $m^2$

- Calculation for area of the circular track

Radius (R)of the playground including track=21m + 2m

= 23m

Area of the playground including track = $\pi R^2$

=π x 23 × 23 $m^2$

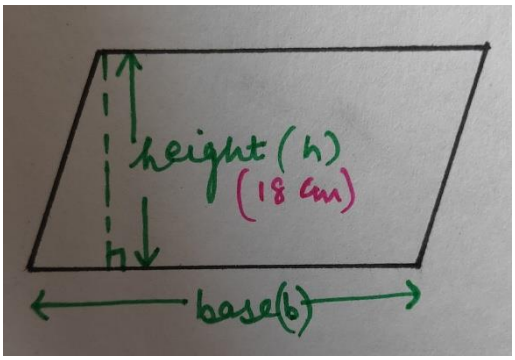
Area of the playground including track =529π$m^2$

Area of the circular track = Area of the playground including track- Area of the playground

= 529π$m^2$-441π $m^2$

=88 π$m^2$

Hence, area of the circular track = 88π$m^2$




**27.Correct option:(D)**

**Given:**

Area of parallelogram =144$cm^2$

Height=18 cm

To find: length of base

Area of parallelogram = base x height

144$cm^2$ = base x 18cm

$\frac{144}{18}$ cm = base

8cm = base

Hence, length of base is 8 cm.

**28. Correct option: (B)**

Given: radius = 2.1 m

And π= 22/7

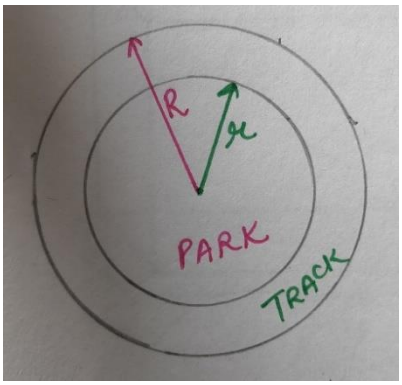
Area of circle = π$r^2$

$=\frac{22}{7} \times 2.1 \times 2.1$

=13.86 $m^2$

**29.Correct option:(A)**

**Given:** difference between the circumference of track and the park =132m




Let the radius of the park = r

The radius of the park including the track = R

Width of the track= R-r

Circumference of the park = 2πr

Circumference of the park including the track = 2πR

Difference between circumference of track and park = 2πR-2πr

=2π(R-r)

Now, given difference between the circumference of track and park=132m

So, we get $2\pi(R-r) = 132m$

$2 \times \frac{22}{7}(R-r) = 132m$

$\frac{44}{7}(R-r) = 132m$

$(R-r) = 132 \times \frac{7}{44}m$

$(R-r) = 132 \times \frac{7}{44}m$

$(R-r) = 21\ m$

So, the width of the track is 21m

**30.Correct option:(B)**

Given: radius = 14cm

And π= 22/7

Area of semicircle = $\frac{\pi r^2}{2}$

$= \frac{1}{2} \times \frac{22}{7} \times 14 \times 14$

$= 308cm^2$

## LEVEL-II

**1. Correct option: (A)**

1hectare = 10000 $m^2$

50hectare = $50 \times 10000\ m^2$

$= 5 \times 10^5\ m^2$

**2. Correct option: (A)**

Area of Triangle = $\frac{1}{2}$ base × height

$= \frac{1}{2} \times 5 \times 8$

$= 20\ m^2$

**3. Correct option: (A)**

Given: radius of circle (r) = 200m

Area of circle = $\pi\ r^2$

$= \pi \times 200^2$

$= \pi \times 200 \times 200$

$= 40000\ \pi$

**4. Correct option: (C)**

Given: side of square = 10 cm

Perimeter of square = 4×side

= 4×10 cm

= 40 cm

**5. Correct option: (A)**

Given:

radius of circle (r) =7 cm

Area of circle = $\pi r^2$

$= \pi \times 7^2$

$= \pi \times 7 \times 7$

$= 49\ \pi cm^2$

**6. Correct option: (A)**

Given: area = $120cm^2$

breadth (b) = 6 cm

To find: length (L)

Area of rectangle =length ×breadth

120 = L×6

120/6 = L

20 = L

Or length = 20 cm

**7. Correct option: (B)**

Given: length (L) = 9 cm

breadth (b) = 6cm

Perimeter = 2(length + breadth)

= 2(9+6)

=2(15)

= 30 cm

**8. Correct option: (B)**

Given: area = $120\ m^2$

length (L) = 12 m

To find: breadth(b)

Area of rectangle =length ×breadth

120 = 12×b

120/12 = b

10 = b

Or breadth = 10m

**9.Correct option:(B)**

Given: edge/side of cube =8m

Total surface area of cube = 6 x $(side)^2$

= 6 x $(8m)^2$

= 6 x $64\ m^2$

Total surface area of cube = $384m^2$

**10.Correct option:(D)**

Given:

Total surface area of the cube= $726m^2$

We know,
Total surface area of cube = 6 x $(side)^2$

$726m^2 = 6 \times (side)^2$

$\frac{726}{6} m^2 = (side)^2$

$121m^2 = (side)^2$

$\sqrt{121}\, m = side$

$11m = side$

Edge/side of cube = 11m

**11.Correct option:(A)**

**Given:**

Perimeter of square = 16cm

To find: area of square

We know

Perimeter of square = 4 x side

$16\ cm = 4 \times side$

$\frac{16}{4} cm = side$

$4cm = side$

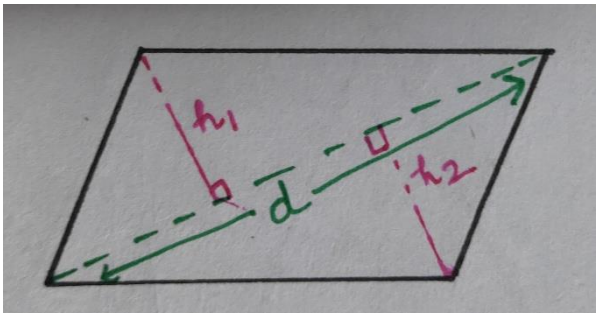
Area of square = $(side)^2$

$= (4cm)^2$

$= 16cm^2$

**Hence, area** of square = $16cm^2$

**12.Correct option:(D)**




**Given:** area of quadrilateral= $420cm^2$

Altitude $(h_1) = 9cm$

Altitude $(h_2) = 12cm$

Let the diagonal=d

Area of quadrilateral= $\frac{1}{2} x\ d\ x(h_1 + h_2)$

$420 = \frac{1}{2} x\ d\ x\ (9 + 12)$

$420 \times 2 = d \times 21$

$840 = d \times 21$

$\frac{840}{21} = d$

$40\ cm = d$

**Hence,** Length of diagonal= 40 cm

**13.Correct options:(A), (D)**

**In Standard International system (SI),** the unit of area is square meter $(m^2)$

In short form square meter is written as sq m.

Hectare is a bigger unit of area.

1 hectare $=10000m^2$

**14. Correct options:( A, B, C)**

$1km = 10^3m$

$1km \times 1\ km = 10^3\ m \times 10^3m$

$1km^2 = 10^6m^2$--------(1)

$1m = 10^2cm$

$1m \times 1\ m = 10^2cm \times 10^2cm$

$1m^2 = 10^4cm^2$

Putting in equation 1

$1km^2 = 10^6 \times 10^4\ cm^2$

$1km^2 = 10^{10}cm^2$----------(2)

Also, $1cm = 10\ mm$

$1cm \times 1\ cm = 10mm \times 10\ mm$

$1cm^2 = 10^2mm^2$

Putting in equation 2

$1km^2 = 10^{10} \times 10^2mm^2$

$1km^2 = 10^{12}mm^2$

**15. Correct options:( A, D)**

$1cm = 10^{-2}m$

$1cm \times 1\ cm = 10^{-2}m \times 10^{-2}m$

$1cm^2 = 10^{-4}m^2$

Or $1cm^2 = 1/10000\ m^2$

**16. Correct options:( A, B)**

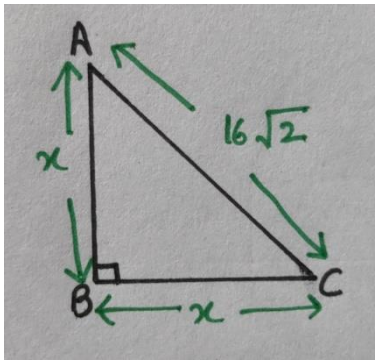
$1m = 10^2cm$

$1m \times 1\ m = 10^2cm \times 10^2cm$

$1m^2 = 10^4cm^2$

Or $1m^2 = 10000\ cm^2$

**17.Correct option:(A)**




- **To find the base and height of triangle**

Triangle ABC is a right-angled isosceles triangle

Given: Hypotenuse (AC)=$16\sqrt{2}$cm

Let Base (BC) = Perpendicular (AB)= x cm

Using Pythagoras theorem in triangle ABC

$(\text{hypotenuse})^2 = (\text{perpendicular})^2 + (\text{base})^2$

$(AC)^2 = (AB)^2 + (BC)^2$

$\left(16\sqrt{2}\right)^2 = x^2 + x^2$

$16 \times 16 \times \sqrt{2} \times \sqrt{2} = 2x^2$

$512 = 2x^2$

$\frac{512}{2} = x^2$

$256=x^2$

$\sqrt{256} = x$

$16=x$

Hence,

Base =height =16 cm

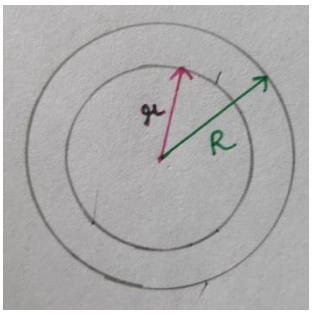
- **To find the area of the triangle**

Area of triangle = $\frac{1}{2}$x base x height

= $\frac{1}{2}$x 16cm x 16cm

Area of triangle =$128cm^2$

**18.Correct option:(A), (D)**

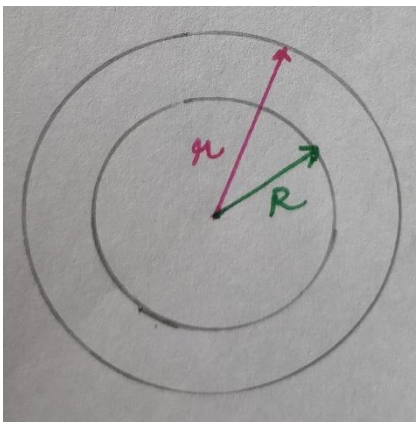



If,

outer radius of the ring= (R)

Inner radius of the ring = (r)

Area of ring = $\pi(R^2 - r^2)$



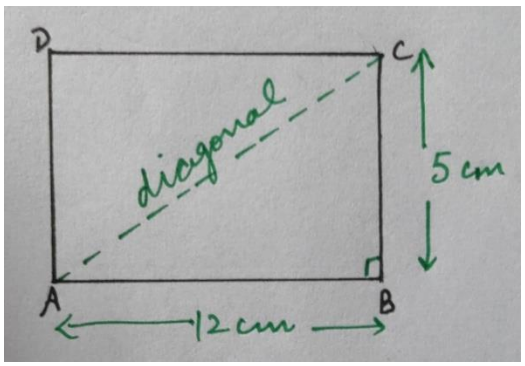

If,

outer radius of the ring= (r)

Inner radius of the ring = (R)

Area of ring = $\pi(r^2 - R^2)$

**19.Correct option:(D)**



Triangle ABC is a right-angled triangle
AB = base = 12cm
BC=perpendicular=5cm
AC= hypotenuse/diagonal
Using Pythagoras theorem in triangle ABC
$(\text{hypotenuse})^2 = (\text{base})^2 + (\text{perpendicular})^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$(AC)^2 = (12)^2 + (5)^2$
$(AC)^2 = 144 + 25$
$(AC)^2 = 169$
$AC = \sqrt{169}$
AC= 13cm
**So,** length of diagonal = 13cm
**20.Correct option:(B)**
Given:
Breadth(b) = 3cm
length (l) = 8cm
To find: area of rectangle
Area of rectangular sheet =length ×breadth
= 8cm x 3cm
Area of rectangular sheet = 24 $cm^2$
**21.Correct option:(A)**
Given: area = 15$m^2$
breadth(b)=3m
To find: length (l)
Area of rectangle =length ×breadth
15 = l × 3
15/3 = l
5m = l
Or
length of rectangular plot = 5m
**22. Correct option:(B)**
Given: area = 120 $m^2$
length (L) = 12 m
To find: breadth(b)
Area of rectangle =length ×breadth
120 = 12×b
120/12 = b
10m = b
Or
breadth = 10m
1m=100cm
10m= 1000cm
So, breadth of rectangular table = 1000cm
**23. Assertion and Reason**
**Assertion (A):** Weight is a force acting on an object due to gravity.
**Reason (R):** Weight depends on the amount of matter in the object.
**Answer: B) Both A and R are true, but R is not the correct explanation of A.**

**Explanation:**

- The assertion is **true** because **weight is the gravitational force** acting on an object.
- The reason is also **true** since weight depends on mass (amount of matter):

$$W = mg$$

- However, the reason does **not explain** why weight is a force due to gravity. It only states one factor affecting weight.

**24. Assertion and Reason**

**Assertion (A):** The area of irregular surfaces can be measured using graph paper.
**Reason (R):** The area is calculated by counting complete and incomplete squares.
**Answer: A) Both A and R are true, and R is the correct explanation of A.**
**Explanation:**

- Irregular figures are commonly measured using graph paper.
- Their area is estimated by counting complete squares and combining incomplete squares.
- Therefore, the reason correctly explains the assertion.

**25. Matrix Matching Type**

| **Column I** | **Column II** | **Explanation** |
|---|---|---|
| a) Area of trapezium is | **Q** | Area of trapezium = $\frac{1}{2}$×(sum of parallel sides)×distance between them |
| b) If the diagonal of a square is $\sqrt{8}$m, then its area is | **S** | Side = $\frac{\sqrt{8}}{\sqrt{2}}$ = 2m, Area = $2^2$ = 4 $m^2$ |
| c) Total surface area of a cube | **R** *(if edge = $\sqrt{10}cm$)* | TSA = $6a^2 = 6 \times 10 = 60$ $cm^2$ |
| d) The diagonal of a square | **P** *(if side = 10 cm)* | Diagonal = $10\sqrt{2}$ |

**Final Matching**

- **a → Q**
- **b → S**
- **c → R**
- **d → P**

**Note:** The image is blurred for parts (c) and (d). The matching above is based on the visible options and the standard formulas.

**26. Matrix Matching Type**

| **Figure** | **Match** | **Explanation** |
|---|---|---|
| a) Circle | **R** | Area of circle = $\pi r^2$ |
| b) Sphere | **T** | Surface area of sphere = $4\pi r^2$ |
| c) Ring (Annulus) | **Q** | Area = $\pi(R^2 - r^2)$ |
| d) Cube | **P** | Total surface area = $6r^2$(here $r$represents the edge length) |
| e) Cylinder | **S** | Total surface area = $2\pi r(r + h)$ |

**Final Matching**

- **a → R**
- **b → T**
- **c → Q**
- **d → P**
- **e → S**

**27. Integer Type**
**Question:** How many square metres are there in 1 hectare?
**Answer: 10000**
**Explanation:** 1 hectare = 10000 $m^2$
**28. Integer Type**
**Question:** How many grams are there in 1 kilogram?
**Answer: 1000**
**Explanation:** 1 kg = 1000 g
**29. Integer Type**
**Question:** How many months in a year have 31 days?
**Answer: 7**
**Explanation:**
The months with 31 days are:

- January
- March
- May
- July
- August
- October
- December

Total = **7 months**.

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## Volume

## LEVEL-I

**1. Correct option:( C)**
Volume of block of wood/cuboid = length x breadth x height
=20 cm x 10 cm x 8 cm
= 1600$cm^3$
**2. Correct option:( D)**
Volume of block of wood/cuboid = length x breadth x height
=30 cm x 15cm x 10 cm
= 4500$cm^3$
**3. Correct option:( B)**
Tank is in the shape of cuboid
Given:
Length = 1.4m
Width =1m
Height/depth=0.7m
Volume of the tank= length x width x height
=1.4 m x 1m x 0.7m
= 0.98$m^3$
1 $m^3$ = 1000 litres
0.98 $m^3$ = 0.98 x1000 litres
$= \frac{98}{100}$x1000 litres
0.98 $m^3$ = 980 litres
Volume of tank = 980 liters

**4. Correct option:(A)**

Given: radius = 3 cm

Height= 4cm

And $\pi = \frac{22}{7}$

Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 4$

$= 37.7 cm^3$

**5. Correct option:(B)**

1 liter= 1000 ml

We know 1ml=1 $cm^3$

So, 1litre = 1000 $cm^3$

(1 $cm^3 = 10^{-6}$ $m^3$)

1litre= 1000 x $10^{-6}$ $m^3$

1 liter = $10^{-3}$ $m^3$

0r

1 liter = $\frac{1}{1000}$ $m^3$

1 $m^3$ = 1000 litres

**6.Correct option:(A)**

Let the length of the cuboid = x units

According to question

Breadth= $\frac{x}{2}$ units

Height = 2x units

Volume of cuboid = length ×breadth × height

= x units × $\frac{x}{2}$ units× 2x units

Volume of cuboid = **$x^3$ cubic units**

**7.Correct option:(B)**

Given:

Length = 2m= 200cm (1m=100cm)

breadth =10cm

Height=1mm=0.1cm (1mm=0.1cm)

Volume of the cuboid= length x breadth x height

=200cm x 10cm x 0.1cm

= $200 cm^3$

Volume of the cuboid=$200 cm^3$

**8. Correct option:(C)**

Given:

Length = 1.8m= 1.8 X 100cm = 180cm (1m=100cm)

Breadth/width =9cm

Height=60cm

Volume of the box/cuboid= length x breadth x height

=180cm x 9cm x 60cm

= $97200 cm^3$

Volume of the cuboid=$97200 cm^3$

**9. Correct option:(D)**

Given:

Length = 6cm

breadth =4.5cm
Height=40mm= 40x0.1cm =4cm (1mm=0.1cm)
Volume of the soap box/cuboid= length x breadth x height
=6cm x 4.5cm x 4cm
= $108cm^3$
Volume of the soap box=$108cm^3$

**10.Correct option:(B)**

Given: radius = 4cm
Height= 21mm = 21 x0.1 cm = 2.1 cm (1mm=0.1cm)
And $\pi = \frac{22}{7}$
Volume of cone =$\frac{1}{3}\pi r^2 h$
$= \frac{1}{3} \times \frac{22}{7} \times 4 \times 4 \times 2.1$
= $35.2cm^3$
Volume of cone = $35.2cm^3$

**11.Correct option:(B)**

Length of the cuboid(L)= 21cm
Breadth of the cuboid(b)=9cm
Height of the cuboid(h)=8cm
Volume of the wooden block/cuboid= length x breadth x height
=21cm x 9cm x 8cm
= $1512\ cm^3$
Volume of the wooden block =$1512\ cm^3$
Number of cubes that can be cut along the length = $\frac{21}{3} = 7$
Number of cubes that can be cut along the breadth = $\frac{9}{3} = 3$
Number of cubes that can be cut along the height = $\frac{8}{3} = 2.6$
As the number of cubes cut along height is not a natural number so we can just cut 2 cubes along the height and rest of the volume is wasted.
Total number of cubes cut = 7 x 3 x 2
= 42
Volume of each cube = $side^3$
= $(3cm)^3$
Volume of each cube =$27cm^3$
Volume of all 42 cubes =27 x 42 $cm^3$
=$1134cm^3$
Volume of the waste wood = volume of wooden block – volume of all the 42 cut cubes
= $1512\ cm^3$-$1134cm^3$
=$378cm^3$

**12.Correct option:(D)**

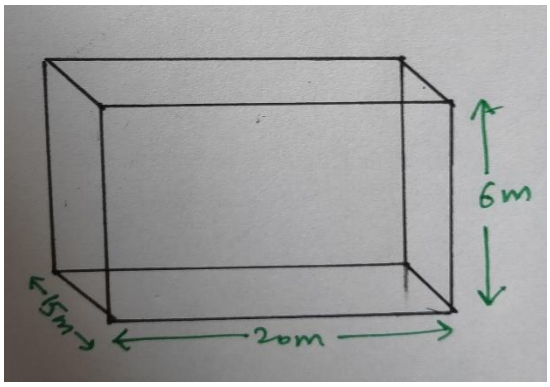
**Given:** ratio of the sides of cubes= 2:5
To find: ratio of volumes
let the side of first cube= 2x
the side of second cube =5x
Volume of cube = $side^3$
Volume of first cube = $(2x)^3$
Volume of first cube = $2\times2\times2\ x^3$
Volume of first cube =$8x^3$

Volume of second cube = $(5x)^3$
Volume of second cube = $5\times5\times5\ x^3$
Volume of second cube = $125x^3$
Ratio of volume of first cube to second cube = $\frac{\text{Volume of first cube}}{\text{volume of second cube}}$

$$= \frac{8x^3}{125x^3}$$
$$= \frac{8}{125}$$
$$= 8:125$$

Hence, Ratio of volume of first cube to second cube=8:125

**13.Correct option:(A)**




Given:
Total population of village =4000
water required by one person=150 liters

**Dimensions of the tank:**

Length =20m
Breadth=15m
Height=6m
To find: number of days for which water is sufficient
Number of days for which the water will be sufficient = $\frac{\text{volume of the tank}}{\text{total water required per day}}$----------(1)

- Calculation for volume of the tank

Volume of the tank/cuboid = length x breadth x height
=20m x 15m x 6m
= $1800m^3$

**Volume of the tank=$1800m^3$**

- Calculation for water required per day

water required by 1 person=150 liters
water required by 4000 persons per day=150x4000 liters
=600000litres

1 liter = $\frac{1}{1000}\ m^3$
600000 liters = $600000x\frac{1}{1000}\ m^3$
=$600m^3$
water required by 4000 persons per day=$600m^3$

**Total water required by 4000 persons per day=$600m^3$**

**From equation (1)**

Number of days for which the water will be sufficient = $\frac{\text{volume of the tank}}{\text{total water required per day}}$

Number of days for which the water will be sufficient = $\frac{1800m^3}{600m^3}$

Number of days for which the water will be sufficient = 3 days

**14.Correct option:(D)**

Given:

Length = 8.2m

breadth =5.3m

Height=2.6m

Volume of cuboid= length x breadth x height

=8.2m x 5.3m x 2.6m

= $112.996m^3$

Volume of the cuboid=$112.996m^3$

**15.Correct option:(C)**

Given:

Length of cuboid = 12m

breadth of cuboid=6m

Height of cuboid=3m

To find: side of the cube

When cuboid is recast into cube its volume remains same

Volume of cube= volume of cuboid

$Side^3$ = length x breadth x height

$Side^3 = 12m \times 6m \times 3m$

$Side^3 = 216m^3$

$side = \sqrt[3]{216}\ m$

**side= 6m**

so,

side of the cube is 6m

**16. Correct option:(B)**

Volume of cuboid = length x breadth x height

=24 cm x 12cm x 5 cm

= $1440cm^3$

**17. Correct option: (B)**

Given: radius of the cylinder = 5.5 cm

Height of the cylinder = 14 cm

Side of cube = l cm

Also π= 22/7

When cylinder is recast into cube its volume remains same

Volume of cube = volume of cylinder

$Side^3 = \pi r^2 h$

$l^3 = \frac{22}{7} \times 5.5^2 \times 14$

$l^3 = \frac{22}{7} \times 5.5 \times 5.5 \times 14$

$l^3 = 1331\ cm^3$

$l = 11cm$

**18.Correct option:(C)**

Given:

Area of the square =$100\ cm^2$

$(side)^2 = 100\ cm^2$

$(side)^2 = (10cm)^2$

Side of square = 10cm

**According to the question**

height of cuboid = side of the square = 10cm

length of cuboid = height + 2cm= 10cm+2cm = 12cm

breadth of cuboid = height -2cm= 10cm-2cm = 8cm

Volume of cuboid= length x breadth x height

=12cm x 8cm x 10cm

= $960cm^3$

Volume of the cuboid=$960cm^3$

**19.Correct option:(A)**

Substitute the values:

$$a^3 = \frac{1}{3} \times \frac{22}{7} \times 22^2 \times 21$$

Since $\frac{21}{3} = 7$,

$$a^3 = \frac{22}{7} \times 22^2 \times 7 = 22^3$$

Therefore,

$$a = \sqrt[3]{22^3} = 22 \text{ m}$$

**20. Correct option:(B)**

**Given:** volume of cube= 512 $cm^3$

To find: length of edge/side of cube

Also Volume of cube = $side^3$

$512cm^3 = (side)^3$

$\sqrt[3]{512}$ cm = side

8cm=side/edge

512= 2x2x2x2x2x2x2x2x2

512= $2^3 \times 2^3 \times 2^3$

Hence, Length of edge of cube=8cm

512= $(2 \times 2 \times 2)^3$

$\sqrt[3]{512} = 8$

**21. Correct option:(B)**

Given: radius = 3 cm

Height= 7cm

And $\pi = \frac{22}{7}$

Volume of cone = $\frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 7$

= $66cm^3$

## LEVEL-II

**1. Correct option: (C)**

(A) A meter scale is a one-meter-long scale used to measure length.

(B) Graph paper or coordinate paper is a writing paper printed with fine lines making up a regular grid. These lines help in plotting graphs.

(C) A measuring jar is used primarily to measure the volume of liquids. A measuring jar has scale markings at varying heights used to measure fixed volume of liquids.

(D) A physical balance is used to measure mass of a body.

**2. Correct option: (B)**

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence,

C.G.S unit of volume is $cm^3$

**3. Correct option: (B)**

1 ml = 1 $cm^3$

The reason for this is, volume of liquid that is present in 1 ml of a liquid is equal to the volume of water that fits in a unit cube of dimensions (1 cm length x 1 cm breadth x 1 cm height)

**4. Correct option: (B)**

Volume is the amount of space occupied by any liquid.

**5. Correct option: (B)**

Length (l)=10m

Breadth(b)=7m

Height(h)=5m

Total surface area of cuboid = 2(lb+bh+hl)

=2(10x7+7x5+5x10)

=2(70+35+50)

=2(155)

= 310 $m^2$

**6. Correct option:( B)**

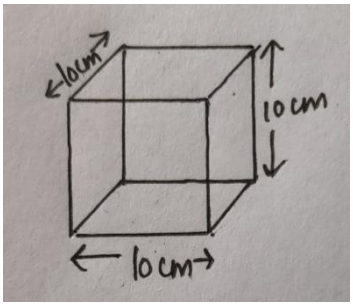
Given: side of the cube = 10 cm

We know

Volume of cube = $side^3$

$= (10cm)^3$

$= 10 \times 10 \times 10\ cm^3$




$=1000\ cm^3$

**7.Correct option:(C)**

**Volume of cube**= $(side)^3$

**As same cube is cut into x cubes so**

Volume of cube of edge 9cm = x × volume of cube of edge 3cm

$(9cm)^3 = x \times (3cm)^3$

$9cm \times 9cm \times 9cm = x \times 3cm \times 3cm \times 3cm$

$729cm^3 = x \times 27cm^3$

$\frac{729}{27} = x$

**27 = x**

**Or** x=27

**8.Correct option:(A)**

**Given:** volume of cube= 343 $cm^3$

To find: total surface area of cube

Also

Volume of cube = $side^3$

$343cm^3 = (side)^3$

$\sqrt[3]{343}\ cm = side$

$7cm = side$

$343 = 7 \times 7 \times 7 = (7)^3$

$\sqrt[3]{343} = 7$

Or side of cube = 7cm

Total surface area of cube = $6 \times side^2$

$= 6 \times (7cm)^2$

$= 6 \times 7cm \times 7cm$

$= 294cm^2$

Total surface area of cube $= 294cm^2$

**9. Answer: A, B and D**

- 1 litre = 1000 cc (1000 $cm^3$)
- 1 litre = $10^{-3}$ $m^3$
- 1 litre = $10^6$ $mm^3$ (because 1 $cm^3$ = 1000 $mm^3$)

**10. Answer: A and C**

- 1000 $cm^3$
- $10^6$ $mm^3$

**11. Answer: A, B, C and D**

- Measuring jar Measures liquid volume.
- Burette Measures and dispenses accurate volumes.
- Pipette Measures/transfers fixed volumes.
- Measuring flask (Volumetric flask) Used to prepare and contain a fixed volume of liquid.

**12. Answer: A, B, C and D**

- $cm^3$ Unit of volume.
- cc (cubic centimetre) Same as $cm^3$.
- $m^3$ SI unit of volume.
- millilitre (mL) Common unit for liquid volume (1 mL = 1 $cm^3$ = 1 cc).

**13. Assertion and Reason**

**Assertion (A):** Volume is measured in cubic metre in the S.I. system.

**Reason (R):** Volume represents the amount of three-dimensional space occupied by a substance.

**Answer: A) Both A and R are true and R is the correct explanation of A.**

**Explanation:**

**Assertion is true:** The SI unit of volume is **cubic metre ($m^3$)**.

**Reason is also true:** Volume is the measure of the three-dimensional space occupied by an object or substance.

Since volume measures three-dimensional space, its SI unit is derived from the SI unit of length (metre) as metre × metre × metre = $m^3$.

Therefore, **R correctly explains A**.

**14. Assertion and Reason**

**Answer: A) Both A and R are true and R is the correct explanation of A.**

**Explanation:**

**Assertion is true:** Measuring jars (graduated cylinders) are commonly used to measure the volume of liquids.

**Reason is true:** They have clearly marked graduations that allow the liquid volume to be

read accurately.
The graduations are the reason why a measuring jar can be used for measuring liquid volume.
Hence, R correctly explains A.

**15. Matrix Matching**

- **Final Matching**

| **Column I** | **Column II** |
|---|---|
| **a)** Cube (10 cm) | **Q (1000 $cm^3$)** |
| **b)** Cuboid (10 × 5 × 2 cm) | **P (100 $cm^3$)** |
| **c)** Sphere (r = 7 cm) | **S (1437 $cm^3$)** |
| **d)** Hemisphere (r = 7 cm) | **R (718.6 $cm^3$)** |

**Answer:**
**a → Q**
**b → P**
**c → S**
**d → R**
**Note:** Option T (0.00143 $m^3$) is approximately equal to 1437 $cm^3$ since $1437\,cm^3 = 0.001437\,m^3$. However, the exact match requested in the matrix is S (1437 $cm^3$).

## Density and Relative Density

## LEVEL-I

### Density:

**1.Correct option:(A)**
Density of a substance is defined as the mass per unit volume of the substance, i.e.,

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

**2. Correct option:(C)**

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

SI unit of mass is kg
SI unit of volume is $m^3$
So,
SI unit of density is $kg/m^3$

**3. Correct option:(A)**

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

C.G.S unit of mass is g
C.G.S unit of volume is $cm^3$
So,
C.G.S unit of density is $g/cm^3$

**4. Correct option:(A), (B)**
$1kg = 1000g$
$1kgm^{-3} = 1000gm^{-3}$
Also,
$1m = 100cm$
$1m^3 = 1000000cm^3$

$1\text{kgm}^{-3} = \frac{1000}{1000000}\text{gcm}^{-3}$

$1\text{kgm}^{-3} = \frac{1}{1000}\text{gcm}^{-3}$

**5. Correct option:(B)**

**Density** of water at 4°C(277K) is $1.0 \times 10^3 \text{kg/m}^3$

**6. Correct option:(A)**

**Density** of ice is $9 \times 10^2 \text{kg/m}^3$

**7. Correct option:(C)**

1kg = 1000g

Also,

1m = 100cm

$1\text{m}^3 = 1000000\text{cm}^3$

$1\text{kgm}^{-3} = \frac{1000}{1000000}\text{gcm}^{-3}$

$1\text{kgm}^{-3} = \frac{1}{1000}\text{gcm}^{-3}$

**Or**

$1\text{gcm}^{-3} = 1000\text{kgm}^{-3}$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

**8. Correct option:(B)**

Given:

Volume of liquid = 500cc

mass of liquid= 1kg=1000g

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{1000}{500}\text{g/cc}$

Density =2 g/cc

**9. Correct option:(C)**

Given:

Volume of liquid = 18cc

mass of liquid= 7.2kg=7.2 X 1000g= 7200 g (1kg=1000g)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{7200}{18}\text{g/cc}$

Density =400g/cc

**10. Correct option:(D)**

M.K.S system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

**Given:** Density =0.75g/cc

We know,

1cc= $1\text{cm}^3$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

$1\text{g/cc} = 10^3\text{kg/m}^3$

$0.75\text{g/cc} = 0.75 \times 10^3\text{kg/m}^3$

$= \frac{75}{100} \times 1000\text{kg/m}^3$

$= 750\text{kg/m}^3$

So, Density in M.K.S system is $750\text{kg/m}^3$

**11. Correct option:(B)**

**C.G.S** system is based on centimeter, **gram** and second as the fundamental units of length, **mass** and time respectively.

Given: Density =450kg/$m^3$

$1kg/m^3 = \frac{1}{1000} g/cc$

$450kg/m^3 = 450 \times \frac{1}{1000} g/cc$

$450kg/m^3 = \frac{45}{100} g/cc$

$450kg/m^3 = 0.45\ g/cc$

**Density** of substance in C.G.S system is $0.45\ g/cc$

**12. Correct option:(C)**

Given: mass of substance= 0.6kg=600g (1kg=1000g)

Density =1.2g/cc

To find: Volume of substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Volume} = \frac{\text{mass}}{\text{density}}$

$\text{Volume} = \frac{600}{1.2} cc$

$= \frac{600 \times 10}{12} cc$

**volume** =500cc

**13. Correct option:(A)**

Given: Volume of liquid= 2litre=2000cc (1litre = 1000cc)

Density =2g/cc

To find: mass of liquid

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

2g/cc x 2000cc = mass

4000g = mass

**Mass = 4000g**

**14. Correct option:(D)**

SI system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

**Given:** Density =$4gcm^{-3}$

We know,

$1gcm^{-3} = 10^3 kgm^{-3}$

$4gcm^{-3} = 4 \times 10^3 kgm^{-3}$

$4gcm^{-3} = 4000kgm^{-3}$

So, Density in SI system is $4000kgm^{-3}$

**15. Correct option:(B)**

Given:

Volume of liquid = $20m^3$

mass of liquid= 5kg

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density $= \frac{5}{20}\text{kgm}^{-3}$

Density $=0.25\text{kgm}^{-3}$

**16. Correct option:(D)**

Given: Volume of liquid= 5litre=5000cc (1litre = 1000cc)

Mass of the liquid= 2 kg=2000g (1kg=1000g)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{2000}{5000}\text{g/cc}$

Density $=0.4\text{g/cc}$

**17. Correct option:(D)**

Given: Volume of body= $2\text{cm}^3$

Density $= 0.52\ \text{gcm}^{-3}$

To find: mass of body

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

$0.52\ \text{gcm}^{-3}$ x $2\text{cm}^3$= mass

1.04g= mass

**Mass= 1.04 g**

$1\text{g} = 10^{-3}\text{kg}$

$1.04\text{g} = 1.04\text{x}10^{-3}\text{kg}$

**So,** mass of body in SI system is $1.04\text{x}10^{-3}\text{kg}$

**18. Answer: (B) No units**

Relative density is the ratio of the density of a substance to the density of water. Since it is a ratio of two similar quantities, the units cancel out. Hence, it is unitless.

**19. Answer: (A) 0.45**

$$\text{Relative Density} = \frac{\text{Density of substance}}{\text{Density of water}} = \frac{0.45}{1} = 0.45$$

**20. Correct option:(B)**

**Given:** Relative density =0.8

**We know,**

**Density** of water at 4°C(277K) is $1.0 \text{ x } 10^3\text{kg/m}^3$ or $10^3\text{kg/m}^3$

$1\text{kg/m}^3 = \frac{1}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = \frac{10^3}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = \frac{1000}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = 1\text{g/cc}$

$\text{Relative density} = \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.8 = \frac{\text{density of substance}}{1\text{g/cc}}$

0.8 x 1g/cc= density of substance

0.8g/cc= density of substance

Density of substance =0.8g/cc

**21. Correct option:(C)**

**Specific gravity/Relative density=** $\frac{\text{density of substance}}{\text{density of water at 4°C}}$

- **To find the density of substance**

**Given:** specific gravity=0.75

**We know,**

**Density** of water at 4°C(277K) is $10^3 kg/m^3$ or $1g/cc$

$1kg/m^3 = \frac{1}{1000} g/cc$

$10^3 kg/m^3 = \frac{10^3}{1000} g/cc$

$10^3 kg/m^3 = \frac{1000}{1000} g/cc$

$10^3 kg/m^3 = 1g/cc$

Specific gravity $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.75 = \frac{\text{density of substance}}{1g/cc}$

0.75 x 1g/cc= density of substance

Density of substance = 0.75 g/cc

- **To find the mass of the substance**

Given: Volume of substance= 800cc

Density of substance = 0.75 g/cc (calculated above)

To find: mass of substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

0.75g/cc x 800cc = mass

$\frac{75}{100}$ x 800 g = mass

600g = mass

600g = mass

**Mass = 600g**

**22. Correct option:(A)**

- **To find the density of substance**

**Given:** Relative density =0.2

**We know,**

**Density** of water at 4°C(277K) is $1.0 \times 10^3 kg/m^3$ or $10^3 kg/m^3$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.2 = \frac{\text{density of substance}}{10^3 kg/m^3}$

0.2 x $10^3 kg/m^3$= density of substance

0.2 x$10^3 kg/m^3$= density of substance

Density of substance =0.2 x$10^3 kg/m^3$

- **To find the mass of the substance**

Given: Volume of substance= $20m^3$

Density of substance =0.2 x$10^3 kg/m^3$ (calculated above)

To find: mass of substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

0.2 x$10^3 kg/m^3$ x $20m^3$ = mass

$\frac{2}{10}$ x$10^3$ x 20 kg = mass

2 x $10^3$ kg = mass

2000kg = mass

**Mass = 2000kg**

**23. Correct option:(B)**

- **To find the density of substance**

**Given:** Relative density =0.2

**We know,**

**Density** of water at 4°C(277K) is $1.0 \times 10^3 kg/m^3$ or $10^3 kg/m^3$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.2 = \frac{\text{density of substance}}{10^3 kg/m^3}$

$0.2 \times 10^3 kg/m^3$= density of substance

$0.2 \times 10^3 kg/m^3$= density of substance

Density of substance $=0.2 \times 10^3 kg/m^3$

- **To find the volume of the substance**

Given: mass of substance= 500kg

Density of substance $=0.2 \times 10^3 kg/m^3$ (calculated above)

To find: volume of substance

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{Volume} = \frac{\text{mass}}{\text{density}}$$

$$\text{Volume} = \frac{500kg}{0.2 \times 10^3 kg/m^3}$$

$$\text{Volume} = \frac{500 \times 10}{2 \times 1000} m^3$$

$$\text{Volume} = 2.5\ m^3$$

**24. Correct option:(B)**

**Given:** density of iron= $7.85 gcm^{-3}$

$1 gcm^{-3} = 10^3 kgm^{-3}$

$7.85 gcm^{-3} = 7.85 \times 10^3 kgm^{-3}$

**Density** of water at 4°C(277K) is $10^3 kgm^{-3}$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density $= \frac{7.85 \times 10^3 kgm^{-3}}{10^3 kgm^{-3}}$

Relative density =7.85

## LEVEL-II

**1. Correct option:(C)**

Density of a substance is defined as the mass per unit volume of the substance, i.e.,

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

**2. Correct option:(D)**

**Specific gravity=** $\frac{\text{density of substance}}{\text{density of water at 4°C}}$

**3. Correct option:(B)**

**Specific gravity=** $\frac{\text{density of substance}}{\text{density of water at 4°C}}$

- **To find the density of substance**

**Given:** specific gravity=0.35

**We know,**

**Density** of water at 4°C(277K) is 1g/cc

Specific gravity $=\frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.35=\frac{\text{density of substance}}{1\text{g/cc}}$

0.35 x 1g/cc= density of substance

Density of substance = 0.35 g/cc

**4. Correct option:(C)**

Given:

Volume of liquid = $49\text{m}^3$

mass of liquid= 7kg

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{7}{49}\text{kgm}^{-3}$

$\text{Density} = 0.14\frac{\text{kg}}{\text{m}^3}$

**5. Correct option:(D)**

M.K.S system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

**Given:** Density =0.85g/cc

We know,

$1\text{cc} = 1\text{cm}^3$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

$1\text{g/cc} = 10^3\text{kg/m}^3$

$0.85\text{g/cc} = 0.85 \text{ x } 10^3\text{kg/m}^3$

$= \frac{85}{100} \text{ x } 1000\text{kg/m}^3$

$= 850\text{kg/m}^3$

So, Density in M.K.S system is $850\frac{\text{kg}}{\text{m}^3}$

**6. Correct option:(A)**

**Given:** mass of iron piece= 7020g

Length = 3cm

Breadth=15cm

Height=20cm

**To find:** density of the iron piece

Volume of iron piece/cuboid = length x breadth x height

=3cm x 15cm x 20cm

$= 900\text{cm}^3$

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{7020}{900}\text{gcm}^{-3}$

$\text{Density} = 7.8\ \text{gcm}^{-3}$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

$7.8\ \text{gcm}^{-3} = 7.8 \text{ x } 10^3\text{kgm}^{-3}$

**7. Correct option:(B)**

**Given:** mass of box= 30kg=30x1000g= 30000g (1kg=1000g)

Length = 6cm

Breadth=3cm

Height=12cm

**To find:** density of the box

Volume of cubical box = length x breadth x height

=6cm x 3cm x 12cm

= 216$cm^3$

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{30000}{216}\frac{g}{cm^3}$

Density =138.8 $\frac{g}{cm^3}$

**8.Correct options:(A, B, C, D)**

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kg

SI unit of volume is $m^3$

So,

SI unit of density is $kgm^{-3}$ or $\frac{kg}{m^3}$

C.G.S unit of mass is g

C.G.S unit of volume is $cm^3$

So,

C.G.S unit of density is $gcm^{-3}$ or $\frac{g}{cm^3}$

**9. Correct options:(A, B, D)**

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kg

SI unit of volume is $m^3$

So,

SI unit of density is $kgm^{-3}$

C.G.S unit of mass is g

C.G.S unit of volume is $cm^3$

So,

C.G.S unit of density is $gcm^{-3}$

**10 . Correct options: (A)**

Given: mass of body=100g

Density = $1gcm^{-3}$

To find: Volume of body

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Volume} = \frac{\text{mass}}{\text{density}}$

$\text{Volume} = \frac{100g}{1gcm^{-3}}$

=100$cm^3$

**volume** =100$cm^3$

**11. Correct options: (A, B)**

$\text{Relative density} = \frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density has no units because it is equal to the ratio of same physical quantity (density).

**12.Correct option: (A)**

**Given:** density of substance = 0.5g/cc

**We know,**

**Density** of water at 4°C(277K) is 1g/cc

Relative density $=\frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density $=\frac{0.5\text{g/cc}}{1\text{g/cc}}$

Relative density =0.5

**13.Correct option: (B)**

**Given:** Relative density =0.4

**We know,**

**Density** of water at 4°C(277K) is $10^3\text{kg/m}^3$ or 1g/cc

$1\text{kg/m}^3 = \frac{1}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = \frac{10^3}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = \frac{1000}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = 1\text{g/cc}$

Relative density $=\frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.4 = \frac{\text{density of substance}}{1\text{g/cc}}$

0.4 x 1g/cc= density of substance

0.4 g/cc= density of substance

Density of substance =0.4g/cc

**14.Correct option: (A)**

**Given:** density of substance = $5000\text{kg/m}^3$

**We know,**

**Density** of water at 4°C(277K) is $10^3\text{kg/m}^3$

Relative density $=\frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density $=\frac{5000\text{kg/m}^3}{10^3\text{kg/m}^3}$

$=\frac{5000\text{kg/m}^3}{1000\text{kg/m}^3}$

Relative density = 5

**15. Correct Matching is: a-PQ; b-R; c-P; d-T**

(a) Density $= \frac{\text{mass}}{\text{volume}}$

C.G.S unit of mass is g

C.G.S unit of volume is $\text{cm}^3$

So,

C.G.S unit of density is $\text{g/cm}^3$

(a) Relative density $=\frac{\text{density of substance}}{\text{density of water at 4°C}}$

(b) $1\text{kg} = 1000\text{g}$

Also,

$1\text{m} = 100\text{cm}$

$1\text{m}^3 = 1000000\text{cm}^3$

$1\text{kgm}^{-3} = \frac{1000}{1000000}\text{gcm}^{-3}$

$1\text{kgm}^{-3} = \frac{1}{1000}\text{gcm}^{-3}$

**Or**

$1\text{g/cm}^3 = 1000\text{kg/m}^3$

(d) Weight is defined as the force exerted on a body by gravity.
Weight is often expressed as
W= mg
Where, w= weight of the body
m = mass of the body
g= acceleration due to gravity
SI unit of mass is kg
SI unit of acceleration due to gravity is $m/s^{-2}$
**SI unit** of weight is same as force that is newton (N).
**1N =** $1kgm/s^{-2}$

**16. Correct Matching is: a-Q; b-R; c-P; d-ST**

(a)Density $= \frac{\text{mass}}{\text{volume}}$
SI unit of mass is kg
SI unit of volume is $m^3$
So,
SI unit of density is $kg/m^3$
(b)
Density $= \frac{\text{mass}}{\text{volume}}$
C.G.S unit of mass is g
C.G.S unit of volume is $cm^3$
So,
C.G.S unit of density is $g/cm^3$
(c) Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$
Relative density has no units because it is equal to the ratio of same physical quantity (density).
(d) **Density** of water at 4°C(277K) is $1000kgm^{-3}$ or $1gcm^{-3}$
($1000kgm^{-3} = 1gcm^{-3}$)

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# Dimensional Formula

## LEVEL-I

**1.Correct option:(B)**

The seven fundamental or the base quantities chosen in SI are called the seven dimensions of the physical world, they are denoted with square brackets [].
Length =[L]
Mass =[M]
Time= [T]
Electric current =[A]
Thermodynamic temperature =[K] or [θ]
Luminous intensity=[Cd]
Amount of substance=[mol]
The expression which shows how and which of the base quantities represent the dimensions of a physical quantity is called the dimensional formula of the given physical quantity.
The dimensions of a physical quantity are the powers to which the base quantities are raised to represent a derived unit of that quantity.
For example,
Velocity $= \frac{\text{displacement}}{\text{time}} = \frac{[L]}{[T]}$
Velocity = $[L^1\ T^{-1}] = [M^0L^1\ T^{-1}]$

So, $[M^0L^1\ T^{-1}]$ is the dimensional formula of velocity.
It tells that unit of velocity depends on [L] and [T] and does not depend on [M].

**2. Correct option:(A)**

The dimensions of a physical quantity are the powers to which the base quantities are raised to represent a derived unit of that quantity.

**For example,**

Volume = length x breadth x height
Length is represented as [L]
Breadth and height also have the units of length.
So,
Volume =[L] x [L] x [L]
= $[L^3]$
Thus, to represent volume we have to raise the [L] to the power 3. Therefore, volume is said to have 3 dimensions in length.

**3. Correct option:(B)**

Velocity $=\frac{\text{displacement}}{\text{time}} = \frac{[L]}{[T]}$ Length/displacement is represented as [L]
Velocity = $[L^1\ T^{-1}] = [M^0L^1\ T^{-1}]$
So, $[M^0L^1\ T^{-1}]$ is the dimensional formula of velocity.
or
$[M^0L\ T^{-1}]$ is the dimensional formula of velocity.

**4. Correct option:(C)**

Area = length x breadth
Length is represented as [L]
Breadth also has the units of length.
So,
Area =[L] x [L]
Area = $[L^2]$= $[M^0L^2\ T^0]$
$[M^0L^2\ T^0]$ is the dimensional formula of area.

**5. Correct option:(B)**

Volume = length x breadth x height
Length is represented as [L]
Breadth and height also have the units of length.
So,
Volume =[L] x [L] x [L]
Volume = $[L^3]$ =$[M^0L^3\ T^0]$
$[M^0L^3\ T^0]$ is the dimensional formula of volume.

**6. Correct option:(D)**

The general formula for acceleration is:
Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

**7. Correct option:(D)**

The general formula for momentum is:
momentum= mass x velocity

**8. Correct option:(A)**

The general formula for force is:
force= mass x acceleration

**9. Correct option:(C)**

Density $= \frac{\text{mass}}{\text{volume}}$

Mass is represented as [M]

$[M^0L^3\ T^0]$ is the dimensional formula of volume.

Density = $\frac{[M]}{[M^0L^3\ T^0]}$

Density $= [M^1L^{-3}\ T^0]$

$[M^1L^{-3}\ T^0]$ is the dimensional formula of density.

Or $[ML^{-3}\ T^0]$ is the dimensional formula of density.

**10. Correct option:(D)**

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Time is represented as [T]

$[M^0L^1T^{-1}]$ is the dimensional formula of velocity.

Acceleration = $\frac{[M^0L^1T^{-1}]}{[T]}$

Acceleration $=[M^0L^1T^{-1-1}]$

Acceleration $=[M^0L^1T^{-2}]$

$[M^0L^1T^{-2}]$ is the dimensional formula of acceleration.

or $[M^0LT^{-2}]$ is the dimensional formula of acceleration.

**11. Correct option:(D)**

Length is represented as [L] $=[M^0LT^0]$

**So, length** has 0 dimensions in time.

Dimensional formula of velocity = $[M^0L^1T^{-1}]$ ( -1 dimensions in time)

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$ (-2 dimensions in time)

Dimensional formula of force = $[M^1L^1T^{-2}]$ (-2 dimensions in time)

**12. Correct option:(B)**

force= mass x acceleration

Mass is represented as [M]

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$

Force = [M]x $[M^0L^1T^{-2}]$

Force = $[M^{1+0}L^1T^{-2}]$

Force = $[M^1L^1T^{-2}]$

Or Force = $[MLT^{-2}]$

Dimensional formula of force = $[MLT^{-2}]$

**13. Correct option:(B)**

Length is represented as [L]

$[M^0L^1T^0]$ is the dimensional formula of length.

**14. Correct option:(D)**

Frequency = number of vibrations/second

Second is the unit of time

And time is represented as [T]

Number of vibrations have no dimensions

So,

Frequency = $\frac{1}{[T]}$

Frequency = $[T^{-1}]$

**So,** dimensions of time in frequency are -1

**15. Correct option:(A)**

Impulse = force x time

Time is represented as [T]

Dimensional formula of force = $[MLT^{-2}]$

So,

Impulse = $[MLT^{-2}]$ x [T]

Impulse = $[MLT^{-2+1}]$

Impulse = $[MLT^{-1}]$

Dimensional formula of impulse = $[MLT^{-1}]$

Dimensions of mass in impulse is 1

**16. Correct option:(A)**

Acceleration due to gravity(g)= $\frac{\text{change in velocity}}{\text{time taken}}$

Time is represented as [T]

$[M^0L^1T^{-1}]$ is the dimensional formula of velocity.

Acceleration due to gravity= $\frac{[M^0L^1T^{-1}]}{[T]}$

Acceleration due to gravity =$[M^0L^1T^{-1-1}]$

Acceleration due to gravity=$[M^0L^1T^{-2}]$

$[M^0L^1T^{-2}]$ is the dimensional formula of acceleration due to gravity.

or $[M^0LT^{-2}]$ is the dimensional formula of acceleration due to gravity.

**17. Correct option:(C)**

(A)Dimensional formula of Electric charge = $[M^0L^0T^1A^1]$

Dimensional formula of Electrical conductivity = $[M^{-1}L^{-3}T^3A^2]$

(B)Dimensional formula of Impulse = $[M^1L^1T^{-1}]$

Dimensional formula of specific heat = $[M^0L^2T^{-2}K^{-1}]$

(C)Dimensional formula of work = $[ML^2T^{-2}]$

Dimensional formula of Heat = $[ML^2T^{-2}]$

(D)Relative density has no dimensions, as it has no units.

Dimensional formula of Temperature = [K] or [θ]

**18. Correct option:(C)**

- Work and energy both are scalars means they have magnitude only and no direction.
- Impulse and linear momentum both are vectors means they have both magnitude and direction.
- Speed is a scalar quantity, it has magnitude only whereas velocity is a vector quantity, it has both magnitude and direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

Dimensional formula of speed and velocity is same that is = $[M^0L^1T^{-1}]$

- Force is a vector quantity and energy are a scalar quantity.

**19. Correct option:(B)**

Given: Force, length and time are fundamental quantities and

**Mass depends on force, length and time**

Mass =$F^aL^bT^c$----------(1)

$[M] = [M^1L^1T^{-2}]^a[L]^b[T]^C$ ( Dimensional formula of force=$[MLT^{-2}]$)

$[M] =[M]^a[L]^{a+b}[T]^{-2a+c}$

Comparing the dimensions of fundamental quantities on both sides

1=a -------(2)

0= a + b -------(3)

0= -2a + c----------(4)
put **a=1** in equation 3
0=1 + b
**b = -1**
put a =1 in equation 4
0=-2x1 + c
0=-2 + c
**c = 2**
**putting values of a, b, c in equation (1)**
**we get**
Mass $=F^1L^{-1}T^2$
Hence, dimensional formula of Mass = $[FL^{-1}T^2]$

**20. Correct option:(A)**
Given: Force, acceleration and time are fundamental quantities and
**Energy depends on force, acceleration and time**
Energy $=F^aA^bT^c$----------(1)
$[M^1L^2T^{-2}]= [M^1L^1T^{-2}]^a[M^0L^1T^{-2}]^b[T]^C$ ( Dimensional formula of force=$[M^1LT^{-2}]$)
$[M^1L^2T^{-2}] = [M]^{a+0}[L]^{a+b}[T]^{-2a-2b+c}$ ( Dimensional formula of energy=$[M^1L^2T^{-2}]$)
(Dimensional formula of acceleration=$[M^0L^1T^{-2}]$
Comparing the dimensions of fundamental quantities on both sides
1=a -------(2)
2= a + b -------(3)
-2= -2a -2b + c----------(4)
put **a=1** in equation 3
2=1 + b
2-1= b
**b =1**
put a =1 in equation 4
-2=-2x1 -2x1 + c
-2=-2 -2+ c
-2+2+2=c
**c = 2**
**putting values of a, b, c in equation (1)**
**we get**
Energy $=F^1A^1T^2$
Hence, dimensional formula of energy = $[FAT^2]$

## LEVEL-II

**1.Correct option:(A)**
Weight is defined as the force exerted on a body by gravity.
Weight is often expressed as
W= mg
Where, w= weight of the body
m = mass of the body
g= acceleration due to gravity
**SI unit** of weight is same as force that is newton (N).
so, its dimensional formula is also same as force
Dimensional formula of force = $[M^1L^1T^{-2}]$
So, Dimensional formula of weight= $[M^1L^1T^{-2}]$

**2. Correct option:(D)**

Frequency = number of vibrations/second

Second is the unit of time

And time is represented as [T]

Number of vibrations have no dimensions

So,

Frequency = $\frac{1}{[T]}$

Frequency = $[T^{-1}] = [M^0L^0T^{-1}]$

Dimensional formula of frequency is $[M^0L^0T^{-1}]$

**3. Correct option:(C)**

Dimensional formula of intensity = $[ML^0T^{-3}]$

**4. Correct option:(B)**

- Dimensional formula of angular momentum and Planck's constant = $[ML^2T^{-1}]$
- Dimensional formula of moment of inertia = $[ML^2T^0]$ which is same as Dimensional formula of moment of force= $[ML^2T^{-2}]$
- Dimensional formula of work and torque = $[ML^2T^{-2}]$
- Dimensional formula of impulse and momentum = $[ML^1T^{-1}]$

**5. Correct option:(D)**

Gravitational constant(G) = $\frac{Fr^2}{m_1m_2}$

Where, F is the force between masses $m_1$ and $m_2$ at a distance r

Dimensional formula of force(F) = $[MLT^{-2}]$

Length/distance is represented as [L]

Dimensions of $r^2$ = [L] x [L] =$[L^2]$

Mass is represented as [M]

So,

Gravitational constant = $\frac{[MLT^{-2}]x[L^2]}{[M]x[M]}$

Gravitational constant = $[M^{1-1-1}L^{1+2}T^{-2}]$

Gravitational constant = $[M^{-1}L^3T^{-2}]$

So; Dimensional formula of gravitational constant =$[M^{-1}L^3T^{-2}]$

**6. Correct option:(B)**

Kinetic energy = $\frac{1}{2}mv^2$

Mass(m) is represented as [M]

Dimensional formula of velocity (v)= $[M^0L^1T^{-1}]$

Dimensions of $v^2$ = $[M^0L^1T^{-1}]$ x $[M^0L^1T^{-1}]$

Dimensions of $v^2$ = $[M^0L^{1+1}T^{-1-1}]$

Dimensions of $v^2$ = $[M^0L^2T^{-2}]$

Kinetic energy = [M] x $[M^0L^2T^{-2}]$

Kinetic energy = $[M^{0+1}L^2T^{-2}]$

Kinetic energy = $[M^1L^2T^{-2}]$

Or Kinetic energy =$[ML^2T^{-2}]$

Dimensional formula of kinetic energy =$[ML^2T^{-2}]$

**7. Correct option:(C)**

Light year is an important unit of long distances. One light year is the distance travelled by light in vacuum in one year.

1 light year = $9.46 \times 10^{15}$m

as light year has the units of length

so, dimensions of light year are [L]

**8.Correct option:(A)**

Dimensional analysis gives no information about the dimensional constants in the formula. For example, 1, 2, 3, π etc.

**9.Correct options:(A, B, C, D)**

Dimensional formula of speed = $[M^0L^1T^{-1}]$ ( 0 dimensions in mass)

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$ (0 dimensions in mass)

Dimensional formula of volume = $[M^0L^3\ T^0]$ ( 0 dimensions in mass)

Dimensional formula of area = $[M^0L^2\ T^0]$ ( 0 dimensions in mass)

**10.Correct options:(A, B)**

Speed = $\frac{\text{distance}}{\text{time}} = \frac{[L]}{[T]}$

Length/displacement/distance is represented as [L] Velocity = $\frac{\text{displacement}}{\text{time}} = \frac{[L]}{[T]}$

Velocity/speed = $[L^1\ T^{-1}] = [M^0L^1\ T^{-1}]$

So, $[M^0L^1\ T^{-1}]$ is the dimensional formula of velocity/speed.

or $[M^0L\ T^{-1}]$ is the dimensional formula of velocity/speed.

Hence, velocity and speed have same dimensional formula.

**11.Correct options: (A, B, D)**

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$ (-2 dimensions in time)

Dimensional formula of force = $[MLT^{-2}]$ (-2 dimensions in time)

Dimensional formula of Work=$[ML^2T^{-2}]$ (-2 dimensions in time)

Dimensional formula of velocity=$[M^0L\ T^{-1}]$ (-1 dimensions in time)

Hence, acceleration, force, work have same dimensions in time

**12.Correct option: (A)**

work =force x displacement

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Work=$[MLT^{-2}]$x [L]

Work =$[ML^{1+1}T^{-2}]$

Work=$[ML^2T^{-2}]$

Dimensions of length in work is 2

**13.Correct option:(A)**

Dimensional formula of Work=$[ML^2T^{-2}]$

Dimensions of mass in work is 1

**14.Correct option:(D)**

Dimensional formula of Work=$[ML^2T^{-2}]$

Dimensions of time in work is -2

**15.Correct option:(B)**

Power = $\frac{\text{work}}{\text{time}}$

Now,

work =force x displacement

Power = $\frac{\text{force x displacement}}{\text{time}}$

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Time is represented as [T]

So,

Power = $\frac{[MLT^{-2}]x[L]}{[T]}$

Power =$[ML^{1+1}T^{-2-1}]$
Power =$[ML^{2}T^{-3}]$
dimensions of length in power are 2

**16. Correct option:(A)**

Dimensional formula of Power =$[ML^{2}T^{-3}]$
dimensions of mass in power are 1

**17. Correct option:(D)**

Dimensional formula of Power =$[ML^{2}T^{-3}]$
dimensions of time in power are -3

**18. Correct Matching is: a-T; b-S; c-QR; d-P**

- Mass is represented by [M].
- Length is represented by [L].
- Strength of electric current is represented by [A] or [I].
- Thermodynamic temperature is represented by [θ].

**19. Correct Matching is: a-R; b-Q; c-S; d-T**

**(a)**

$\text{Density} = \frac{\text{mass}}{\text{volume}}$
Mass is represented as [M]
$[M^{0}L^{3}\ T^{0}]$ is the dimensional formula of volume.
$\text{Density} = \frac{[M]}{[M^{0}L^{3}\ T^{0}]}$
$\text{Density} = [M^{1}L^{-3}\ T^{0}]$
$[M^{1}L^{-3}\ T^{0}]$ is the dimensional formula of density.
0r $[ML^{-3}\ T^{0}]$ is the dimensional formula of density.

**(b)** work =force x displacement
Length/displacement is represented as [L]
Dimensional formula of force = $[MLT^{-2}]$
Work=$[MLT^{-2}]$x [L]
Work =$[ML^{1+1}T^{-2}]$
Work=$[ML^{2}T^{-2}]$
Dimensional formula of Work=$[ML^{2}T^{-2}]$

**(c)** $\text{Power} = \frac{\text{work}}{\text{time}}$
Now,
work =force x displacement
$\text{Power} = \frac{\text{force x displacement}}{\text{time}}$
Length/displacement is represented as [L]
Dimensional formula of force = $[MLT^{-2}]$
Time is represented as [T]
So,
Power =$\frac{[MLT^{-2}]x[L]}{[T]}$
Power =$[ML^{1+1}T^{-2-1}]$
Power =$[ML^{2}T^{-3}]$
Dimensional formula of Power =$[ML^{2}T^{-3}]$

**(d)** Linear momentum= mass x velocity
Mass is represented as [M]
Dimensional formula of velocity=$[M^{0}L\ T^{-1}]$

So,
Linear momentum = [M] x $[M^0LT^{-1}]$
Linear momentum = $[M^{1+0}LT^{-1}]$
Linear momentum = $[MLT^{-1}]$
Dimensional formula of Linear momentum = $[MLT^{-1}]$

# 2. Kinematics

## Solutions

## Basic Concepts of Motion and Types of Motion

### LEVEL-I

**1. Correct option: (B)**

**Motion:** An object is said to be in motion if it changes its position with time, with respect to its surroundings or some reference point.

**For example,** A bird flying in air is in motion because it changes its position continuously with respect to a person standing on the ground.

**2. Correct option: (D)**

**Oscillatory motion:** It is that motion in which a body moves/fluctuates to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

**3. Correct option: (D)**

A moving object is said to have a random motion if it does not move in any particular direction and its motion has no specific path.

A butterfly flying in a garden is an example of Random motion.

**4. Correct option: (B)**

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

A butterfly flying in a garden is an example of Random motion.

**5. Correct option: (D)**

**Rectilinear motion:** It is also known as linear motion. It is that motion in which a particle or an object /body moves along a straight line or well-defined straight path. For example, the motion of train along a straight railway track, an object dropped from a height above the ground etc.

**6. Correct option: (C)**

**Curvilinear motion:** The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

When a car takes a turn, it follows a curved path. Hence, a car taking a turn executes curvilinear motion.

**7. Correct option: (D)**

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

A kite flying in air is an example of Random motion.

**8. Correct option: (D)**

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

The motion fishes in water are an example of Random motion.

**9. Correct option: (D)**

**Oscillatory motion:** It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

Hence, motion of pendulum of wall clock is an example of oscillatory motion as it moves To and fro between two fixed points about a mean position.

**10. Correct option: (C)**

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

The motion fishes in water are an example of Random motion.

**11. Correct option: (B)**

**Translatory motion:** It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.

hence, marching of soldiers on a straight road is an example of translator motion as the soldiers along parallel straight lines.

**12. Correct option: (C)**

**Translatory motion:** It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.

**Hence,** the free fall of fruits from the tree under the gravity is an example of translatory motion because the fruit moves downwards along a straight path.

**13. Correct option: (A)**

Distance: It is the actual path traversed by the object in the given time.

Distance travelled along a circular path is calculated with the help of circumference of the circle.

Distance travelled by the particle when it completes full circle = circumference of the circle

Distance travelled by the particle when it completes half circle (ACB)

$= \frac{(circumference\ of\ the\ circle)}{2}$

$= \frac{2\pi r}{2}$

$= \pi r$ ( r= radius of the semicircle=7m)

$(\pi = 22/7)$

Distance travelled by the particle when it completes half circle (ACB) = $\frac{22}{7}$ x 7 m

= 22m

**14. Correct option: (C)**

Distance is the actual path traversed/covered by the object in the given time.

**15. Correct option: (D)**

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence,

C.G.S unit of distance is centimeter.

**16. Correct option: (B)**

**SI** system is an internationally accepted system of units for measurements.

According to this system unit of length is meter.

As distance is also the length of path.so, SI unit of distance is meter.

**17. Correct option: (A)**

**Displacement:** It is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

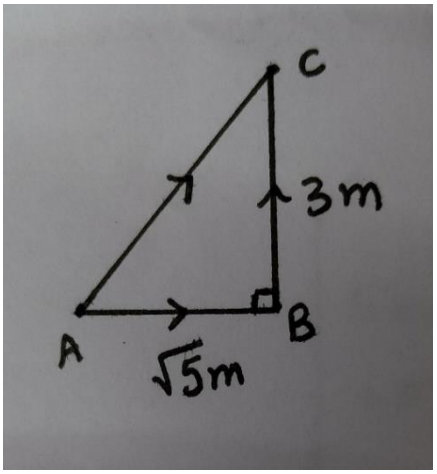
Here, initial position is A and final position is C. So, displacement is the line joining Initial and final position of the body.

**18. Correct option: (A)**

**Displacement** is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

In the given figure, the object is moving from point A to B to C.

Here, initial position is A and final position is C.




When we join A to C, we get the shortest distance AC= displacement

Triangle ABC is a right-angled triangle

AB = base = $\sqrt{5}m$

BC=perpendicular=3m

AC= hypotenuse

Using Pythagoras theorem in triangle ABC

$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$

$(AC)^2 = (AB)^2 + (BC)^2$

$(AC)^2 = \left(\sqrt{5}\right)^2 + (3)^2$

$(AC)^2 = 5 + 9$

$(AC)^2 = 14$

$AC = \sqrt{14}\, m$

Hence, displacement (AC)= $\sqrt{14}\, m$

**19. Correct option: (C)**

Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. Hence it is distance measured in a particular direction. If the distance travelled by the body is 0 then displacement is also 0.

**20. Correct option: (A)**

**Displacement** is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

In the given figure, the particle is moving from point A to B through P and another particle through Q.

Here, initial position is A and final position is B.

**Calculation of displacement for first particle travelling from A TO B through P**

shortest distance AB= displacement

Triangle ABP is a right-angled triangle

AP = base = 8m

BP=perpendicular=6m

AB= hypotenuse

Using Pythagoras theorem in triangle ABC

$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$

$(AB)^2 = (AP)^2 + (BP)^2$

$(AB)^2 = (8)^2 + (6)^2$

$(AB)^2 = 64 + 36$

$(AB)^2 = 100$

AB = $\sqrt{100}$ m

AB= 10m

Hence, displacement (AB)= 10m

**Calculation of displacement for first particle travelling from A TO B through Q**

shortest distance AB= displacement

Triangle ABQ is a right-angled triangle

BQ= base = 8m

AQ=perpendicular=6m

AB= hypotenuse

Using Pythagoras theorem in triangle ABC

$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$

$(AB)^2 = (BQ)^2 + (AQ)^2$

$(AB)^2 = (8)^2 + (6)^2$

$(AB)^2 = 64 + 36$

$(AB)^2 = 100$

AB = $\sqrt{100}$ m

AB= 10m

Hence, displacement (AB)= 10m

**21. Correct option: (C) Both A and B**

Displacement is a **vector quantity**, so it has **both magnitude and direction**. It represents the shortest distance from the initial position to the final position in a specific direction.

**22. Correct option: (B)**

The horse travels **12 m north** and **5 m east**, forming a right triangle.

Displacement $= \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ m

**23. Correct option: (C)**

The bird starts at the **tree top** and returns to the **same tree top** after each trip. Since the initial and final positions are the same, the displacement is **zero**.

## LEVEL – II

**1. Answer:** A) Random motion

House flies move in an unpredictable and irregular manner without following any specific path, which is the definition of random motion.

**2. Answer:** D) Rest

A chair lying stationary in a room does not change its position with respect to its surroundings as time passes, meaning it is in a state of absolute rest.

**3. Answer:** C) Time

The most fundamental characteristic of all moving bodies is that their position continuously changes with respect to time.

**4. Answer:** C) Curvilinear motion

A flyover bridge generally follows an arched or curved trajectory. Therefore, a bus moving along it traces a curved path, which is an example of curvilinear motion.

**5. Answer:** B) 14 m

Distance is the total length of the actual path covered by the moving body. The man walks 8 m towards the east and then 6 m towards the north. The total distance is the simple addition of these path lengths: 8 m + 6 m = 14 m.

**6. Answer:** A) $\sqrt{34}$ km

Displacement is the shortest straight-line distance between the initial position A and the final position C. Based on a standard grid interpretation from the figure where length is 5 km and width is 3 km, the displacement is the diagonal of the rectangle. Using the Pythagorean theorem: Displacement = $\sqrt{5^2 + 3^2} = \sqrt{25 + 9} = \sqrt{34}$ km.

**7. Answer:** B) 28 m

The particle travels along a semi-circular path from point A to point B, with a given radius of 14 m. Displacement is the shortest straight-line distance between the initial and final points, which in the case of a semi-circle is its diameter. Therefore, Displacement = 2 × radius = 2 × 14 = 28 m.

**8. Answer:** A) 12 m

Distance is the actual length of the path traveled from point A through B and C, ending at D. Adding the individual path segments shown in the figure: 5 m (A to B) + 5 m (B to C) + 2 m (C to D) = 12 m.

**9.** **Answer:** A) 3 m
The particle moves 5 m towards the east (which we can consider as a +5 m direction) and then moves 8 m towards the west (a -8 m direction). The net displacement from the starting point is +5 m - 8 m = -3 m. The magnitude of this displacement is 3 m.

**10.** **Answer:** B) $\pi r$
The total distance of a full circular path is defined by its circumference, $2\pi r$. After completing exactly half a circle, the distance covered is half of the total circumference, which is $\frac{2\pi r}{2} = \pi r$.

**11.** **Answer:** A) If the car is moving on straight road
The distance moved by a body and the magnitude of its displacement are only exactly equal when the body travels in a single, straight line without reversing or changing its direction.

**12.** **Answer:** C) 40 m, 0
The boy walks 20 m to the bookshop and travels the same 20 m distance back. The total distance covered is 20 m + 20 m = 40 m. Because his final position is exactly the same as his starting position, his net change in position (displacement) is 0.

**13.** **Answer:** A, B, D
Periodic motion is any motion that repeats itself in equal, regular intervals of time. The hands of a wristwatch, the Earth rotating on its axis and the Earth revolving around the sun all complete cycles in strictly fixed time intervals.

**14.** **Answer:** A, B, C, D
Rotatory motion occurs when a rigid body rotates or spins around an internal fixed axis. A spinning top, the blades of a windmill, a ceiling fan, and the wheels of a moving car all turn about their respective central axes.

**15.** **Answer:** A, B, C, D
Oscillatory motion is defined as a back-and-forth or to-and-fro motion about a fixed mean position. A clock pendulum, a see-saw, a sewing needle moving up and down, and a compressed spring vibrating back and forth all exhibit oscillatory motion.

**16.** **Answer:** B)
Translatory motion When considering the car as a whole unit moving down a straight road, every part of the vehicle travels the same distance in the same amount of time, which characterizes translatory (specifically rectilinear) motion. Note that while the wheels individually perform rotatory motion, the standard primary motion of the vehicle body is translatory.

**17.** **Answer:** A, D
The minute hand of a clock completes exactly one full circle in one hour. Because it returns directly to its starting point, its net displacement is zero. The actual path length (distance) covered by the tip is the circumference of the circle it traces out, which is $2\pi R$.

**18.** **Answer:** A, B, C, D
Displacement is fundamentally a measure of length. Metre (the standard S.I. unit), centimetre, millimetre, and kilometre are all valid units of length used to measure displacement.

**19.** **Answer:** A) Scalar quantity
Distance is simply the total path length covered and requires only a magnitude (numerical value and unit) for its complete description, lacking any specific directional component.

**20.** **Answer:** A) 30km
Distance is the total length of the path traveled, completely disregarding the direction. The car covers a 20 km path followed by a 10 km path. The total distance is 20 km + 10 km = 30 km.

**21.** **Answer:** A)
Circular motion is considered a special case of rotatory motion because an object moves around a fixed axis while maintaining a constant distance (radius) from the axis. Therefore, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

**22.** **Answer:** Both the Assertion and the Reason are fundamentally false. The distance traveled by a moving body is a cumulative scalar quantity that continually adds up; it can never decrease as time passes. Furthermore, distance is independent of the direction of motion, making the reason false as well.

**23.** **Answer:** C) Straight line
The visual path drawn from point A to point B is perfectly linear, indicating that the insect is moving in a straight line (rectilinear motion).

**24.** **Answer:** A) Curvilinear
The path connecting point B to point C is drawn as a smooth curve, indicating curvilinear motion.

**25.** **Answer:** D) Random
The path from point C to point D follows an erratic, zig-zag trajectory with continuous and unpredictable direction changes, defining random motion.

**26.** **Answer:** C) 800 m
The track length for a single round is given as 200 m. Since the athletes take four full rounds, the total distance is calculated by multiplying the track length by the number of rounds: $4 \times 200$ m = 800 m.

**27.** **Answer:** C) 0
Because the athletes complete full rounds where the starting line and finish line are identical, their final physical position is exactly the same as their initial position. Consequently, the shortest straight-line distance between start and finish (displacement) is 0.

**28.** **Answer:** A) Distance
By definition, the actual total length of the path covered by any moving object is called its distance.

**29.** **Answer:** a) → R, b) → S or T, c) → P, d) → Q
For a circular playground, the distance covered in one revolution is its circumference, $2\pi r$ (matching R). The displacement across a semi-circular playground is a straight line through the center, which is the diameter or 2r (matching S). The magnitude of displacement can never exceed distance, so it is always less than or equal to the distance (matching P). The distance traveled by a moving body is always a positive value (matching Q).

**30.** **Answer:** a) → R, b) → Q, c) → S, d) → P
A swinging cradle pivots back and forth, executing oscillatory motion (R). A spinning top spins rapidly around its own vertical axis, illustrating rotatory motion (Q). Fishes swim in water using irregular, wandering paths, exhibiting random motion (S). A car driving forward on a purely straight line is executing rectilinear motion (P).

**31.** **Answer:** 22 m
The distance traveled along a semi-circular path is exactly half of the circumference of a circle, calculated using the formula $\pi r$. Given a radius of 7 m, the distance is $\frac{22}{7} \times 7 = 22$ m.

**32.** **Answer:** 88 m
A body completing one full revolution on a circular path covers a distance equal to the circle's circumference, $2\pi r$. Given the radius r = 14 m, the distance is $2 \times \frac{22}{7} \times 14 = 88$ m.

**33.** **Answer:** 0.314 m
For the displacement to become exactly zero after moving, the body must travel one complete circle to return to its origin. The minimum distance required is one full circumference, $2\pi r$. Given a radius of 5 cm, the distance is $2 \times \pi \times 5$ cm $= 10\pi$ cm. To provide the answer in meters, we convert this to $0.1\pi$ m, which numerically evaluates to approximately 0.314 m.

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## Speed, Velocity and Acceleration

## LEVEL – I

**1.** **Answer:** D) 40
Speed is 72 km/h, which converts to $72 \times \frac{5}{18}$ = 20 m/s. The distance is calculated as speed multiplied by time, which gives 20 * 2 = 40 m.

**2.** **Answer:** B) Distance / Time
The fundamental formula for speed is the ratio of distance traveled to the time taken.

**3.** **Answer:** B) m/s
The standard International System (S.I.) unit used to measure speed is meters per second (m/s).

**4.** **Answer:** C) 30 m/s
To convert a speed from km/h to m/s, multiply it by $\frac{5}{18}$. Multiplying $108 \times \frac{5}{18}$ yields 30 m/s.

**5.** **Answer:** A) High speed
Because speed is distance divided by time, covering a large distance (large numerator) in a short time (small denominator) results in a high speed.

**6.** **Answer:** B) 6
For an object traveling equal distances at different speeds, the average speed formula is $v_{avg} = \frac{2v_1v_2}{v_1+v_2}$. Substituting the known values gives $3 = \frac{2\times2\times v_2}{2+v_2}$. This simplifies to $6 + 3v_2 = 4v_2$, which means $v_2$ is 6 m/s.

**7.** **Answer:** C) 100 m/s
Velocity is defined as displacement divided by time. Dividing 200 m by 2 s gives a velocity of 100 m/s towards the west.

**8.** **Answer:** B) 54
To convert a speed from m/s to km/h, multiply by $\frac{18}{5}$. Multiplying $15 \times \frac{18}{5}$ gives 54 km/h.

**9.** **Answer:** B) Odometer
An odometer is the specific instrument used to measure the total distance moved by a vehicle.

**10.** **Answer:** C) 5
Speed is equal to distance divided by time. Dividing 10 m by 2 s gives a speed of 5 m/s.

**11.** **Answer:** C) 30
Distance is calculated by multiplying speed and time. For the first segment, $60 \times \frac{20}{60}$ = 20 km. For the second segment, $30 \times \frac{20}{60}$ = 10 km. The total distance is 20 + 10 = 30 km.

**12.** **Answer:** C) 50kmph
A time of 120 minutes is exactly equal to 2 hours. Average speed is the total distance divided by the total time, so 100 km / 2 h = 50 km/h.

**13.** **Answer:** B) 12.5
Multiply by $\frac{5}{18}$ to convert from km/h to m/s. Multiplying $45 \times \frac{5}{18}$ yields 12.5 m/s.

**14.** **Answer:** B) 20 minutes
Time is determined by dividing the distance by the speed. Dividing 20 km by 60 km/h yields $\frac{1}{3}$ of an hour, which equals 20 minutes.

**15.** **Answer:** A) 100 m
Displacement is the product of velocity and time. Multiplying 5 m/s by 20 s results in 100 m.

**16.** **Answer:** A) Total distance / Total time
Average speed is mathematically defined as the total distance traveled divided by the total time taken to travel that distance.

**17.** **Answer:** A) 10
When a body moves with a uniform (constant) speed, its average speed over any time interval is identical to that constant speed, which is 10 km/h.

**18.** **Answer:** A) Uniform velocity or C) Uniform speed
Velocity is defined by both speed and a specific direction. Because the speed is uniform and the direction is constant (towards the east), the body possesses uniform velocity and uniform speed.

**19.** **Answer:** C) 0.216 km/h
To convert m/s into km/h, multiply the value by $\frac{18}{5}$ (or 3.6). Multiplying $0.06 \times 3.6$ gives 0.216 km/h.

**20.** **Answer:** B) Vector
Velocity requires both a magnitude (speed) and a specific direction, which defines it as a vector quantity.

**21.** **Answer:** C) Displacement / Time
Velocity is the rate of change of displacement, meaning it is calculated as displacement divided by time.

**22.** **Answer:** A) m/s
The standard S.I. unit for measuring velocity is meters per second (m/s).

**23.** **Answer:** A) Its instantaneous speed
The speedometer of a car measures the vehicle's specific speed at any exact instant in time (instantaneous speed).

**24.** **Answer:** B) 6
Using the average speed formula for equal distances, $3 = \frac{2 \times 2 \times v_2}{2 + v_2}$. Solving for $v_2$ yields 6 m/s.

**25.** **Answer:** B) Speed
Speed and velocity share identical dimensional units for measurement, such as m/s, because both represent a distance (or displacement) measured over time.

**26.** **Answer:** D) m^2/s^2
Acceleration represents a change in velocity over time, which gives units of distance per time squared (e.g., $m/s^2$ or $km/s^2$). Option D represents velocity squared, not acceleration.

**27.** **Answer:** A) Uniform acceleration
Experiencing equal changes in velocity over equal intervals of time is the precise definition of moving with uniform acceleration.

**28.** **Answer:** D) 4m/s
Because acceleration is the change in velocity divided by time, the change in velocity can be found by multiplying acceleration by time. Thus, $2 \times 2 = 4$ m/s.

**29.** **Answer:** B) Acceleration
The physical rate of a continuous increase (or change) in a body's velocity over time is called acceleration.

**30.** **Answer:** C) 4s
Utilizing the first kinematic equation, $v = u + at$. Plugging in the provided values results in $25 = 5 + 5t$. This simplifies to $5t = 20$, yielding $t = 4$ seconds.

**31.** **Answer:** C) +0.5 m/s^2
Acceleration is calculated as $\frac{v-u}{t}$, which is $\frac{0-10}{20}$ = -0.5 $m/s^2$. Because retardation is the negative of acceleration, the retardation is +0.5 $m/s^2$.

**32.** **Answer:** D) 2m/s^2
The vehicle's acceleration is $\frac{0-10}{5}$ = -2 $m/s^2$. Deceleration refers to the magnitude of this negative acceleration, which is 2 $m/s^2$.

**33.** **Answer:** C) 4 m/s^2
The acceleration is $\frac{0-20}{5}$ = -4 $m/s^2$. Retardation is the positive magnitude of the velocity decrease, yielding 4 $m/s^2$.

**34.** **Answer:** D) 10 m/s^2, 20 m/s^2
First, convert both velocities to m/s: 18 km/h = 5 m/s, and 36 km/h = 10 m/s. Using the formula $a = \frac{v-u}{t}$, the first car's acceleration is $\frac{5-0}{0.5}$ = 10 $m/s^2$, and the second car's acceleration is $\frac{10-0}{0.5}$ = 20 $m/s^2$.

**35.** **Answer:** C) 0 m/s
The third equation of motion is $v^2 - u^2 = 2as$, which rearranges to $v^2 = u^2 + 2as$. If $u^2 = -2as$, then substituting this gives $v^2 = -2as + 2as = 0$. This means the final velocity $v$ is 0 m/s.

**36.** **Answer:** A) Velocity - time
The first equation of motion, formulated as $v = u + at$, provides a direct relationship between an object's velocity and the time elapsed.

**37.** **Answer:** A) S = ut + 1/2 at^2
The standard formula for the second equation of motion, representing displacement under uniform acceleration, is $S = ut + \frac{1}{2}at^2$.

**38.** **Answer:** A) S = ((u+v)/2) * t
Displacement can be determined by taking the average velocity, which is $\frac{u+v}{2}$, and multiplying it by the time $t$.

**39.** **Answer:** B) Displacement - time
The second equation of motion ($S = ut + \frac{1}{2}at^2$) specifically relates an object's displacement to the time elapsed.

## LEVEL – II

**1.** **Answer:** C) 2.5
First, convert the velocities into m/s: initial velocity $u$ = 108 km/h = 30 m/s, and final velocity $v$ = 72 km/h = 20 m/s. The acceleration is $\frac{20-30}{4}$ = -2.5 m/s². The deceleration (the magnitude of the negative acceleration) is 2.5 m/s².

**2.** **Answer:** A) 3 m/s^2
Convert the final velocity into m/s: 54 km/h = 15 m/s. Since the scooter starts from rest, $u$ = 0. Acceleration is calculated as $\frac{15-0}{5}$ = 3 m/s².

**3.** **Answer:** C) 2.5 sec
First, determine the uniform acceleration: $a = \frac{30-10}{5}$ = 4 m/s². To find the time taken to accelerate from 30 m/s to 40 m/s, use $t = \frac{v-u}{a}$, yielding $\frac{40-30}{4}$ = 2.5 seconds.

**4.** **Answer:** C) 6 m/s^2
Acceleration is the total change in velocity divided by time. Therefore, $a = \frac{35-5}{5}$ = 6 m/s².

**5.** **Answer:** A) 2 m/s^2
First, convert the velocities: 36 km/h = 10 m/s, and 72 km/h = 20 m/s. The acceleration is $\frac{20-10}{5}$ = 2 m/s².

**6.** **Answer:** A) 0 m/s^2
Uniform velocity indicates that there is absolutely no change in velocity over time. Since acceleration measures the rate of change of velocity, it must be 0 m/s².

**7.** **Answer:** C) 1 ms^-2
Converting the speeds gives an initial speed of 10 m/s and a final speed of 20 m/s. The acceleration is $\frac{20-10}{10}$ = 1 m/s².

**8.** **Answer:** A) 4 ms^-2
The initial velocity is 20 m/s and it is brought to rest (0 m/s) in 5 seconds. Acceleration is $\frac{0-20}{5}$ = -4 m/s². De-acceleration is the positive magnitude of this value, which is 4 m/s². Note: Option B would only be correct if the question asked for acceleration.

**9.** **Answer:** C) 125 m
Determine the initial velocity using $v = u + at$. Plugging in the values gives $0 = u - (0.4 \times 25)$, meaning $u$ = 10 m/s. Next, use the displacement formula $S = ut + \frac{1}{2}at^2$. This becomes $(10 \times 25) - (0.5 \times 0.4 \times 625)$ = 250 - 125 = 125 m.

**10.** **Answer:** A) -6 m/s^2
Employing the third equation of motion, $v^2 - u^2 = 2as$. Substituting the knowns gives $0 - 6^2 = 2 \times a \times 3$, which simplifies to $-36 = 6a$. Solving for $a$ gives -6 m/s².

**11.** **Answer:** C) 9a/2
The distance traveled exactly in the $n$-th second is given by the formula $S_n = u + \frac{a}{2}(2n - 1)$. Since the body starts from rest ($u = 0$) and $n = 5$, $S_5 = 0 + \frac{a}{2}(10 - 1) = \frac{9a}{2}$.

**12.** **Answer:** A) 37.5 m/s^2
Using the relation $v^2 - u^2 = 2as$, we establish $0 - 15^2 = 2 \times a \times 3$. This equates to $-225 = 6a$, resulting in $a = -37.5$ m/s². Retardation is the positive magnitude, which is 37.5 m/s².

**13.** **Answer:** A) Displacement = Average velocity × time
Because average velocity is determined by dividing total displacement by total time, rearranging the formula yields Displacement = Average velocity × time.

**14.** **Answer:** C) 15
Using the $n$-th second formula, $S_n = u + \frac{a}{2}(2n - 1)$. With $u = 0$, $a = 6$, and $n = 3$, the calculation is $S_3 = 0 + \frac{6}{2}(6 - 1) = 3 \times 5 = 15$ m.

**15.** **Answer:** A) 78 m
Applying the kinematic displacement equation $S = ut + \frac{1}{2}at^2$. Substituting the provided values gives $S = (4 \times 6) + (0.5 \times 3 \times 6^2) = 24 + 54 = 78$ m.

**16.** **Answer:** C) 30 m
Using the distance formula for the $n$-th second, $S_n = u + \frac{a}{2}(2n - 1)$. With $u = 3$, $a = 6$, and $n = 5$, we obtain $S_5 = 3 + \frac{6}{2}(10 - 1) = 3 + 27 = 30$ m.

**ONE OR MORE THAN ONE CORRECT OPTION TYPE**

**17.** **Answer:** A, B, C, D
All listed options (m/s, cm/s, kmph, and km/s) correctly represent distance divided by time, making them all valid units of speed.

**18.** **Answer:** A, B
Velocity requires both magnitude and direction, which makes it a vector quantity (Option A). Velocity is fundamentally defined as the rate of change of displacement (Option B). Options C and D are factually incorrect.

**19.** **Answer:** A, B, C
Units for velocity must denote displacement over a period of time. Therefore, m/s, cm/s, and km/h are all valid velocity units.

**20.** **Answer:** A, B, C
Velocity is mathematically a vector quantity (Option A). Its standard S.I. unit is m/s (Option B). Because average velocity is displacement over time, a zero displacement dictates a zero average velocity (Option C).

**21.** **Answer:** B) Less than or equal to one
Displacement represents the shortest straight-line path between two points, meaning it is mathematically always less than or equal to the total distance traveled. Consequently, the ratio of velocity to speed will always be $\leq 1$.

**22.** **Answer:** A) 15 m/s
Convert km/h to m/s by multiplying by $\frac{5}{18}$. Doing so yields $54 \times \frac{5}{18} = 15$ m/s.

**23.** **Answer:** A, B
Acceleration measures how quickly velocity changes over time, meaning its units represent length per time squared, such as m/s² and cm/s².

**24.** **Answer:** A, B,
Acceleration is triggered by any alteration in velocity. This encompasses a change strictly in magnitude, strictly in direction or a simultaneous change in both (Options A and B).

**25.** **Answer:** A, C
Starting entirely from rest ($u = 0$), the velocity is determined by $v = at = 4 \times 5 = 20$ m/s. Converting this result to km/h by multiplying by $\frac{18}{5}$ yields 72 km/h.
Therefore, both 20 m/s and 72 km/h are correct.

**26.** **Answer:** B, C, D
Acceleration is dimensionally distance per time squared. Options B, C, and D are standard units for velocity, not acceleration.

**27.** **Answer:** C) 18 km/h²
Acceleration is given by $a = \frac{v-u}{t}$. Substituting the provided parameters yields $\frac{30-5}{5} = \frac{25}{5} = 5$ m/s². $\frac{5\ x\ 60x60}{1000} = 18\ km/h^2$

**28.** **Answer:** A, B, C, D
All given statements are true. An object can momentarily have zero velocity while experiencing acceleration (e.g., changing direction) (A). A body moving with uniform or constant velocity undergoes zero change, hence zero acceleration (B, D). Lastly, an object can possess a non-zero velocity while having zero acceleration (C).

**29.** **Answer:** B) 57.4m
Using the equation $S = ut + \frac{1}{2}at^2$, insert the variables: $S = (4 \times 7) + (0.5 \times 1.2 \times 7^2)$. This results in $28 + (0.6 \times 49) = 28 + 29.4 = 57.4$ m.

**30.** **Answer:** A, B
First, convert 36 km/h to 10 m/s. Acceleration is $\frac{0-10}{2}$ = -5 m/s², which means the retardation is 5 m/s² (Option B). The distance covered is $S = ut + \frac{1}{2}at^2 = (10 \times 2) - (0.5 \times 5 \times 4) = 20 - 10 = 10$ m (Option A).

**31.** **Answer:** A, C
Convert the final velocity into m/s: 72 km/h = 20 m/s. Starting from rest, the acceleration is $\frac{20-0}{10}$ = 2 m/s². Option C (200 cm/s²) is mathematically equivalent to 2 m/s², meaning both A and C are correct.

**32.** **Answer:** A, B
Employing $v^2 - u^2 = 2as$, we find $0^2 - 4^2 = 2 \times (-8) \times S$. This equation simplifies to $-16 = -16S$, leaving $S = 1$ meter (Option A). Because 1 m is exactly equal to 100 cm, Option B is also correct.

**ASSERTION AND REASON TYPE**

**33.** **Answer:** A) Both A and R are true and R is correct explanation
Average speed is calculated utilizing total distance, whereas average velocity utilizes displacement. Because displacement can be geometrically less than distance traveled, average velocity can consequently be less than average speed. The reason accurately explains the assertion.

**34.** **Answer:** A) Both A and R true and R explains A
The second equation of motion ($S = ut + \frac{1}{2}at^2$) yields displacement. It is directly derived mathematically by substituting average velocity ($\frac{u+v}{2}$) multiplied by time, making the reason a valid justification for the assertion.

**COMPREHENSION TYPE**

**35.** **Answer:** C) 324 km-south
Displacement is a vector and must include both a distance and a specific direction. Since the aeroplane covers 324 km strictly heading south, the displacement is precisely 324 km-south.

**36.** **Answer:** A) 972
Velocity is displacement divided by the time elapsed. The given time is 20 minutes, which equals $\frac{1}{3}$ of an hour. Velocity is therefore $324 \div \frac{1}{3} = 972$ km/h.

**37.** **Answer:** C) 60 km/hr
Velocity equals total displacement divided by total time. Dividing the 90 km displacement by 1.5 hours results in 60 km/h (heading east).

**38.** **Answer:** A) -2.5 m/s^2
According to the formula, acceleration $a = \frac{v-u}{t}$. Inserting the values gives $\frac{0-50}{20}$ = -2.5 m/s². Retardation by definition is a positive magnitude, but given the negative signs in the options, Option A represents the mathematical acceleration value.

**39.** **Answer:** D) 7.5
Convert the given velocities into m/s: 9 km/h = 2.5 m/s, and 36 km/h = 10 m/s. The total change in velocity is $10 - 2.5 = 7.5$ m/s.

**40.** **Answer:** C) 0.75
Acceleration is the measured change in velocity divided by the time. Taking the velocity change from the prior calculation (7.5 m/s) and dividing it by the given time (10 s) gives an acceleration of 0.75 m/s².

**MATRIX MATCHING TYPE**

**41.** **Answer:** a - Q, b - S, c - R, d - T
A speed of 1 km/h converts directly to $\frac{5}{18}$ m/s (a - Q). A speedometer is a device that measures instantaneous speed (b - S). Average velocity uses the formula of total displacement over total time taken (c - R). Uniform speed indicates an object covers equal distances within equal intervals of time (d - T).

**42.** **Answer:** a - R, b - Q, c - S, d - P
The primary velocity-time relation is $v = u + at$ (a - R). The primary displacement-time relation is $S = ut + \frac{1}{2}at^2$ (b - Q). The velocity-displacement relation is $v^2 - u^2 = 2as$ (c - S). The displacement formula for the $n$-th second is $S_n = u + \frac{a}{2}(2n - 1)$ (d - P).

**INTEGER TYPE**

**43.** **Answer:** 10
Average speed is calculated as total distance divided by total time. The total distance is $60 + 40 = 100$ m, and the total time is $5 + 5 = 10$ s. The average speed is $\frac{100}{10} = 10$ m/s.

**44.** **Answer:** 10
Employing the formula for distance traveled during the $n$-th second, $S_n = u + \frac{a}{2}(2n - 1)$. Starting from rest ($u = 0$) with $a = 4$ and $n = 3$, the math is $S_3 = 0 + \frac{4}{2}(6 - 1) = 2 \times 5 = 10$ m.

**45.** **Answer:** 13.33
When equal distances are traversed at distinct speeds, the average speed formula is $\frac{2v_1v_2}{v_1+v_2}$. Substituting the speeds yields $\frac{2\times10\times20}{10+20} = \frac{400}{30}$. This simplifies down to $\frac{40}{3}$, which evaluates to approximately 13.33 m/s.

# 3. Newton's Laws of Motion
# Solutions

**TOPIC-1:**

**Force, Newton's First Law of Motion and Inertia**

**1. Answer: A) Field forces**

**Explanation:**

Field forces, such as gravitational, electrostatic, and magnetic forces, act without direct physical contact between objects. They can exert their effect over a distance.

**2. Answer: B) Weight**

**Explanation:**

Weight is the gravitational force exerted by the Earth on an object. It always acts vertically downward toward the Earth's center.

**3. Answer: A) $mg$**

**Explanation:**

The weight of an object is given by the formula:

$W = mg$

where:

$W$= weight,

$m$= mass of the object,

$g$= acceleration due to gravity.

**4. Answer: A) Normal reaction**

**Explanation:**

The normal reaction force acts only when two surfaces are in contact. Since it requires physical contact, it is a contact force. The other options are non-contact (field) forces.

**5. Answer: A) Vertically downwards**

**Explanation:**

The weight of an object always acts vertically downward, directed toward the center of the Earth.

**6. Answer: B) Contact force**

**Explanation:**

A force produced due to direct physical contact between two objects is called a contact force.

**7. Answer: C) Unbalanced**

**Explanation:**

Only unbalanced forces produce a net force that can change the state of motion of an object. Balanced forces cancel each other and do not affect motion.

**8. Answer: C) Inertia**

**Explanation:**

Newton's First Law describes inertia, the natural tendency of an object to resist any change in its state of rest or uniform motion.

**9. Answer: B) Move forward**

**Explanation:**

When a moving car stops suddenly, the lower part of the body stops with the car, while the upper part continues moving forward due to the inertia of motion.

**10. Answer: A) Move backward**

**Explanation:**

When a stationary car starts suddenly, the lower part of the body moves forward with the car, while the upper part tends to remain at rest due to the inertia of rest, making the passenger jerk backward.

**11. Answer: C) Mass of the object**

**Explanation:**

Inertia depends only on the mass of an object. A greater mass means greater inertia and greater resistance to changes in motion.

**12. Answer: B) Inertia of motion**

**Explanation:**

Even after the power is switched off, the blades of a fan continue rotating for some time because they tend to remain in motion due to inertia of motion.

**13. Answer: B) Mass**

**Explanation:**

Mass is the measure of the inertia of a body. An object with greater mass possesses greater inertia.

**14. Answer: B) 4 kg**

**Explanation:**

Since inertia is directly proportional to mass, the 4 kg object has greater inertia than the 1 kg object.

**15. Answer: C) Both A and B**

**Explanation:**

According to Newton's First Law, an object remains:

- at rest, or
- in uniform motion in a straight line,

unless acted upon by an external unbalanced force.

**16. Answer: C) Inertia**

**Explanation:**

The tendency of a body to continue in its state of rest or uniform motion unless acted upon by an external force is called inertia.

**17. Answer: C) Force**

**Explanation:**

A force is a push or pull that changes or tends to change the state of rest or uniform motion of an object.

**18. Answer: A) Law of inertia**

**Explanation:**

Newton's First Law of Motion is also known as the Law of Inertia because it explains the property of inertia.

**19. Answer: C) Inertia of rest keeps the upper part of the body at rest whereas the lower part moves forward with the horse**

**Explanation:**

When a horse suddenly starts moving forward, the rider's lower body moves with the horse, while the upper body tends to remain at rest due to inertia of rest, causing the rider to fall backward.

**20. Answer: D) Unable to change by itself the state of rest and uniform linear motion**

**Explanation:**

Inertia is the property of a body by which it cannot change its own state of rest or

uniform motion in a straight line without the action of an external unbalanced force.

**21. Answer: B) Due to inertia of motion, the upper part of the body continues moving forward while the feet come to rest**

**Explanation:**

When a person jumps from a moving bus, the feet stop immediately on touching the ground, but the upper body continues moving forward due to inertia of motion, causing the person to fall forward.

**22. Answer: D) Exactly in the hand which threw it up**

**Explanation:**

Inside a car moving with constant velocity, a ball thrown vertically upward retains the car's horizontal velocity. Therefore, it returns to the thrower's hand.

**23. Answer: A) Inertia of direction**

**Explanation:**

When a bus takes a sharp turn, passengers tend to continue moving in their original direction due to the inertia of direction, making them feel as though they are thrown outward.

**24. Answer: D) By running he gives himself large inertia of motion**

**Explanation:**

A long jumper runs before jumping to gain greater forward speed. Due to the resulting inertia of motion, the athlete continues moving forward for a longer distance, helping achieve a longer jump.

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**LEVEL-II**

**1. Answer: C)**

As the mud-covered wheel rotates, the mud particles fly off along the tangent to the circular path because they tend to maintain their instantaneous straight-line direction due to the inertia of direction.

**2. Answer: C)**

When the carpet is beaten, it suddenly moves forward. The dust particles tend to remain in their initial state of rest due to inertia (Newton's First Law of Motion), causing them to separate from the carpet.

**3. Answer: D):**

Gravitational force is the mutual force of attraction exerted by the Earth on all objects, pulling them toward its surface.

**4. Answer: A*)**

An electric fan operates using an electric motor. At the school level, the intended answer is Electric force (A) because the motor runs on electric current. *(Technically, the rotation is produced by electromagnetic/magnetic forces.)*

**5. Answer: A):**

Muscular force is the contact force exerted by the contraction and relaxation of muscles in living organisms.

**6. Answer: C):**

When the train stops suddenly, the lower body in contact with the seat stops immediately due to friction, while the upper body continues to move forward due to the inertia of motion.

**7 Answer: A):**

A passenger jumping from a moving bus already has the forward velocity of the

bus. Running forward after landing gradually reduces this motion and helps prevent falling.

**8 Answer: A):**

When a tree branch is shaken, the branch moves suddenly, but the ripe fruits tend to remain at rest due to inertia of rest. As a result, their stalks snap and the fruits fall.

**9 Answer: C):**

A body remains stationary when all the forces acting on it balance one another. In this condition, the net force is zero (static equilibrium).

**10 Answer: B):**

When a horse suddenly leaps forward, the rider's lower body moves with the horse, while the upper body tends to remain at rest due to inertia of rest, making the rider appear to fall backward.

**ONE OR MORE THAN ONE CORRECT ANSWE TYPE**

**11. Correct Answer: B**

**Explanation:**

When the water tanker is moving with uniform velocity, the water inside also moves with the same velocity. When the brakes are applied suddenly, the tanker slows down, but the water tends to continue moving forward due to the inertia of motion. As a result, the water surges toward the front of the tanker.

**12. Correct Answers: B, C, D**

**Explanation:**

Magnetic force is a non-contact force because it acts without direct physical contact between objects. It is also known as a force at a distance since its effect can be experienced even when the interacting objects are separated. Magnetic force acts through a magnetic field, making it a field force.

**13. Correct Answers: A, B**

**Explanation:**

As a bucket is pulled up from a well:

- The Earth exerts a downward gravitational force on the bucket, known as its weight. (A)
- The rope exerts an upward pulling force on the bucket, called tension. (B)

Thus, both weight and tension act on the bucket simultaneously.

**14. Correct Answers: A, B, C, D**

**Explanation:**

An external force can change the state of an object in several ways. It can:

- Set a stationary object into motion. (A)
- Bring a moving object to rest. (B)
- Change the direction of motion of an object. (C)
- Change the shape or size (deform) of an object. (D)

Therefore, all the given options are correct.

**15. Correct Answers: A, B, C, D**

**Explanation:**

All the given options represent recognized types of forces:

- Muscular force – exerted by the muscles of living organisms. (A)
- Mechanical force – exerted by machines or mechanical devices. (B)
- Electrostatic force – acts between electrically charged objects. (C)
- Magnetic force – acts between magnets or moving electric charges. (D)

Hence, all the options are correct.

**16. Correct Answers: C, D**

**Explanation:**

- Mass is a measure of inertia. An object with greater mass has greater resistance to changes in its state of rest or motion. (C)
- Newton's First Law gives a qualitative definition of force by describing its effect on an object's motion, whereas Newton's Second Law provides a quantitative relationship between force, mass, and acceleration through the equation $F = ma$. (D)

**Assertion and Reason Type**

**17 Answer: A):**

Both the Assertion and the Reason are true, and the Reason correctly explains the Assertion. When a bus starts suddenly, the passenger's lower body moves with the bus, while the upper body tends to remain at rest due to inertia of rest, causing the passenger to tilt backward.

**18 Answer: B):**

Both the Assertion and the Reason are true, but the Reason is not the correct explanation of the Assertion. The tendency of a body to remain in uniform motion is due to its inertia, whereas the "Law of Inertia" (Newton's First Law) merely describes this behavior.

**Comprehension Type**

**19 Answer: A):**

When a car starts suddenly, the passenger's body appears to move backward relative to the car because it resists the change from its state of rest due to inertia of rest.

**20 Answer: C):**

When the string breaks, the centripetal force acting on the rotating stone disappears. The stone continues to move along the tangent to the circular path due to the inertia of direction.

**Matrix Matching Type**

**21**

| Column I | Column II | Explanation |
|---|---|---|
| a → T | Muscular force | Pushing a lawn roller requires muscular effort. |
| b → P, S | Mechanical force | Automobile and steam engines convert energy into mechanical work. |
| c → R | Gravitational force | A stone falls toward the Earth due to gravity. |
| d → Q | Electrostatic force | A rubbed plastic comb attracts paper bits due to static electricity. |

**22**

| Column I | Column II | Explanation |
|---|---|---|
| P → 2 | Change in speed | Pedaling harder increases the bicycle's speed. |
| Q → 1 | Change in dimensions | Stretching a rubber band changes its length and shape. |
| R → 4 | Change in direction | Heading a football changes its direction of motion. |

| | | |
|---|---|---|
| S → 3 | Inertia of rest | Dust remains at rest while the carpet moves during beating. |

**Integer Type**

**23 Answer: 8)**

The increase in speed is:
Increase in Speed = Final Speed – Initial Speed
= 20 m/s – 12 m/s = 8 m/s
Answer: 8

**24 Answer: 24)**

Since the car starts from rest,
Initial Velocity = 0 m/s
The change in velocity is:
Change in Velocity = Final Velocity – Initial Velocity
= 24 m/s – 0 m/s = 24 m/s
Answer: 24

## TOPIC-2: Momentum and Newton's Second Law of Motion

**LEVEL-I**

**1. Answer: (A) Force**

**Explanation:** According to Newton's Second Law of Motion, the force acting on an object is equal to the product of its mass and acceleration.

$F = ma$

where:

- $F$= Force
- $m$= Mass
- $a$= Acceleration

**2. Answer: (A) 9.8**

**Explanation:** One kilogram-force (kgf) is the gravitational force acting on a mass of 1 kg.

$F = mg$

$1\ \text{kgf} = 1\ \text{kg} \times 9.8\ \text{m/s}^2 = 9.8\ \text{N}$

Hence,

1 kgf = 9.8 N

**3. Answer: (B) Momentum**

**Explanation:** Linear momentum is defined as the product of the mass and velocity of a body.

$p = mv$

where:

- $p$= Momentum
- $m$= Mass
- $v$= Velocity

**4. Answer: (C) Linear momentum**

**Explanation:** Sir Isaac Newton introduced the concept of momentum, which is known as the quantity of motion possessed by a moving object.

**5. Answer: (B) $\vec{p}$**

**Explanation:** Linear momentum is a vector quantity and is represented by the symbol

$\vec{p}$

**6. Answer: (B) Momentum**

**Explanation:** Impulse is equal to the change in momentum.

Impulse = $F\Delta t = \Delta p$

Therefore, the unit Newton-second (Ns) is also the unit of momentum.

**7. Answer: (B) m/s²**

**Explanation:** Acceleration is the rate of change of velocity with time.

$a = \frac{\Delta v}{\Delta t}$

Its SI unit is

$\boxed{\text{m/s}^2}$

**8. Answer: (C) $10^5$**

**Explanation:** The SI unit of force is Newton and the CGS unit is dyne.

$1\ \text{N} = 1\ \text{kg} \cdot \text{m/s}^2$

Converting units:

- $1\ \text{kg} = 10^3\ \text{g}$
- $1\ \text{m} = 10^2\ \text{cm}$

Therefore,

$1\ \text{N} = 10^3 \times 10^2 = 10^5$ dyne

**9. Answer: (A) Ns**

**Explanation:** The SI unit of momentum is

$\text{kg} \cdot \text{m/s}$

Since

$1\ \text{N} = 1\ \text{kg} \cdot \text{m/s}^2$

$1\ \text{Ns} = 1\ \text{kg} \cdot \text{m/s}$

Hence, Newton-second (Ns) is also the SI unit of momentum.

**10. Answer: (B) An acceleration of 1 m/s² on a body of mass 1 kg**

**Explanation:** One Newton is defined as the force required to produce an acceleration of

$1\ \text{m/s}^2$

in a body of mass

1 kg.

**11. Answer: (A) Mass and (C) Speed**

**Explanation:** Momentum is given by

$p = mv$

Therefore, momentum is directly proportional to:

- Mass (when velocity is constant)
- Speed/Velocity (when mass is constant)

Hence, both A and C are correct.

**12. Answer: (B) 25 kg m/s**

**Explanation:** Using

$p = mv$

$p = 5 \times 5 = 25\ \text{kg m/s}$

Hence, the momentum is

$\boxed{25\ \text{kg m/s}}$

**13. Answer: (A) 2 kg**

**Explanation:** Using Newton's Second Law,

$F = ma$

$m = \frac{F}{a}$

$m = \frac{20}{10} = 2 \text{ kg}$

**14. Answer: (B) 1 $\text{m/s}^2$**

**Explanation:** First convert kilogram-force into Newton.

Using

$1 \text{ kgf} \approx 10 \text{ N}$

$2 \text{ kgf} = 20 \text{ N}$

Now,

$a = \frac{F}{m}$

$a = \frac{20}{20} = 1 \text{ m/s}^2$

**15. 1Answer: (B) Spring balance**

**Explanation:** A beam balance measures mass, whereas a spring balance measures weight, which is the gravitational force acting on an object.

**16. Answer: (A) Mass × Velocity**

**`Explanation:** Momentum is defined as

$p = mv$

Therefore, momentum equals mass × velocity.

**17. Answer: (A) Mass × Acceleration**

**Explanation:** According to Newton's Second Law,

$F = ma$

Thus, force equals mass × acceleration.

**18. Answer: (B) Applied force**

**Explanation:** Newton's Second Law states that the rate of change of momentum of a body is directly proportional to the net external (applied) force acting on it.

**19. Answer: (C) kg·m/s**

**Explanation:** Since

$p = mv$

the SI unit of momentum is

$\boxed{\text{kg} \cdot \text{m/s}}$

**20. Answer: (D) Force acting on it**

**Explanation:** From Newton's Second Law,

$a = \frac{F}{m}$

For a constant mass, acceleration is directly proportional to the applied force. Therefore, increasing the force increases the acceleration.

**21. Answer: (B) 75 kg m/s**

**Explanation:** Using

$p = mv$

$p = 15 \times 5 = 75 \text{ kg m/s}$

Hence, the momentum of the body is

$\boxed{75 \text{ kg m/s}}$

**LEVEL-II**

**Single Correct Answer Type**

**1: Answer – D**

**Explanation:**

A force can:

- Change the magnitude of speed (increase or decrease speed),
- Change the direction of motion (as in circular motion),
- Change the shape or size of an object (deformation).

Hence, all the given statements are correct.

**2: Answer – B**

**Explanation:**

According to Newton's Second Law,

$F = ma$

For a constant force,

$a = \frac{F}{m}$

Acceleration is inversely proportional to mass. Therefore, a lighter vehicle experiences greater acceleration than a heavier one when the same force is applied.

**3: Answer – B**

**Explanation:**

A fielder pulls their hands backward while catching a ball to increase the time taken to stop the ball.

Using the impulse relation,

$F = \frac{\Delta p}{\Delta t}$

Increasing the stopping time ($\Delta t$) decreases the impact force, making the catch safer.

**4: Answer – B**

**Explanation:**

Given:

- Weight $W = 9.8$ N
- Applied force $F = 5$ N

First, calculate the mass:

$m = \frac{W}{g} = \frac{9.8}{9.8} = 1$ kg

Now,

$a = \frac{F}{m} = \frac{5}{1} = 5\ \text{m/s}^2$

Therefore, the acceleration is 5 m/s².

**5: Answer – C**

**Explanation:**

An object continues to accelerate as long as an unbalanced (net) force acts on it. Acceleration becomes zero only when the net external force becomes zero.

**6: Answer – A**

**Explanation:**

Impulse equals the change in momentum.

$\Delta p = F \times t$

$= 12 \times 4 = 48$ kg\cdotpm/s

Hence, the change in momentum is 48 kg·m/s.

**7: Answer – C**

**Explanation:**

Convert all quantities into SI units.

Mass:

$20\ \text{g} = 0.02\ \text{kg}$

Acceleration:

$5\ \text{cm/s}^2 = 0.05\ \text{m/s}^2$

Now,

$F = ma$

$= 0.02 \times 0.05$

$= 0.001\ \text{N}$

$= 10^{-3}\ \text{N}$

Hence, the correct answer is $10^{-3}$N.

**8: Answer – A**

**Explanation:**

For Object 1,

$F_1 = 2 \times 2.5 = 5\ \text{N}$

For Object 2,

$F_2 = 5 \times 2 = 10\ \text{N}$

Therefore,

$F_1 = 5\ \text{N}, F_2 = 10\ \text{N}$

**9: Answer – D**

**Explanation:**

The applied force is

$F = ma = 2 \times 5 = 10\ \text{N}$

For a new mass of 1.5 kg,

$a = \frac{F}{m} = \frac{10}{1.5} \approx 6.67\ \text{m/s}^2$

Rounded off,

$a \approx 6.6\ \text{m/s}^2$

**ONE OR MORE THAN ONE CORRECT ANSWER TYPE**

**10. Answer: A is correct.**

**Explanation:**

The change in linear momentum ($\Delta p$)is calculated using:

$\Delta p = m(v - u)$

Given:

- Mass, $m = 2\ \text{kg}$
- Initial velocity, $u = 4\ \text{m/s}$
- Final velocity, $v = 8\ \text{m/s}$

$\Delta p = 2 \times (8 - 4)$

$\Delta p = 2 \times 4 = 8\ \text{kg} \cdot \text{m/s}$

**11. Answer: A and B are correct.**

**Explanation:**

- Statement A: Correct. Linear momentum is the quantity of motion possessed by a moving body and is given by $p = mv$.
- Statement B: Correct. According to the Impulse–Momentum Theorem, the impulse acting on an object is equal to its change in momentum.

- Statement C: Incorrect. Impulse, acceleration, and the change in momentum are all in the same direction as the net applied force, not in the opposite direction.
- Statement D: Incorrect. In uniform circular motion, although the speed remains constant, the direction of velocity changes continuously. Hence, the direction of linear momentum also changes continuously, so momentum is not conserved.

**12. Explanation:**

According to Newton's Second Law,

$F = ma$

Since,

$a = \frac{v - u}{t}$

Therefore,

$F = m\left(\frac{v - u}{t}\right)$

Given:

- Force, $F = 0.6$ N
- Initial velocity, $u = 5$ m/s
- Final velocity, $v = 6$ m/s
- Time, $t = 2$ s

Substituting the values,

$0.6 = m\left(\frac{6 - 5}{2}\right)$

$0.6 = \frac{m}{2}$

$m = 1.2$ kg

Converting into grams,

1.2 kg = 1200 g

Conclusion:

The mass of the object is 1.2 kg or 1200 g. Therefore, Options A (1200 g) and C (1.2 kg) are correct.

**13. Options A, B, and C are correct.**

**Explanation:**

Newton's Second Law of Motion states:

The rate of change of momentum of a body is directly proportional to the net external force acting on it and takes place in the direction of that force.

**From this statement:**

- Statement A: Correct. In the SI system, the constant of proportionality is 1, so the net force is equal to the rate of change of momentum.
- Statement B: Correct. The rate of change of momentum is directly proportional to the applied net force.
- Statement C: Correct. The change in momentum occurs in the same direction as the applied net force.
- Statement D: Incorrect (if provided), since it does not agree with Newton's Second Law.

**Assertion–Reason Type**

**14: Answer – A**

**Explanation:**

Both the Assertion and the Reason are true.

Momentum is given by

$p = mv$

For a constant mass, momentum increases directly with velocity. Therefore, the Reason correctly explains the Assertion.

**15: Answer – A**

**Explanation:**

Both the Assertion and the Reason are true.

Newton's Second Law states that

$F = \frac{dp}{dt}$

For constant mass,

$F = ma$

Thus, the Reason correctly explains the Assertion.

**Comprehension Type**

**16: Answer – A**

**Explanation:**

$F = ma$

$= 1.5 \times 2.4 = 3.6$ N

Hence, the force is 3.6 N.

**17: Answer – D**

**Explanation:**

$a = \frac{F}{m} = \frac{15}{2} = 7.5\ \text{m/s}^2$

Therefore, the acceleration is 7.5 m/s².

**18: Answer – C**

**Explanation:**

Convert mass into SI units:

600 g = 0.6 kg

Now,

$F = ma$

$= 0.6 \times 10 = 6$ N

Hence, the correct answer is 6 N.

**19: Answer – C**

**Explanation:**

Initial momentum,

$p_i = mu$

$= 3 \times 10 = 30$ kg.m/s

Therefore, the initial momentum is 30 kg·m/s.

**20: Answer – D**

**Explanation:**

Final momentum,

$p_f = mv$

$= 3 \times 20 = 60$ kg.m/s

Hence, the final momentum is 60 kg·m/s.

**21: Answer – B**

**Explanation:**

Change in momentum,

$\Delta p = p_f - p_i$

$= 60 - 30 = 30$ kg.m/s

Therefore, the change in momentum is 30 kg·m/s.

**Matrix Matching Type**

**22. Answer:**

**a – Q**
**b – S**
**c – P**
**d – R**

Explanation:

- a → Q: The SI unit of force is Newton (N), equivalent to kg·m/s².
- b → S: Newton's First Law is also known as the Law of Inertia.
- c → P: Newton's Second Law is represented by $F = ma$.
- d → R: Momentum is defined as mass × velocity.

**23. Answer:**

**P – 1**
**Q – 2**
**R – 3**
**S – 4**

**Explanation:**

- P → 1: A fast cricket ball has high velocity and therefore high momentum.
- Q → 2: A karate player stops the hand in a very short time, producing a large impact force.
- R → 3: A heavy truck has large mass, giving it greater momentum and causing more damage.
- S → 4: An object at rest has zero velocity; hence its momentum is zero.

**Integer Type**

**24. Answer: 12**

**Explanation:**

$m = \frac{F}{a} = \frac{24}{2} = 12$ kg

**25. Answer: 150**

**Explanation:**

$F = ma$

$= 100 \times 1.5 = 150$ N

**26. Answer: 100**

**Explanation:**

$p = mv$

$= 10 \times 10 = 100$ kg.m/s

Therefore, the momentum of the object is 100 kg·m/s.

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**TOPIC-3: Newton's Third Law of Motion and Conservation of Momentum**

**1. Correct Answer: (B) Law of conservation of momentum**

**Explanation:**

Newton's third law states that for every action, there is an equal and opposite reaction. In an isolated system, these equal and opposite internal forces cancel each other, resulting in no net external force. Therefore, the total linear momentum of the system remains constant. Hence, the law of conservation of momentum is a direct consequence of Newton's third law.

**2. Correct Answer: (C) Newton's third law of motion**

**Explanation:**

According to Newton's third law, internal forces between interacting bodies are equal in magnitude and opposite in direction. In the absence of external forces,

these internal forces cancel out, so the total linear momentum of the system remains conserved**.**

**3. Correct Answer: (D) Opposite to the direction of jump, option A also correct**
**Explanation:**
When a person jumps forward from a boat, they push the boat backward. According to Newton's third law, the boat experiences an equal and opposite force and moves in the direction opposite to the jump.

**4. Correct Answer: (C) Third law of motion**
**Explanation:**
A swimmer pushes water backward with their hands and feet (action). The water pushes the swimmer forward with an equal and opposite force (reaction), enabling the swimmer to move.

**5. Correct Answer: (C) Only in the absence of external force**
**Explanation:**
The law of conservation of linear momentum states that the total momentum of a system remains constant only when no external force acts on the system.

**6. Correct Answer: (D) Forces**
**Explanation:**
Action and reaction are a pair of equal and opposite forces that act simultaneously on two different interacting bodies.

**7. Correct Answer: (C) Newton's third law**
**Explanation:**
When a bullet is fired, the gun exerts a forward force on the bullet (action). The bullet exerts an equal and opposite force on the gun (reaction), causing the gun to recoil backward.

**8. Correct Answer: (C) Newton's third law**
**Explanation:**
A rocket expels hot gases downward at high speed (action). The gases exert an equal and opposite force on the rocket (reaction), causing it to move upward.

**9. Correct Answer: (D) Reaction**
**Explanation:**
When the foot exerts a force on the stone, the stone simultaneously exerts an equal and opposite reaction force on the foot. This reaction force causes pain.

**10. Correct Answer: (C) 0**
**Explanation:**
A body moving with constant velocity has zero acceleration. According to Newton's second law,
$F = ma$
Since $a = 0$,
$F = 0$.
Therefore, the net external force acting on the body is zero.

**11. Correct Answer: (B) –0.5 m/s**
**Explanation:**
Using the law of conservation of momentum,
Initial momentum = Final momentum
$0 = m_b v_b + m_g v_g$
Given,

- Bullet mass = 100 g = 0.1 kg

- Bullet velocity = 100 m/s
- Gun mass = 20 kg

$0 = (0.1 \times 100) + (20 \times v_g)$

$0 = 10 + 20v_g$

$20v_g = -10$

$v_g = -0.5$ m/s

The negative sign indicates that the gun moves backward.

**12. Correct Answer: (D) All the above**

**Explanation:**

Jet aeroplanes, rockets, and recoiling guns all operate based on Newton's third law of motion, where every action is accompanied by an equal and opposite reaction.

**13. Correct Answer: (A) Action**

**Explanation:**

In an action–reaction pair, the force exerted by the first body on the second is called the action force, while the equal and opposite force exerted by the second body on the first is called the reaction force.

**14. Correct Answer:** 1.2 kg

**Explanation:**

Using the law of conservation of momentum,

$m_1u_1 + m_2u_2 = (m_1 + m_2)v$

Given,

- $m_1 = 2$kg
- $u_1 = 40$m/s
- $u_2 = 0$
- $v = 25$m/s

$2 \times 40 = (2 + m_2) \times 25$

$80 = 50 + 25m_2$

$30 = 25m_2$

$m_2 = \frac{30}{25} = 1.2$ kg

Since 1.2 kg is not among the given options, the question is invalid due to incorrect answer choices.

**15. Correct Answer: (C) Equal, opposite**

**Explanation:**

Newton's third law states:

`"To every action, there is an equal and opposite reaction."

Thus, every action force has a reaction force that is equal in magnitude and opposite in direction.

**16. Correct Answer: (B) The ground on the horse**

**Explanation:**

The horse pushes the ground backward with its hooves. In response, the ground exerts an equal and opposite forward force on the horse. This forward frictional force from the ground enables the horse to move and pull the cart forward.

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**LEVEL-II**

**Single Correct Option Type**

**1. Correct Option: (A)**

**Explanation:**

When a person jumps out of a boat, they push the boat backward with their feet

(action). According to Newton's Third Law of Motion, the boat exerts an equal and opposite force on the person (reaction), pushing the person forward. As a result, the boat moves backward.

**2. Correct Option: (C)**

**Explanation:**

When a gun is fired, it exerts a force on the bullet, propelling it forward (action). Simultaneously, the bullet exerts an equal and opposite force on the gun (reaction), causing the gun to recoil backward. This is a direct application of Newton's Third Law of Motion.

**3. Correct Option: (C)**

**Explanation:**

According to the Law of Conservation of Linear Momentum,

$m_A u_A + m_B u_B = m_A v_A + m_B v_B$

Substituting the given values,

$1500 \times 25 + 1000 \times 15 = 1500 \times 20 + 1000 \times v_B$

$37500 + 15000 = 30000 + 1000 v_B$

$52500 = 30000 + 1000 v_B$

$22500 = 1000 v_B$

$v_B = 22.5 \text{ m/s}$

Therefore, the speed of car B after the collision is $22.5$ m/s.

**4. Correct Option: (C)**

**Explanation:**

A cannon recoils when it fires a cannonball because the forward motion of the cannonball is accompanied by an equal and opposite backward force on the cannon. This recoil is explained by Newton's Third Law of Motion.

**5. Correct Option: (D)**

**Explanation:**

According to Newton's First Law of Motion, an object moving with uniform velocity in a straight line continues to do so unless acted upon by an external unbalanced force. Therefore, no force is required to maintain uniform motion.

**6. Correct Option: (D)**

**Explanation:**

Newton's Third Law states that action and reaction forces are equal in magnitude and opposite in direction. Since these are internal forces within a system, they cancel each other. Therefore, when no external force acts on the system, the total linear momentum remains conserved.

**7. Correct Option: (C)**

**Explanation:**

To completely balance a force acting towards the North-East, an equal force must act in the South-West direction because the balancing force must have the same magnitude but the opposite direction.

**ONE OR MORE THAN ONE CORRECT ANSWER TYPE**

**8. Correct Options: (A), (B), (D)**

**Explanation:**

- (A) is correct: When a gun fires, it exerts a forward force on the bullet (action). Simultaneously, the bullet exerts an equal and opposite force on the gun, causing it to recoil backward (reaction).

- (B) is correct: A swimmer pushes the water backward with their hands and feet. In response, the water exerts an equal and opposite force on the swimmer, pushing them forward.
- (C) is incorrect: Action and reaction forces do not act on the same body. They always act on two different interacting bodies.
- (D) is correct: A force always exists as part of an action–reaction pair. Therefore, a single isolated force cannot exist in nature.

**9. Correct Options: (A), (B), (C), (D)**

**Explanation:**

All the given situations are examples of Newton's third law of motion, which states that for every action, there is an equal and opposite reaction.

- (A) Rowing a boat: The oars push the water backward, and the water pushes the boat forward.
- (B) Walking: A person's foot pushes the ground backward, and the ground pushes the person forward.
- (C) Rocket propulsion: The rocket expels hot gases backward, and the gases exert an equal and opposite force that propels the rocket forward.
- (D) Firing a gun: The gun pushes the bullet forward, while the bullet exerts an equal and opposite force on the gun, causing recoil.

**10. Correct Options: (B), (C)**

**Explanation:**

- (B) is correct: According to Newton's third law, action and reaction forces are equal in magnitude.
- (C) is correct: Action and reaction forces always act in opposite directions.
- (A) is incorrect: Action and reaction forces act on different bodies, not on the same body.
- (D) is incorrect: Action and reaction forces do not cancel each other, because they act on different objects. Therefore, they can produce motion.

**Assertion and Reason Type**

**11. Correct Option: (A)**

**Explanation:**

Assertion: A person walks by pushing the ground backward. (True)

Reason: The ground pushes the person forward with an equal and opposite force. (True)

The reason correctly explains the assertion according to Newton's Third Law of Motion.

**12. Correct Option: (A)**

**Explanation:**

Assertion: Rockets accelerate upward in space. (True)

Reason: The rocket ejects hot gases downward, and the gases exert an equal and opposite force on the rocket, producing upward thrust. (True)

The reason correctly explains the assertion.

**Comprehension Type**

**Passage – I**

**13. Correct Option: (B)**

**Explanation:**

When the man pushes the car, the force applied by the man on the car is the action force.

**14. Correct Option: (A)**

**Explanation:**

According to Newton's Third Law, action and reaction forces are always equal in magnitude and opposite in direction. Therefore, if the action force is 5 N, the reaction force is also 5 N.

**15. Correct Option: (D)**

**Explanation:**

The passage states that the man applies a force of 5 N to push the car. Hence, the magnitude of the action force is 5 N.

**Passage – II**

**16. Correct Option: (D)**

**Explanation:**

When a horse pulls a cart:

- The horse pushes the ground backward.
- The ground pushes the horse forward.

Both statements are correct. The forward force exerted by the ground enables the horse-cart system to move forward.

**17. Correct Option: (D)**

**Explanation:**

According to Newton's Third Law, action and reaction forces:

- are equal in magnitude,
- opposite in direction,
- act simultaneously, and
- always occur in pairs.

Therefore, all the given statements are correct.

**18. Correct Option: (B)**

**Explanation:**

Action and reaction forces are equal in magnitude but opposite in direction. Hence, $\vec{F}_{\text{action}} = -\vec{F}_{\text{reaction}}$

**Matrix Matching Type**

**19. Correct Matches**

| Column I | Column II | Explanation |
|---|---|---|
| a → P | $mv - mu$ | Represents the change in momentum. |
| b → S | $W = mg$ | Formula for the weight of an object. |
| c → Q | $\vec{F}_{action} = -\vec{F}_{reaction}$ | Newton's Third Law in vector form. |
| d → R | Rocket | Rocket propulsion is based on conservation of momentum. |

**20. Correct Matches**

| Column I | Column II | Explanation |
|---|---|---|
| P → 2 | Swimming | Swimmer pushes water backward; water pushes swimmer forward. |
| Q → 1 | Walking | Feet push the ground backward; the ground pushes the person forward. |

| | | |
|---|---|---|
| R → 4 | Rocket | Exhaust gases move downward, producing upward thrust. |
| S → 3 | Newton's Third Law | States that every action has an equal and opposite reaction. |

**Integer Type**

**21. Correct Answer: 2**

**Explanation:**

Impulse is equal to the change in momentum.

Impulse = $m(v-u)$

Given,

- Mass $m = 2$ kg
- Initial velocity $u = 1$ m/s
- Final velocity $v = 0$ m/s

Impulse = $2(0-1) = -2$ N.S

The magnitude of impulse is

$\boxed{2 \text{ N.S}}$

Therefore, the correct answer is 2.

**22. Correct Answer: 3**

**Explanation:**

In a completely closed cage, a flying bird pushes the air downward. The air transfers this force to the bottom of the cage. Therefore, the weighing scale still measures the combined weight of the cage and the bird.

Total reading = 2 kg + 1 kg = 3 kg

Hence, the correct answer is 3.

**23. Correct Answer: 180**

**Explanation:**

The action force (book on the table) acts vertically downward, while the reaction force (table on the book) acts vertically upward.

Since the two forces act in exactly opposite directions, the angle between them is

$\boxed{180°}$

Hence, the correct answer is 180.

# 4. Work, Energy and Power
# Solutions

## TOPIC-1: WORK
### LEVEL-I

**1. Correct Option: C) When a force produces motion**

**Explanation:**

In physics, work is done only when a force acting on an object causes it to move (produces displacement). If there is no displacement, no work is done.

**2. Correct Option: A) Without transfer of energy**

**Explanation:**

Work is the process of transferring energy from one object or system to another through the action of a force. Therefore, work cannot be done without energy transfer.

**3. Correct Option: A) Joule**

**Explanation:**

The SI unit of work is the joule (J).

- SI unit: Joule (J)
- CGS unit: erg
- Newton (N) is the SI unit of force, not work.

**4. Correct Option: C) Zero**

**Explanation:**

The formula for work done is:

$W = Fd\cos\theta$

When the force and displacement are perpendicular,

$\theta = 90°, \cos 90° = 0$

Therefore,

$W = 0$

**5. Correct Option: A) Force × distance moved in the direction of the applied force**

**Explanation:**

Work is the product of the force and the displacement in the direction of the force.

$W = F \times d$

or, more generally,

$W = Fd\cos\theta$

**6. Correct Option: B) Statement I is true and Statement II is false**

**Explanation:**

- Statement I is true because
  1 erg = 1 dyne × 1 cm
- Statement II is false because
  1 Joule = 1 Newton × 1 metre
  It defines one joule, not one erg.

**7. Correct Option: C) 9800 J**

**Explanation:**

Total mass

$m = 80 + 20 = 100$ kg

Height climbed

$h = 40 \times 0.25 = 10$ m
Using
$W = mgh$
$W = 100 \times 9.8 \times 10$
$W = 9800$ J
(The given time is required only for calculating power.)

**8. Correct Option: C) 1200 J**
**Explanation:**
$W = Fd$
$W = 60 \times 20$
$W = 1200$ J

**9. Correct Option: B) No work is done**
**Explanation:**
Since the angle between force and displacement is
$90°$
$W = Fd\cos 90° = 0$
Hence, no work is done.

**10. Correct Option: A) 50 J**
**Explanation:**
$W = Fd$
$W = 10 \times 5$
$W = 50$ J

**11. Correct Option: C) Zero**
**Explanation:**
The centripetal force always acts towards the centre, while the displacement is tangential to the path.
Thus,
$\theta = 90°$
$W = Fd\cos 90° = 0$

**12. Correct Option: A) Positive**
**Explanation:**
When force and displacement are in the same direction,
$\theta = 0°$
$W = Fd\cos 0° = +Fd$
Hence, the work done is positive.

**13. Correct Option: B) Negative**
**Explanation:**
The braking force acts opposite to the direction of motion.
$\theta = 180°$
$W = Fd\cos 180° = -Fd$
Therefore, the work done is negative.

**14. Correct Option: C) 12000 J**
**Explanation:**
Using
$g = 10\ \mathrm{m/s^2}$
$W = mgh$
$W = 200 \times 10 \times 6$
$W = 12000$ J

**15. Correct Option: C) 10 J**

**Explanation:**

Convert

25 cm = 0.25 m

$W = 40 \times 0.25$

$W = 10$ J

**16. Correct Option: C) –40 J**

**Explanation:**

Gravity acts downward while the displacement is upward.

$W = -mgh$

$W = -(1 \times 10 \times 4)$

$W = -40$ J

**17. Correct Option: B) 300 J**

**Explanation:**

Total mass

$m = 12 \times 0.25 = 3$ kg

$W = mgh$

$W = 3 \times 10 \times 10$

$W = 300$ J

**18. Correct Option: $W = 5 \times 10^6$ J**

**Explanation:**

Distance

$d = 1000$ m

Force

$F = 5 \times 10^3$ N

$W = Fd$

$W = (5 \times 10^3)(1000)$

$W = 5 \times 10^6$ J

**19. Correct Option: B) 30 kJ**

**Explanation:**

Total mass

$m = 20 + 10 = 30$ kg

$W = 30 \times 10 \times 100$

$W = 30000$ J

$W = 30$ kJ

**20. Correct Option: B) 10 J**

**Explanation:**

$W = Fd$

$W = 5 \times 2$

$W = 10$ J

**21. Correct Option: C) 225 J**

**Explanation:**

Using

$g = 10\ \mathrm{m/s^2}$

$W = mgh$

$W = 15 \times 10 \times 1.5$

$W = 225$ J

**LEVEL-II**

**Single Correct Option Type**

**1. Answer: C) Both A and B**

**Explanation:**

Work done is given by:

$W = Fs\cos\theta$

where:

- $F$= applied force
- $s$= displacement
- $\theta$= angle between force and displacement

For work to be done:

- There must be a component of displacement along the direction of the force (Statement A).
- The angle between force and displacement should not be 90°, because $\cos 90° = 0$(Statement B).

Therefore, both A and B are correct.

**2. Answer: B) Positive**

**Explanation:**

During free fall, both the gravitational force and the displacement act vertically downward.

Since,

$\theta = 0°, \cos 0° = 1$

$W = Fs\cos 0° = Fs$

Hence, the work done by gravity is positive.

**3. Answer: A) Zero**

**Explanation:**

A person standing at a bus stop does not undergo any displacement.

Since,

$W = F \times s$

and $s = 0$,

$W = 0$

Therefore, the work done is zero.

**4. Answer: C) Both x and y are same**

**Explanation:**

The work done in lifting an object vertically against gravity is

$W = mgh$

where:

- $m$= mass
- $g$= acceleration due to gravity
- $h$= vertical height

The time taken does not affect the work done; it only affects the power.

Hence, both persons do the same amount of work.

**5. Answer: A) Is not equal to zero**

**Explanation:**

In uniformly accelerated circular motion, the speed changes continuously due to tangential acceleration.

Since kinetic energy changes,

$W = \Delta K$

where $W$is the net work done and $\Delta K$is the change in kinetic energy.
Therefore, the net work done is not zero.

**6. Answer: A) Total energy**

**Explanation:**

According to the work-energy theorem for a system,
$W_{\text{net}} = \Delta E$
where $E$represents the total mechanical energy (kinetic + potential energy).
Hence, the net work done equals the change in the total mechanical energy.

**7. Answer: A) 35 J**

**Explanation:**

The total work done from A to C is the algebraic sum of the work done along each part of the path.
$W_{AC} = W_{AB} + W_{BC}$
$= 20 + 15 = 35$ J
Hence, the work done is 35 J.

**8. Answer: B) 0**

**Explanation:**

The coolie applies an upward force to support the load, while his displacement is horizontal.
Thus,
$\theta = 90°$
Since,
$\cos 90° = 0$
$W = Fs\cos 90° = 0$
Hence, the work done by the coolie on the load is zero.

**9. Answer: A) When an object is thrown upwards**

**Explanation:**

Gravity always acts downward.
When an object moves upward,
$\theta = 180°$
Since,
$\cos 180° = -1$
the work done by gravity becomes negative.

**10. Answer: A) 0**

**Explanation:**

For a satellite moving in a circular orbit, gravity acts toward the centre of the Earth, while the satellite's displacement at every instant is tangential.
Thus,
$\theta = 90°$
Since,
$\cos 90° = 0$
the work done by gravity is zero.

**ONE OR MORE THAN ONE CORRECT ANSWER TYPE**

**11. Correct Options: A, B**

**Explanation:**

- Option A: When the metal ball is resting on the plate, there is no displacement of the ball. Since work is done only when a force produces displacement, no mechanical work is done.

- Option B: When the ball is raised to a height, it gains gravitational potential energy ($PE = mgh$). As it falls, this stored energy is converted into kinetic energy, enabling the ball to do work on the plate (such as producing sound or causing slight deformation).

**12. Correct Options: A, B, C, D**

**Explanation:**

- Option A: Potential energy is the energy possessed by an object due to its position or configuration.
- Option B: Kinetic energy is the energy possessed by an object due to its motion.
- Option C: When work is done on an object, energy is transferred to or from the object. Thus, work and energy are closely related.
- Option D: Energy is defined as the capacity to do work.

Note: Some textbooks emphasize Option A when specifically defining potential energy, but considering the general concept of energy, all four statements are correct.

**13. Correct Options: A, B**

**Explanation:**

Mechanical work is done only when a force causes displacement in its direction or in a direction having a component along the force ($W = Fd\cos\theta$).

- Option A: The cement bags are lifted into the lorry. Since a force is applied and the bags are displaced upward, work is done.
- Option B: The toy car (or trolley) moves when pulled. The applied force causes displacement, so work is done.
- Option C: Although the boy applies force to the rock, it does not move. Therefore, displacement is zero and no mechanical work is done.
- Option D: The wall does not move when pushed, so displacement is zero. Hence, no mechanical work is done, even though muscular effort is involved.

**14. Correct Options: A, C, D**

**Explanation:**

- Option A: If the force is always perpendicular to the direction of motion ($\theta = 90°$), then
  $W = Fd\cos 90° = 0.$
  Hence, no work is done (e.g., centripetal force in circular motion).
- Option C: If an object remains stationary, there is no displacement. Therefore, no mechanical work is done on the object.
- Option D: Work done by a force depends on the displacement of its point of application. If the point of application does not move, the displacement is zero and the work done by that force is zero.

**Assertion and Reason Type**

**15. Answer: A) Both A and R are true, and R is the correct explanation of A.**

**Explanation:**

The assertion correctly states that displacement is necessary for work to be done.
The reason is also correct because
$W = Fs\cos\theta$
If displacement $s = 0$,
$W = 0$
Hence, the reason correctly explains the assertion.

**16. Answer: A) Both A and R are true, and R is the correct explanation of A.**

**Explanation:**

During free fall, both the gravitational force and displacement are directed downward.

Therefore,

$\theta = 0°$

and

$W = Fs\cos 0° = Fs$

Thus, gravity does positive work, and the reason correctly explains the assertion**.**

**Comprehension Type**

**17. Answer: C) The body is displaced in the direction of the applied force**

**Explanation:**

Work done is positive when the displacement has a component in the direction of the applied force.

This occurs when the angle between force and displacement is less than 90°.

**18. Answer: B) The body is displaced in the direction opposite to the direction of the force**

**Explanation:**

Work done is negative when displacement is opposite to the direction of the applied force.

Here,

$\theta = 180°$

and

$\cos 180° = -1$

Hence, the work done is negative.

**19. Answer: Defect (Both A and D are correct)**

**Explanation:**

Work done is

$W = Fs\cos\theta$

The work becomes zero if:

- No force is applied ($F = 0$), or
- There is no displacement ($s = 0$).

Therefore, both Options A and D satisfy the condition for zero work, making the defective.

**20. Answer: B) Zero**

**Explanation:**

If a person remains stationary,

$s = 0$

Therefore,

$W = Fs = 0$

Hence, the work done is zero.

**21. Answer: B) Zero**

**Explanation:**

The supporting force is vertical, while the displacement is horizontal.

Since,

$\theta = 90°$

$W = Fs\cos 90° = 0$

Hence, the work done is zero.

**22. Answer: A) 20,580 J**

**Explanation:**

The man lifts both himself and the load.

Given:

- Man's mass = 50 kg
- Load mass = 20 kg

Total mass,

$m = 50 + 20 = 70$ kg

Height,

$h = 30$ m

Acceleration due to gravity,

$g = 9.8\ \text{m/s}^2$

Work done,

$W = mgh$

$= 70 \times 9.8 \times 30$

$= 20{,}580$ J

Hence, the correct answer is 20,580 J.

**23. Matrix Match Type**

| Column I | Correct Match | Explanation |
|---|---|---|
| a | S, T | When displacement is in the same direction as the force, $\theta = 0°$. Hence, work is positive and $W = Fs$. |
| b | R | While moving upward, displacement is opposite to gravity. Therefore, $\theta = 180°$, and gravity does negative work. |
| c | R | Kinetic friction always acts opposite to the direction of motion. Hence, it performs negative work. *(Note: Column II option "−Fe" is a printing typo; it should be "−Fs".)* |
| d | P | When force is perpendicular to displacement ($\theta = 90°$), work done is zero. |
| e | P | If a rock does not move, displacement is zero. Therefore, the work done is zero. |

**24. a → T, b → S, c → R, d → P**

**Explanation:**

- a → T:

  1 erg is the CGS unit of work and is defined as:

  1 erg = 1 dyne × 1 cm
- b → S:

  Since

  $1\ \text{eV} = 1.6 \times 10^{-19}$ J,

  therefore,

  $1\ \text{J} = \frac{1}{1.6 \times 10^{-19}}\ \text{eV} = 6.25 \times 10^{18}\ \text{eV}.$
- c → R:

  One kilogram-weight metre is equal to:

  1 kgwt m = 9.8 N × 1 m = 9.8 J.

  Since

  $1\ \text{J} = 10^7$ erg,

therefore,
$9.8\ \text{J} = 9.8 \times 10^7$ erg.

- d → P:
The prefix kilo (k) denotes $10^3$. Hence,
$1\ \text{kW} = 10^3\ \text{W} = 1000\ \text{W}$.

**25. Answer :** 19600 J
*(or 20000 J if $g = 10\ \text{m/s}^2$ is used).*
Explanation:
The work done by the crane in lifting the car is equal to the gain in gravitational potential energy.
$W = mgh$
Given:

- $m = 500$ kg
- $h = 4$ m
- $g = 9.8\ \text{m/s}^2$

Substituting the values,
$W = 500 \times 9.8 \times 4$
$W = 19600$ J
Therefore, the work done by the crane is 19600 J.

**26. Answer : 3234 J**
**Explanation:**
The work done by the boy while climbing the stairs is equal to the increase in gravitational potential energy.
First, calculate the total vertical height:
$h = 40 \times 0.15 = 6$ m
Using the formula,
$W = mgh$
Given:

- $m = 55$ kg
- $g = 9.8\ \text{m/s}^2$
- $h = 6$ m

$W = 55 \times 9.8 \times 6$
$W = 3234$ J
Hence, the work done by the boy is 3234 J.

**27. Answer Key:**
When the object moves vertically upwards ($\theta = 180°$).
Explanation:
The work done by a force is given by:
$W = Fd\cos\theta$
The force due to gravity always acts vertically downward.
When an object is moved or thrown vertically upward, its displacement is opposite to the direction of gravity. Therefore,
$\theta = 180°$
Since,
$\cos 180° = -1$,
the work done by gravity becomes
$W = Fd\cos 180° = -Fd$.

Thus, gravity does negative work when an object moves vertically upward because the force and displacement are in opposite directions.

## TOPIC-2: Energy and Power

**LEVEL-I**

**1. Answer: (A) 1 m**

**Explanation:**

Work done is given by:

$W = F \times s$

where $F$is the force and $s$is the displacement in the direction of the force.

By definition, 1 joule (J) is the work done when a force of 1 newton (N) moves an object through a distance of 1 metre (m).

**2. Answer: (B) Joule**

**Explanation:**

The SI unit of work and energy is the joule (J).

- Joule (J) – SI unit
- Erg – CGS unit
- Newton (N) – Unit of force
- Kilojoule (kJ) – Multiple of the joule

**3. Answer: (D) Chemical**

**Explanation:**

Fuels store chemical energy in their molecular bonds. During combustion, this chemical energy is converted mainly into heat energy and, in many cases, light energy.

**4. Answer: (D) Potential**

**Explanation:**

Water stored at a height possesses gravitational potential energy due to its elevated position above the ground.

**5. Answer: (B) Point 2**

**Explanation:**

- Point 1: Only potential energy (KE = 0)
- Point 2: Both potential and kinetic energy
- Point 3: Maximum kinetic energy and minimum potential energy (taking the ground as the reference level)

Hence, the ball possesses both potential and kinetic energy at Point 2.

**6. Answer: (A) Maximum at the middle of the motion**

**Explanation:**

During oscillatory motion,

- At the extreme positions, the velocity is zero, so kinetic energy is zero.
- At the equilibrium (middle) position, the velocity is maximum.

Therefore, the kinetic energy is maximum at the middle of the motion**.**

**7. Answer: (B) P, R and S**

**Explanation:**

**Statements P, R and S are correct.**

Statement Q is incorrect because all hot objects emit thermal radiation, but many hot objects do not emit visible light.

**8. Answer: (C) Its mass as well as its speed**

**Explanation:**

The kinetic energy of a moving body is given by

K.E. $= \frac{1}{2}mv^2$

Hence, it depends on both the mass and the speed of the body.

**9. Answer: (A) Increases, decreases**

**Explanation:**

As the object moves upward,

- Its height increases, so its potential energy increases.
- Its speed decreases, so its kinetic energy decreases.

At the highest point, the kinetic energy becomes zero.

**10. Answer: (C)**

$\boxed{\textbf{K.E.} = \frac{1}{2}mv^2}$

**Explanation:**

This is the standard expression for the kinetic energy of a body of mass $m$moving with speed $v$.

**11. Answer: (A) 25 W**

**Explanation:**

Work done,

$W = Fs = 600 \times 20 = 12000$ J

Time,

8 min = 480 s

Power,

$P = \frac{W}{t} = \frac{12000}{480} = 25$ W

**12. Answer: (A) 1 J/s**

**Explanation:**

Power is defined as

$P = \frac{W}{t}$

Hence,

$\boxed{1 \text{ W} = 1 \text{ J s}^{-1}}$

**13. Answer: (C) 200 W**

**Explanation:**

Work done,

$W = mgh$

$= 60 \times 10 \times 10 = 6000$ J

Power,

$P = \frac{6000}{30} = 200$ W

**14. Answer: (B)**

$\boxed{\frac{mv^2}{2t}}$

**Explanation:**

Work done equals the gain in kinetic energy,

$W = \frac{1}{2}mv^2$

Average power,

$P = \frac{W}{t} = \frac{\frac{1}{2}mv^2}{t} = \frac{mv^2}{2t}$

**15. Answer: (C) 180 W**

**Explanation:**

Mass flow rate,

$\frac{m}{t} = 40 \text{ kg s}^{-1}$

Power supplied as kinetic energy,

$P = \frac{1}{2}\left(\frac{m}{t}\right)v^2$

$= \frac{1}{2} \times 40 \times 3^2 = 20 \times 9 = 180 \text{ W}$

**16. Answer: (A) 7200 kg**

**Explanation:**

Power,

$P = 12000 \text{ W}$

Time,

$t = 60 \text{ s}$

Energy supplied,

$E = Pt = 12000 \times 60 = 720000 \text{ J}$

Using,

$E = mgh$

$720000 = m \times 10 \times 10$

$m = 7200 \text{ kg}$

**17. Answer: (D) 480 W**

**Explanation:**

Total force,

$500 + 100 = 600 \text{ N}$

Work done,

$W = 600 \times 4 = 2400 \text{ J}$

Power,

$P = \frac{2400}{5} = 480 \text{ W}$

**18. Answer: (A) 2.4 kWh**

**Explanation:**

Total power,

$5 \times 120 = 600 \text{ W} = 0.6 \text{ kW}$

Energy consumed,

$E = Pt = 0.6 \times 4 = 2.4 \text{ kWh}$

**19. Answer: (B) 50 kW**

**Explanation:**

Mass,

90 tonnes = 90000 kg

Time,

1 hour = 3600 s

Power,

$P = \frac{mgh}{t}$

$= \frac{90000 \times 10 \times 200}{3600}$

$= 50000 \text{ W} = 50 \text{ kW}$

**20. Answer: (A) $P_A > P_B$**

**Explanation:**

Both children perform the same amount of work because they climb the same height.

Since,

$P = \frac{W}{t}$

the child who takes less time develops greater power.

As Child A takes 15 s while Child B takes 20 s,

$\boxed{P_A > P_B}$

---

**LEVEL-II**

**Single Correct Option Type**

**1. Correct Option: B**

When a car screeches to a halt, its kinetic energy is converted mainly into thermal energy (due to friction between the brakes/tyres and the road) and sound energy (the screeching noise).

**2. Correct Option: A**

According to the Law of Conservation of Energy, energy can neither be created nor destroyed. It can only be transferred or converted from one form to another**.**

**3. Correct Option: C**

An object raised above the ground possesses gravitational potential energy because of its position in Earth's gravitational field.

**4. Correct Option: C**

Both work and energy have the same SI unit, the joule (J).

- Work → Joule (J)
- Energy → Joule (J)
- Power → Watt (W)
- Efficiency → No unit (dimensionless)

**5. Correct Option: A**

Using the formula,

$U = mgh$

$= 3 \times 10 \times 25 = \boxed{750 \text{ J}}$

**6. Correct Option: D**

A stretched spring or rubber band stores elastic potential energy. When released, this stored energy is converted into kinetic energy.

**7. Correct Option: B**

Power is defined as

$P = \frac{\text{Work}}{\text{Time}}$

Since both work and time are scalar quantities, power is also a scalar quantity.

**8. Correct Option: B**

1 Horsepower (HP) = $\boxed{746 \text{ W}}$

**9. Correct Option: B**

Power is the rate of doing work.

$P = \frac{W}{t}$

**10. Correct Option: B**

$P = \frac{15\text{ J}}{15\text{ s}} = 1\text{ W}$

$\boxed{1\text{ W}}$

**11. Correct Option: B**

The kinetic energy of one bullet is

$\frac{1}{2}mv^2$

If the gun fires $n$bullets every second, then

$P = n\left(\frac{1}{2}mv^2\right)$

Hence,

$\boxed{P = \frac{1}{2}nmv^2}$

**12. Correct Option: D**

Total Power

$= 4 \times 500 = 2000\text{ W} = 2\text{ kW}$

Energy consumed

$= 2\text{ kW} \times 10\text{ h} = \boxed{20\text{ kWh}}$

**13. Correct Option: D**

Mass flow rate

$\frac{dm}{dt} = \rho Av$

$= 1000 \times (10 \times 10^{-4}) \times 20 = 20\text{ kg s}^{-1}$

Theoretical kinetic power

$P = \frac{1}{2}\left(\frac{dm}{dt}\right)v^2$

$= \frac{1}{2} \times 20 \times 20^2 = 4000\text{ W} = \boxed{4\text{ kW}}$

**ONE OR MORE THAN ONE CORRECT ANSWER TYPE**

**14. Correct Options: B, D**

**Explanation:**

Power is the rate at which work is done or kinetic energy changes.

$P = \frac{dK}{dt} = \vec{F} \cdot \vec{v} = Fv\cos\theta$

For the kinetic energy of a body to **increase continuously**, power must be positive.

$\frac{dK}{dt} > 0 \Rightarrow \cos\theta > 0$

This is possible only when the angle between the force and velocity is **acute** ($\theta < 90°$).

- **Option A:** Incorrect. The force need not be exactly parallel to the velocity; any acute angle is sufficient.
- **Option B:** Correct. An acute angle ensures positive power.
- **Option C:** Incorrect.
- **Option D:** Correct. Since

  $K = \frac{p^2}{2m}$,

an increase in kinetic energy (for constant mass) also means an increase in the magnitude of linear momentum.

**15. Correct Options: B, C, D**

**Explanation:**

A force acting on a moving body may have different effects depending on its direction relative to the motion.

- If the force acts along the direction of motion, the body speeds up and its kinetic energy increases.
- If the force acts opposite to the direction of motion, the body slows down and its kinetic energy decreases.
- If the force continues to oppose the motion, the body may eventually come to rest, making its kinetic energy zero.

Therefore:

- Option A: Incorrect. A force does not always increase kinetic energy.
- Option B: Correct. Kinetic energy may decrease.
- Option C: Correct. Kinetic energy may increase.
- Option D: Correct. Kinetic energy may become zero.

**16. Correct Options: A, B**

**Explanation:**

Given:

- Mass of water, $m = 100\,\text{kg}$
- Height, $h = 20\,\text{m}$
- Time, $t = 10\,\text{s}$
- $g = 10\,\text{m/s}^2$

Work done in lifting the water:

$W = mgh$

$W = 100 \times 10 \times 20 = 20\,000\ \text{J}$

Power:

$P = \frac{W}{t}$

$P = \frac{20\,000}{10} = 2000\ \text{W}$

Since

$2000\ \text{W} = 2 \times 10^3\ \text{W} = 2\ \text{kW},$

both expressions represent the same power.

- Option A: Correct.
- Option B: Correct.

**17. Correct Options: A, C**

**Explanation:**

Power is defined as

$P = \frac{W}{t}$

or

$P = \vec{F} \cdot \vec{v}.$

Since work is a scalar quantity and the dot product of two vectors is also a scalar, power is a scalar quantity.

Also, for a fixed time interval,

$P \propto W.$

Therefore:

- Option A: Correct. Power is a scalar quantity.
- Option B: Incorrect.
- Option C: Correct. Power is directly proportional to work done for a fixed time.
- Option D: Incorrect.

**18. Correct Options: A, B, C, D**

**Explanation:**

All the given statements are correct.

- Option A: Correct.

  $P = \frac{W}{t}$

  For a fixed amount of work,

  $P \propto \frac{1}{t}$.
- Option B: Correct.

  The dimensional formula of power is

  $[M^1L^2T^{-3}]$.
- Option C: Correct.

  Instantaneous power is given by

  $P = \vec{F} \cdot \vec{v}$.
- Option D: Correct.

  Power is defined as the work done per unit time.

  Hence, all four statements are correct.

**19. Correct Options: A, B, C, D**

**Explanation:**

Each option represents a valid relation involving power.

- Option A: Correct.

  $P = \frac{W}{t}$

  This is the basic definition of power.
- Option B: Correct.

  $P = \vec{F} \cdot \vec{v}$

  This gives the instantaneous power delivered by a force.
- Option C: Correct.

  For a machine delivering $n$particles per second, each having kinetic energy $\frac{1}{2}mv^2$,

  $P = n\left(\frac{1}{2}mv^2\right)$.
- Option D: Correct.

  The standard conversion is

  1 horsepower (hp) = 746 W.

  Hence, all the given statements are correct.

**Comprehension Type**

**Passage–I**

Given:

- Mass = 90 kg
- Height = 10 m
- $g = 10\ \text{m s}^{-2}$

Work done by each person

$W = mgh$

$= 90 \times 10 \times 10 = \boxed{9000\ \text{J}}$

**20. Correct Option: A**

Power of $M_1$

$P = \frac{9000}{15} = \boxed{600\ \text{W}}$

**21. Correct Option: D**

Power of $M_2$

$P = \frac{9000}{20} = \boxed{450\ \text{W}}$

**22. Correct Option: B**

Ratio of powers

$\frac{600}{450} = \frac{4}{3}$

**Passage–II**

**23. Correct Option: B**

Using conservation of energy,

K.E. gained = P.E. lost = $mgh$

$= 3 \times 10 \times 1 = \boxed{30\ \text{J}}$

**24. Correct Option: A**

K.E. $= \frac{1}{2}mv^2$

$= \frac{1}{2} \times 15 \times 50^2$

$= 7.5 \times 2500 = \boxed{18\,750\ \text{J}}$

**25. Correct Option: B**

K.E. $= \frac{1}{2} \times 10 \times 2^2$

$= 5 \times 4 = \boxed{20\ \text{J}}$

**Matrix Matching Type**

**26. Correct Match**

| Column I | Column II |
|---|---|
| **a** | **Q** |
| **b** | **R** |
| **c** | **P** |
| **d** | **R** |

**Explanation**

For a conservative force,

$F = -\frac{dU}{dx}$

- At A: Positive slope ⇒ Force is negative (Q)
- At B: Zero slope ⇒ Force is zero (R)
- At C: Negative slope ⇒ Force is positive (P)
- At E: Zero slope ⇒ Force is zero (R)

**27. Correct Match**

| Column I | Column II |
|---|---|
| **a** | **S** |
| **b** | **R** |

| Column I | Column II |
|---|---|
| **c** | **Q** |
| **d** | **P** |

**Explanation**

a) Power is a Scalar → S

b) Dimensional formula of power

$\boxed{ML^2T^{-3}}$

→ R

c)

1 HP = 746 W

→ Q

d) C.G.S. unit of power = erg $s^{-1}$ → P

**Integer Type**

**28. Correct Answer: 2**

Energy = Power × Time

= 1000 W × 2 h

= 1 kW × 2 h = $\boxed{2\text{ kWh}}$

**29. Correct Answer: 500**

P.E. = $mgh$

= 5 × 10 × 10 = $\boxed{500\text{ J}}$

**30. Correct Answer: 200**

$P = \frac{W}{t}$

$= \frac{5000}{25} = \boxed{200\text{ W}}$

## PREVIOUS YEAR QUESTIONS

**1. Correct answer: (C)**

We know, if a body is vertically thrown up there is conversion of KE into PE. When kinetic energy becomes half of its original value, the other half is converted into potential energy. Therefore,

$\frac{490}{2} = 2 \times 9.8 \times h \Rightarrow h = 12.5$ m

**2. Correct answer: (D)**

Since, force on the vehicle is constant, it will move with a constant acceleration. For particle starting from rest, velocity,

$v = at$

Thus, kinetic energy,

$E = \frac{1}{2}mv^2 = \frac{1}{2}ma^2t^2$

Graph of $E$versus $t$will be parabola of increasing slope. Thus, graph (iv) is correct.

Power associated with force is equal to,

$P = Fv = Fat$

Thus, graph of $P$versus $t$will be a straight line. Hence, graph (i) is correct. And graphs (ii) and (iii) are incorrect.

**3. Correct answer: (B)**

Kinetic energy decreases when body works against friction. The work done is independent of time, while power of a body varies inversely with time. The work done over a closed path is zero for conservative forces and non-zero for non-

conservative forces.

P–(iv), Q–(i), R–(ii), S–(iii)

**4. Correct answer: (C)**

For conservative force,

Change in potential energy = −Force applied ×Displacement

= −(Area under force-position curve)

For graph I, change in potential energy

$= -\frac{1}{2} \times F_1 x_1$

For graph II, change in potential energy

$= -F_1 x_1$

For graph III, change in potential energy

$= \frac{1}{2} F_1 x_1$

Thus change in potential energy is maximum in graph III and minimum in graph II.

**5. Correct answer: (C)**

At highest point $P$, kinetic energy will transform into potential energy. Thus,

$mgh = 10$ J

$h = \frac{10}{0.5 \times 10} = 2$ m

As in the $OP$part, spherical ball is in free fall. Hence acceleration will be 10 $ms^{-2}$.

From the graph, the average resistive force provided by water from $P$to $Q$,

$= \frac{\text{Change in KE}}{\text{Change in distance}} + \text{Weight of ball } = \frac{5}{0.1} + 0.5 \times 10 = 55$ N

**6. Correct answer: (A)**

Height = 3 m, $v$ =?

Change in PE =KE;

$m \times g \times 3 = \frac{1}{2} m v^2 \Rightarrow \sqrt{2 \times 10 \times 3} = v; v = \sqrt{60} = 7.7$ m/s

**7. Correct answer: (B)**

By work energy theorem, work done =change in KE,

$\Delta KE = \frac{1}{2} \times (12 + 16) \times 5 + \frac{1}{2}(16 + 12) \times 5 = 70 + 70 = 140$ J

**8. Correct answer: (B)**

Force, $F = 60000$ N

Work done by motor,

$W = F \times S = 60000 \times 12 = 720000$ J = 720 kJ

Work done by gravity during this process is −720 kJ.

**9. Correct answer: (B)**

Height attained by both the girls is same, thus work done against gravity will be same by both of them. Also net energy transfer will be from muscular (chemical) to potential energy.

When Divya has fallen $\left(\frac{2}{3}\right)^{\text{rd}}$ then,

$K$ = change in potential energy = $mg\left(\frac{2}{3}h\right)$

$= 50 \times 10\left(\frac{2}{3}\right) \times 12 = 4000$ J

**10. Correct answer: (C) Here**

$m_1 = 2$ kg; $u_1 = 0$; $v_1 = v$

$m_2 = 6$ kg; $u_2 = 0$; $v_2 = 2v$

As, work done =change in kinetic energy,

$$\frac{\text{Work done by force, } F_2}{\text{Work done by force, } F_1} = \frac{\frac{1}{2}m_2(v_2^2 - u_2^2)}{\frac{1}{2}m_1(v_1^2 - u_1^2)}$$

$$\Rightarrow \frac{\text{Work done by force, } F_2}{\text{Work done by force, } F_1} = \frac{6 \times (2v)^2}{2v^2} = \frac{6 \times 4}{2} = \frac{12}{1}$$

Work done by force, $F_2$: Work done by force, $F_1 = 12:1$.

# 5. Pressure
# Solutions

**LEVEL-I**

**1. Answer: B) Thrust**

**Explanation:**

Thrust is the total force acting normally (perpendicularly) on a surface.

**2. Answer: A) Newton (N)**

**Explanation:**

Since thrust is a force, its SI unit is the newton (N).

**3. Answer: B) Dyne**

**Explanation:**

The C.G.S. unit of force (and hence thrust) is the dyne.

**4. Answer: A) Pressure**

**Explanation:**

Pressure is defined as the normal force acting per unit area.

$P = \frac{F}{A}$

**5. Answer: C) Hydrostatic pressure**

**Explanation:**

The pressure exerted by a stationary liquid because of gravity is called hydrostatic pressure.

**6. Answer: D) Mercury**

**Explanation:**

Mercury is preferred because it has:

- High density
- Low vapour pressure
- Does not wet glass

Hence the height of the mercury column is only about 76 cm.

**7. Answer: B) 1 : 10**

**Explanation:**

SI unit:

$1\ \text{Pa} = 1\ \text{N/m}^2$

CGS unit:

$1\ \text{dyne/cm}^2$

Since,

$1\ \text{N} = 10^5\ \text{dyne}$

and

$1\ \text{m}^2 = 10^4\ \text{cm}^2$

Therefore,

$1\ \text{N/m}^2 = \frac{10^5}{10^4} = 10\ \text{dyne/cm}^2$

Hence,

$1\ \text{dyne/cm}^2 : 1\ \text{Pa} = 1 : 10$

**8. Answer: A) $\mathbf{1.01 \times 10^5\ N/m^2}$**

**Explanation:**

Standard atmospheric pressure is

$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$
approximately
$1.01 \times 10^5 \text{ Pa}$

**9. Answer: C) 10 $\text{m}^2$**
**Explanation:**
$P = \frac{F}{A}$
$A = \frac{F}{P} = \frac{40}{4} = 10 \text{ m}^2$

**10. Answer: B) 2 atm**
**Explanation:**
$P_{\text{total}} = P_{\text{atm}} + P_{\text{liquid}}$
$3 = 1 + P_{\text{liquid}}$
$P_{\text{liquid}} = 2 \text{ atm}$

**11. Answer: A) 3200 Pa**
**Explanation:**
Convert area:
$50 \text{ cm}^2 = 50 \times 10^{-4} = 0.005 \text{ m}^2$
Now,
$P = \frac{F}{A} = \frac{16}{0.005} = 3200 \text{ Pa}$

**12. Answer: B) $hdg$**
**Explanation:**
Hydrostatic pressure is
$P = hdg$
where

- $h$= depth
- $d$= density
- $g$= acceleration due to gravity

**13. Answer: C) 6400 Pa**
**Explanation:**
Convert density:
$0.8 \text{ g/cm}^3 = 800 \text{ kg/m}^3$
Now,
$P = hdg$
$= 0.8 \times 800 \times 10 = 6400 \text{ Pa}$

**14. Answer: A) 34 m**
**Explanation:**
$P = hdg$
$333200 = h \times 1000 \times 9.8$
$h = \frac{333200}{9800} = 34 \text{ m}$

**15. Answer: D) 3920 Pa**
**Explanation:**
Convert depth:
$40 \text{ cm} = 0.4 \text{ m}$
$P = 0.4 \times 1000 \times 9.8 = 3920 \text{ Pa}$

**16. Answer: C) 10.3 m**

**Explanation:**

Atmospheric pressure remains the same.

$h_w d_w = h_m d_m$

$h_w \times 1000 = 0.76 \times 13600$

$h_w = 10.336 \text{ m} \approx 10.3 \text{ m}$

**17. Answer: A) $32.6 \times 10^3$ Pa**

**Explanation:**

Weight

$F = mg = 60 \times 9.8 = 588$ N

Area

$180 \text{ cm}^2 = 0.018 \text{ m}^2$

Pressure

$P = \frac{588}{0.018} = 32666.7$ Pa

$P \approx 32.6 \times 10^3$ Pa

**18. Answer: B) $2 \times 10^5$ Pa**

**Explanation:**

$\Delta P = hdg$

$= 20 \times 1000 \times 10 = 2 \times 10^5$ Pa

**19. Answer: A) Barometer**

**Explanation:**

**A barometer is used to measure atmospheric pressure.**

**20. Answer: A) $1.0129 \times 10^5$ Pa**

**Explanation:**

Using CGS units,

$P = hdg$

$= 76 \times 13.6 \times 980 = 1{,}012{,}928 \text{ dyne/cm}^2$

Since

$1 \text{ Pa} = 10 \text{ dyne/cm}^2$

$P = \frac{1{,}012{,}928}{10} = 101{,}292.8$ Pa

$\boxed{P \approx 1.0129 \times 10^5 \text{ Pa}}$

**21. Answer: C) $1.225 \times 10^4$ dyne/cm$^2$**

**Explanation:**

Using CGS units,

$P = hdg$

$= 12.5 \times 1 \times 980 = 12{,}250 \text{ dyne/cm}^2$

$= 1.225 \times 10^4 \text{ dyne/cm}^2$

**22. Answer: A) Pressure**

**Explanation:**

This is the definition of Pascal's law.

**23. Answer: D) Pascal's law**

**Explanation:**

Hydraulic lifts, hydraulic brakes and hydraulic presses all work on Pascal's law, which states that pressure applied to an enclosed fluid is transmitted equally in all directions.

**24. Answer: A)**

$$\boxed{\frac{F_1}{A_1} = \frac{F_2}{A_2}}$$

**Explanation:**

Since pressure is transmitted equally,

$P_1 = P_2$

and

$P = \frac{F}{A}$

Therefore,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

**25. Answer: C) 20 N**

**Explanation:**

Using Pascal's law,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

$\frac{10}{5} = \frac{F_2}{10}$

$2 = \frac{F_2}{10}$

$F_2 = 20$ N

Answer: 20 N

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**LEVEL-II**

**1. Answer: C**

**Explanation:**

Area = $0.5\ \text{cm}^2 = 0.5 \times 10^{-4}\ \text{m}^2$

Using the formula,

$P = \frac{F}{A}$

$P = \frac{500}{0.5 \times 10^{-4}} = 10^7$ Pa

Hence, the correct answer is C.

**2. Answer: D**

**Explanation:**

Both liquids and gases are called fluids because their particles can move freely past one another, allowing them to flow and take the shape of their container.

**3. Answer: C**

**Explanation:**

Using the hydrostatic pressure formula,

$P = hdg$

Substituting the given values,

$100 = h \times 4 \times 10$

$40h = 100$

$h = \frac{100}{40} = 2.5$ m

Hence, the correct answer is C.

**4. Answer: C**

**Explanation:**

The total (absolute) pressure at a depth *h*inside a liquid is given by

$P = P_0 + hdg$

where

- $P_0$= atmospheric pressure,
- $hdg$= pressure due to the liquid column.

Hence, the correct answer is C.

**5. Answer: B**

**Explanation:**

Using

$P = hdg$

$600 = h \times 12 \times 10$

$120h = 600$

$h = \frac{600}{120} = 5$ m

Hence, the correct answer is B.

**ONE OR MORE THAN ONE CORRECT ANSWER TYPE**

**6. Correct Option(s): A, B**

**Explanation:**

Thrust and Weight are both forces, so they have the same SI unit, the Newton (N).

- A. Thrust and Weight, Both are forces (Unit: N).
- B. Weight and Thrust, Same reason as above.

**7. Correct Option(s): A, B, C, D**

**Explanation:**

A hydraulic press works on Pascal's Principle, according to which pressure applied to an enclosed fluid is transmitted equally in all directions.

Its common applications include:

- A. Lifting and servicing vehicles
- B. Crushing seeds to extract oil
- C. Punching holes in metal sheets
- D. Forging and shaping metal sheets

**8. Correct Option(s): A, B**

**Explanation:**

Given,

- Force = 5 N
- Area = 5 m²

$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{5}{5} = 1\ \text{N/m}^2$

Since,

$1\ \text{N/m}^2 = 1$ Pascal (Pa)

Therefore,

- A. 1 N/m²
- B. 1 Pa

**9. Correct Option(s): B, C**

**Explanation:**

The SI unit of pressure is the Pascal (Pa).

Also,

$1\ \text{Pa} = 1\ \text{N/m}^2$

Therefore,

- B. Pascal (Pa) SI unit of pressure.
- C. Newton per square metre ($N/m^2$) Equivalent to Pascal.

**10. Correct Option: D**

Explanation:

Given,

- Force = 400 N
- Area = 0.004 $m^2$

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{400}{0.004} = \frac{400}{4 \times 10^{-3}} = 1 \times 10^5 \text{ N/m}^2$$

Since,

$1 \text{ N/m}^2 = 1 \text{ Pa}$

$\therefore$ Pressure $= 1 \times 10^5$ Pa

Hence, Option D is the correct answer.

**Comprehension Type**

**11. Answer: C**

**Explanation:**

Area

$200 \text{ cm}^2 = 200 \times 10^{-4} = 0.02 \text{ m}^2$

Using

$F = PA$

$F = 27500 \times 0.02 = 550$ N

Hence, the correct answer is C.

**12. Answer: D**

**Explanation:**

Using

$$A = \frac{F}{P}$$

$$A = \frac{300}{1500} = 0.2 \text{ m}^2$$

Converting into $cm^2$,

$0.2 \times 10000 = 2000 \text{ cm}^2$

Hence, the correct answer is D.

**13. Answer: A**

**Explanation:**

Using

$F = PA$

$F = 50000 \times 0.20 = 10000$ N

Hence, the correct answer is A.

**14. Answer: B**

**Explanation:**

Pressure difference,

$\Delta P = 120000 - 30000 = 90000$ Pa

Using

$\Delta P = hdg$

$90000 = h \times 1000 \times 10$

$$h = \frac{90000}{10000} = 9 \text{ m}$$

Hence, the correct answer is B.

**15. Answer: C**

**Explanation:**

Pressure difference,

$\Delta P = 150000 - 30000 = 120000$ Pa

Using

$\Delta P = hdg$

$120000 = h \times 1000 \times 10$

$h = \frac{120000}{10000} = 12$ m

Hence, the correct answer is C.

**16. Answer: B**

**Explanation:**

Using

$P = hdg$

$P = 1 \times 20 \times 10 = 200$ Pa

Hence, the correct answer is B.

**Assertion and Reason Type**

**17. Answer: A**

**Explanation:**

Both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Fluid pressure increases with depth because the height of the liquid column above a point increases, increasing the weight of the liquid and hence the pressure.

**18. Answer: A**

**Explanation:**

Both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

A submerged football rises because the pressure at its lower surface is greater than the pressure at its upper surface, producing a net upward buoyant force.

**19. Answer: A**

**Explanation:**

Both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Since

$P = \frac{F}{A}$,

increasing the contact area decreases the pressure. The camel's broad feet spread its weight over a larger area, preventing it from sinking into sand.

**20. Answer:**

- **a → P**
- **b → R**
- **c → Q**
- **d → S**

**Explanation:**

- a → P: Total pressure at a depth is given by $P = P_0 + hdg$.
- b → R: Hydrostatic pressure depends directly on the depth of the liquid.

- c → Q: Liquid pressure at a given depth is independent of the container's cross-sectional area.
- d → S: Pressure exerted sideways by a liquid is called lateral pressure**.**

**21. Answer:**

- **a → Q, R**
- **b → S**
- **c → P**
- **d → T**

**Explanation:**

- a → Q, R: Pascal (Pa) and N/m² are equivalent SI units of pressure.
- b → S: The CGS unit of pressure is dyne/cm².
- c → P: 1 bar = $10^5$ Pa.
- d → T:

1 millibar = $10^{-3}$ bar = $10^{-3} \times 10^5 = 10^2$ Pa

**22. Answer: 10**

**Explanation:**

Using Pascal's Principle,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

$\frac{F_1}{5} = \frac{1250}{625}$

$\frac{1250}{625} = 2$

$F_1 = 5 \times 2 = 10$ N

Hence, the required force is 10 N.

**23. Answer: 7200 (kg-weight) or 72000 (N)**

**Explanation:**

The force multiplication in a hydraulic press depends on the ratio of piston areas.

$\frac{A_{\text{large}}}{A_{\text{small}}} = \left(\frac{30}{2.5}\right)^2 = 12^2 = 144$

Therefore,

$F_{\text{large}} = 144 \times 50 = 7200$ kg-wt

Since

1 kg-wt ≈ 10 N,

$F_{\text{large}} = 72000$ N

Hence, the answer is 7200 kg-weight or 72000 N.

**24. Answer: 2500**

**Explanation:**

Using the relation,

Thrust = Pressure × Area

$F = 500 \times 5 = 2500$ N

Hence, the thrust is 2500 N.

# 6. Light

**LEVEL-I**

**1. Correct Option: D**

**Explanation:**

Light is an electromagnetic wave and does not require any material medium for its propagation. It can travel through a vacuum. Therefore, the statement that light requires a material medium is false, making Option D the correct answer.

**2. Correct Option: C**

**Explanation:**

When light falls on the surface of water, a part of it is reflected while the remaining part is refracted into the water. Since a significant portion of light enters the water, the reflected light has lower intensity, causing the image seen on the water surface to appear less bright than the actual object.

**3. Correct Option: B**

**Explanation:**

A concave mirror forms an erect, virtual, and magnified image only when the object is placed between the pole (P) and the principal focus (F) (i.e., within the focal length).

**4. Correct Option: A**

**Explanation:**

According to the principle of reversibility of light, when a point source is placed at the principal focus of a concave mirror, the reflected rays emerge parallel to the principal axis, producing a parallel beam.

**5. Correct Option: A**

**Explanation:**

A convex mirror always produces a virtual, erect, and diminished image of a real object, irrespective of the object's position. Hence, it never forms an inverted image of a real object.

**6. Correct Option: C**

**Explanation:**

For spherical mirrors,

$f = \frac{R}{2}$

where $f$ is the focal length and $R$ is the radius of curvature. This relation is valid for both concave and convex mirrors under the paraxial approximation.

**7. Correct Option: B**

**Explanation:**

Among spherical mirrors, only a concave mirror can produce a virtual image larger than the object, provided the object lies between the pole and the principal focus.

**8. Correct Option: D**

**Explanation:**

When an object is placed at the centre of curvature of a concave mirror,

$u = 2f, v = 2f$

The image is formed at the same point as the object. Therefore, the distance between the object and its image is zero.

**9. Correct Option: C**

**Explanation:**

Power of a mirror is given by

$P = \frac{1}{f(\text{in metres})}$

For a convex mirror,

$f = +20$ cm $= +0.2$ m

$P = \frac{1}{0.2} = 5$ D

Hence, the correct answer is 5 D.

**10. Correct Option: C**

**Explanation:**

For a real image magnified $n$times,

$m = -n = -\frac{v}{u}$

Hence,

$v = nu$

Using the mirror formula,

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{nu} + \frac{1}{u} = \frac{1}{f}$

$\frac{n+1}{nu} = \frac{1}{f}$

Therefore,

$u = \frac{n+1}{n} f$

**11. Correct Option: C**

**Explanation:**

A dentist uses a concave mirror to obtain a virtual image magnified four times.

Given,

$m = +4, u = -0.60$ cm

Using

$m = \frac{f}{f-u}$

$4 = \frac{f}{f+0.60}$

$4f + 2.4 = f$

$3f = -2.4$

$f = -0.8$ cm

Radius of curvature,

$R = 2 \mid f \mid = 1.60$ cm

**12. Correct Option: D**

**Explanation:**

An erect image magnified four times can only be formed by a concave mirror.

Let

$u = -x, v = +4x$

The distance between the object and image is

$4x - (-x) = 5x = 90$ cm

$x = 18$ cm

Thus,

$u = -18$ cm, $v = +72$ cm

Using the mirror formula,

$$\frac{1}{72} - \frac{1}{18} = \frac{1}{f}$$

$$\frac{-3}{72} = \frac{1}{f}$$

$f = -24$ cm

**13. Correct Option: D**

**Explanation:**

A concave mirror forms an image twice the size of the object in two positions:

- Object between P and F → Virtual, erect, magnified image.
- Object between F and C → Real, inverted, magnified image.

Since moving the object farther away again produces a magnified image, the initial position must have been between the pole and the principal focus.

**14. Correct Option: B**

**Explanation:**

Given,

$m = -3, u = -20$ cm

Using

$$m = \frac{f}{f - u}$$

$$-3 = \frac{f}{f + 20}$$

$$-3f - 60 = f$$

$$4f = -60$$

$f = -15$ cm

Therefore, the magnitude of the focal length is

$| f | = 15$ cm

**15. Correct Option: A**

**Explanation:**

Given,

$R = 15$ cm

$f = -7.5$ cm, $u = -10$ cm

Using

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{-7.5} - \frac{1}{-10}$$

$$= -\frac{2}{15} + \frac{1}{10}$$

$$= -\frac{1}{30}$$

$v = -30$ cm

Magnification,

$$m = -\frac{v}{u} = -\frac{-30}{-10} = -3$$

Thus, the image is real, inverted, and three times larger.

**16. Correct Option: D**

**Explanation:**

Given,

$m = -4.5 = -\frac{9}{2}, u = -20$ cm

Using

$m = \frac{f}{f-u}$

$-\frac{9}{2} = \frac{f}{f+20}$

$-9f - 180 = 2f$

$11f = -180$

$f = -\frac{180}{11}$ cm

**17. Correct Option: C**

**Explanation:**

A convex mirror always forms a virtual, erect, and diminished image for every real object placed in front of it.

**18. Correct Option: C**

**Explanation:**

Given,

$R = 20$ cm

$f = -10$ cm, $u = -100$ cm

Magnification,

$m = \frac{f}{f-u} = \frac{-10}{-10+100} = -\frac{1}{9}$

If the object height is

$h_o = 5$ cm

then

$h_i = mh_o = -\frac{1}{9} \times 5 = -0.555$ cm

Hence, the image size is approximately

0.55 cm

**19. Correct Option: A**

**Explanation:**

Given,

$f = -50$ cm, $m = -2$

Using

$m = \frac{f}{f-u}$

$-2 = \frac{-50}{-50-u}$

$100 + 2u = -50$

$2u = -150$

$u = -75$ cm

Therefore, the object should be placed 75 cm from the mirror.

**20. Correct Option: A**

**Explanation:**

The object is placed at

$u = -4f$

agnification,

$$m = \frac{f}{f-u} = \frac{-f}{-f+4f} = -\frac{1}{3}$$

If the object height is

6 cm

then

$$| h_i | = \frac{1}{3} \times 6 = 2 \text{ cm}$$

Hence, the image length is 2 cm.

**21. Correct Option: D**

**Explanation:**

For a convex mirror,

$f = +30$ cm, $u = -30$ cm

Using the mirror formula,

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{30}$$

$$\frac{1}{v} = \frac{2}{30} = \frac{1}{15}$$

$v = +15$ cm

The positive sign indicates that the image is formed 15 cm behind the mirror.

**22. Correct Option: D**

**Explanation:**

A convex mirror forms a virtual, erect, and diminished image for every real object placed in front of it, irrespective of whether the object is:

- Between the pole and the focus,
- Beyond the centre of curvature, or
- At infinity.

Hence, all the given ranges satisfy the condition. Therefore, the correct answer is All of the above.

---

**LEVEL-II**

**1. Answer: (A)**

**Explanation:**

The object is placed 40 cm from a concave mirror having a focal length of 20 cm. Since,

$u = 2f = 40$ cm

the object is at the centre of curvature (C). Therefore, the image is formed at the centre of curvature itself. The image is:

- Real
- Inverted
- Same in size as the object

**2. Answer: (D)**

**Explanation:**

For any real object, a convex mirror always forms an image that is:

- Virtual
- Erect

- Diminished

The image is formed behind the mirror between the pole (P) and the focus (F).
Hence, the statement "Image is real" is incorrect.

**3. Answer: (A)**

**Explanation:**

For a convex mirror,

$f = +10$ cm, $m = +\frac{1}{2}$

Using the mirror magnification formula,

$m = \frac{f}{f-u}$

$\frac{1}{2} = \frac{10}{10-u}$

$10 - u = 20$

$u = -10$ cm

Hence, the object is placed 10 cm in front of the mirror.

**4. Answer: (A)**

**Explanation:**

The real image is obtained on a wall, so the magnification is $m = -16$.

Let the flower be at a distance $x$cm from the mirror.

Then,

$u = -x$

Since the wall is 120 cm from the flower,

$v = -(x + 120)$.

Using

$m = -\frac{v}{u}$,

$-16 = -\frac{-(x+120)}{-x}$

$16x = x + 120$

$15x = 120$

$x = 8$ cm

Therefore, the flower is 8 cm from the mirror.

**5. Answer: (C)**

**Explanation:**

Given,

$f = -20$ cm

and the image is twice the size of the object,

$|m| = 2$.

Case 1: Real image

$m = -2$

$u = -30$ cm, $v = -60$ cm

Distance between object and image:

$|-60 - (-30)| = 30$ cm

Case 2: Virtual image

$m = +2$

$u = -10$ cm, $v = +20$ cm

Distance between object and image:

$20 - (-10) = 30$ cm

Thus, in both cases, the distance between the object and the image is $\boxed{30\text{ cm}}$.

**6. Answer: (B)**

**Explanation:**

The mirror forms an erect image whose height is 2 m for a 6 m tall object.
Since the image is erect and diminished, the mirror must be a convex mirror, as it is the only mirror that always produces such an image for a real object.

**ONE OR MORE THAN ONE ANSWER TYPE**

**7. Correct Options: A, B, C, D**

**Explanation:**

- (A) & (B): A convex mirror always forms a virtual, erect image for a real object. However, if the incident rays are already converging (i.e., the mirror has a virtual object), a convex mirror can also produce a real image.
- (C): A concave mirror forms a magnified image when:
  - the object is placed between the focus (F) and the centre of curvature (C) (real, inverted, magnified), or
  - the object is placed between the pole (P) and the focus (F) (virtual, erect, magnified).
- (D): A concave mirror produces an inverted image for every real object placed beyond the focus (F).

**8. Correct Options: A, B**

**Explanation:**

- A plane mirror always forms a virtual and erect image irrespective of the object's position.
- A convex mirror also always forms a virtual and erect image for every real object.
- A concave mirror forms a virtual and erect image only when the object is between the pole and the focus. For all other object positions, it forms a real and inverted image. Therefore, it does not satisfy the given condition.

**9. Correct Options: A, B**

**Explanation:**

A concave mirror can produce both real and virtual images depending on the position of the object.

- When the object is placed beyond the focus (u > f), the image formed is real and inverted.
- When the object is placed between the pole and the focus (u < f), the image formed is virtual and erect.

Hence, the image formed by a concave mirror is not always real or always virtual.
Therefore, statements (A) and (B) are correct, while (C) and (D) are incorrect because they make absolute generalizations.

**10. Correct Options: A, B**

**Explanation:**

Using the mirror magnification formula,

$$m = \frac{f}{f - u}$$

where $u$is the object distance and $f$is the focal length (using the standard Cartesian sign convention).
Since the question does not specify whether the image is real or virtual, both possibilities must be considered.
Case 1: Virtual image ($m = +2$)

$$2 = \frac{f}{f-u}$$
$$2(f-u) = f$$
$$2f - 2u = f$$
$$u = \frac{f}{2}$$

Case 2: Real image ($m = -2$)

$$-2 = \frac{f}{f-u}$$
$$-2(f-u) = f$$
$$-2f + 2u = f$$
$$2u = 3f$$
$$u = \frac{3f}{2}$$

Therefore, the object can be placed at either

$$\boxed{\frac{f}{2} \text{ or } \frac{3f}{2}}$$

depending on whether the image is virtual or real.

**11. Correct Options: A, C, D**

**Explanation:**

The Pole (P), Radius of Curvature (R), and Principal Axis are fixed geometric properties of a spherical mirror. They depend only on the mirror's shape and remain unchanged regardless of how light rays fall on the mirror.

However, the Principal Focus (F) is ideally defined for paraxial rays (rays close to the principal axis). For rays striking the mirror far from the principal axis (marginal rays), spherical aberration causes them to converge at a different point. Thus, the exact location of the focus depends on the nature of the incident rays.

Hence:

- Pole – Fixed geometric point ✓
- Radius of Curvature – Fixed geometric property ✓
- Principal Axis – Fixed geometric line ✓

**12. Answer: (A)**

**Explanation:**

Given,

$u = -30$ cm, $f = -20$ cm

Using the mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$
$$\frac{1}{v} = -\frac{1}{20} + \frac{1}{30}$$
$$\frac{1}{v} = -\frac{1}{60}$$
$$v = -60 \text{ cm}$$

The negative sign indicates that the image is formed 60 cm in front of the mirror.

**13. Answer: (B)**

**Explanation:**

Magnification is

$$m = -\frac{v}{u}$$

$m = -\frac{-60}{-30}$

$m = -2$

Hence, the magnitude of magnification is

$\boxed{2}$.

**14. Answer: (A)**

**Explanation:**

Initially,

$u = 30$ cm

The object is moved 10 cm closer to the mirror.

Therefore,

$u = 20$ cm

Since

$u = f = 20$ cm,

the object is placed at the principal focus of the concave mirror.

Hence, the reflected rays become parallel, and the image is formed at infinity.

**15. Answer: (B)**

**Explanation:**

Given,

$f = +10$ cm, $u = -20$ cm

Using the mirror formula,

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{10} + \frac{1}{20}$

$\frac{1}{v} = \frac{3}{20}$

$v = \frac{20}{3}$ cm

Since $v$is positive, the image is formed

$\boxed{\frac{20}{3} \text{ cm}}$

behind the mirror.

**16. Answer: (A)**

**Explanation:**

Magnification is

$m = -\frac{v}{u}$

$m = -\frac{20/3}{-20}$

$m = \frac{1}{3}$

Hence, the magnification is

$\boxed{\frac{1}{3}}$.

**17. Answer: (C)**

**Explanation:**

Given,

$h_o = 0.5$ cm, $m = \frac{1}{3}$

Image height,

$h_i = m \times h_o$

$= \frac{1}{3} \times 0.5$

$= 0.167$ cm

Hence,

$\boxed{h_i \approx 0.167 \text{ cm}}$.

**18. Answer: (B)**

**Explanation:**

Assertion (A): A plane mirror always has a magnification of +1. (True)

Reason (R): A plane mirror forms a virtual and erect image. (True)

However, being virtual and erect alone does not explain why the image is exactly the same size as the object. Therefore, both statements are true, but R is not the correct explanation of A.

**19. Answer: (A)**

**Explanation:**

As the object distance approaches zero ($u \to 0$), the radius of curvature of the mirror becomes much larger than the object distance ($R \gg u$).

Under this condition, a very small portion of the curved mirror behaves like a plane mirror. Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion**.**

**20. Answer Key:**

**a → Q, b → S, c → P, d → T**

**Detailed Explanation**

a) Object at the focal length of a convex mirror

For a convex mirror,

- Object distance, $u = -f$
- Focal length, $f = +f$

Using the mirror formula,

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} + \frac{1}{-f} = \frac{1}{f}$

$\frac{1}{v} = \frac{2}{f}$

$v = \frac{f}{2}$

Magnification,

$m = -\frac{v}{u} = -\frac{f/2}{-f} = \frac{1}{2} = 0.5$

Hence, it matches Q (Magnification = 0.5).

b) Object at the centre of curvature of a concave mirror

For a concave mirror,

$u = -2f, v = -2f$

Magnification,

$m = -\frac{v}{u} = -\frac{-2f}{-2f} = -1$

Hence, it matches S (Magnification = –1).

c) Object at the principal focus of a concave mirror

When the object is placed at the principal focus,

- Reflected rays become parallel.
- Image is formed at infinity.

Thus,

$v = -\infty$

Magnification,

$m = -\frac{v}{u} = -\frac{-\infty}{-f} = -\infty$

Hence, it matches P (Magnification = –∞).

d) Object at the radius of curvature of a convex mirror

For a convex mirror,

$u = -2f, f = +f$

Using the mirror formula,

$\frac{1}{v} + \frac{1}{-2f} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{f} + \frac{1}{2f} = \frac{3}{2f}$

$v = \frac{2f}{3}$

Magnification,

$m = -\frac{v}{u} = -\frac{2f/3}{-2f} = \frac{1}{3} \approx 0.33$

Hence, it matches T (Magnification ≈ 0.33).

**21. Answer Key**

- **a → Q, T**
- **b → P, S**
- **c → P, T**
- **d → R, T**

**Detailed Explanation**

a) $m = \frac{1}{4}$

Since

- $m > 0$, the image is virtual and erect.
- $|m| < 1$, the image is diminished.

A diminished, virtual image of a real object is produced only by a convex mirror.

Therefore,

- Q – Convex mirror
- T – Virtual image

b) $m = -1$

Since

- $m < 0$, the image is real and inverted.
- $|m| = 1$, the image is the same size as the object.

This occurs when the object is placed at the centre of curvature of a concave mirror.

Therefore,

- P – Concave mirror
- S – Real image

c) $m = 2$

Since

- $m > 0$, the image is virtual and erect.

- $|m| > 1$, the image is magnified.
  A magnified virtual image is formed only by a concave mirror when the object lies between the pole and the focus.
  Therefore,
- P – Concave mirror
- T – Virtual image

  d) $m = 1$
  Since
- $m > 0$, the image is virtual and erect.
- $|m| = 1$, the image is the same size as the object.
  This is the characteristic property of a plane mirror.
  Therefore,
- R – Plane mirror
- T – Virtual image

**Integer Type Questions**

**22. Answer: 2**

**Solution**

Given,

$u = -5$ cm, $f = -10$ cm

Using the magnification formula,

$$m = \frac{f}{f-u}$$

Substituting the values,

$$m = \frac{-10}{-10-(-5)} = \frac{-10}{-5} = 2$$

**Answer:** $\boxed{2}$

**23. Answer: −30cm**

**Solution**

Given,

$u = -30$ cm, $f = -15$ cm

Since

$R = 2f = 2(-15) = -30$ cm

the object is placed at the centre of curvature.

For a concave mirror, when the object is at the centre of curvature,

$v = u = -30$ cm

Therefore,

$\boxed{v = -30 \text{ cm}}$

*(Magnitude of the image distance = 30 cm.)*

**24. Answer: −2**

**Solution**

Given,

$h_o = 5$ cm, $h_i = -10$ cm

Magnification is

$$m = \frac{h_i}{h_o}$$

Substituting,

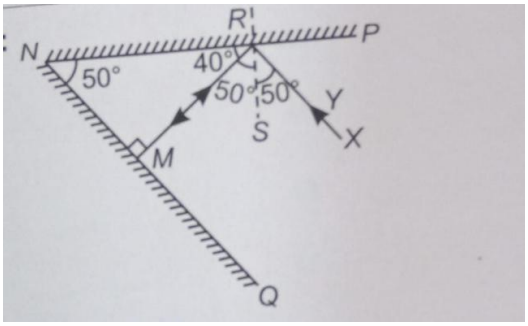
$$m = \frac{-10}{5} = -2$$

Therefore,

$\boxed{m = -2}$

## PREVIOUS YEAR QUESTIONS

**1. Correct Answer: is B)**




By law of reflection,

∠XRS = ∠MRS = 50°

(∵ ∠i = ∠r)

In ΔRNM,

∠MRN = 40°; ∠RNM = 50° (given)

∴ ∠RMN = 90°

As the light ray is incident normally on the mirror Q, it will get retrace the same path.

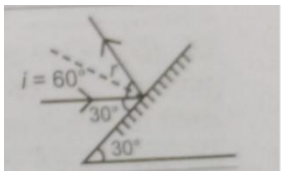
Hence, angle of reflection is 0°.

**2. Correct Answer is B): The rays of light coming**

from the object P only can reach the eye after reflected back by the mirror, so only object P is seen in the mirror.

**3. Correct Answer: (B)**

Angle of incidence,




$\angle i = 90° - 30° = 60°$

By law of reflection,

$\angle r = \angle i = 60°$

Angle of incidence = 60°

Angle of reflection = 60°

**4. Correct Answer: is (B) : The object at larger**

distance from the mirror that is closer to the obstacle will disappear first. Thus R will disappear first.

**5. Correct Answer: is (A) : In case of spherical**

mirrors (concave or convex) the incident parallel rays after reflection may converge or diverge. It is only possible in case of plane mirror that the parallel incident rays reflected as a parallel beam.

**6. Correct Answer: is (B): Image ($I_1$) formed by $M_2$**

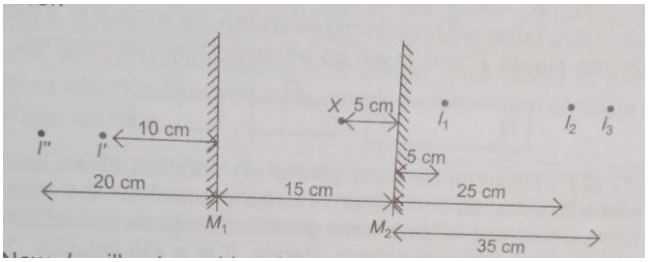
will be at 5 cm behind the mirror.

Distance between mirror $M_1$ and object X is 10 cm.

So, image (I′) formed by $M_1$ will be 10 cm behind $M_1$.

This image will act as object for $M_2$. Now, distance between this object and $M_2$ = 10 + 15 = 25 cm.

Second image ($I_2$) formed by $M_2$ will be 25 cm behind the mirror.



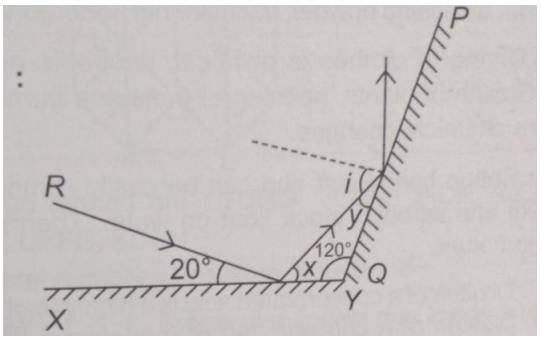

Now, $I_1$ will act as object for $M_1$ and the distance between $I_1$ and $M_1$ is 20 cm. So the image (I″) formed by $M_1$ will be 20 cm behind the mirror.

Image I″ will act as object for $M_2$ and distance between this object (I″) and $M_2$ = 20 + 15 = 35 cm.

Third image ($I_3$) formed by $M_2$ will be 35 cm behind $M_2$.

**7. Correct Answer: is B)**

Angle $x = 20°$




$y = 180° - 120° - 20°$

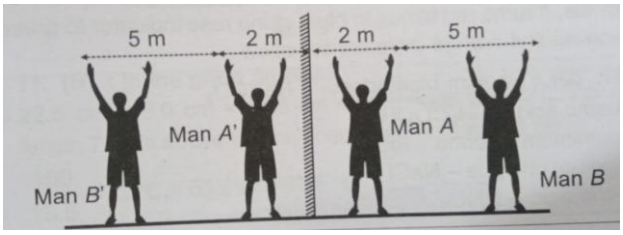
$y = 40°$

$i = 90° - y$

$i = 90° - 40° = 50°$

**8. Correct Answer is D) : The image is at a same**

distance as that of object infront of plane mirror,




So distance = 5 + 2 + 2 = 9 m

**9. Correct Answer : is (D) : If the mirror is rotated**

by an angle θ, the new reflected ray is rotated by an angle 2θ in same direction. So, it rotates 20° anticlockwise.

# 7. Electricity and its effects

## Topic-1:
## Electric Current and Chemical Effects of Electricity
## LEVEL – I

**1. Answer: A**

**Explanation:** When a glass rod is rubbed with a silk cloth, electrons are transferred from the glass rod to the silk cloth. As the glass rod loses electrons, it becomes positively charged, while the silk cloth gains electrons and becomes negatively charged.

**2. Answer: A**

**Explanation:** Electrons carry a negative charge. When a neutral body loses electrons, it has more protons than electrons, resulting in a positive charge**.**

**3. Answer: C**

**Explanation:** Charging by induction occurs without direct physical contact. A charged object brought near a conductor causes redistribution of charges within the conductor. Charging by friction and conduction both require physical contact.

**4. Answer: C**

**Explanation:** An electroscope is used to detect the presence of electric charge. If it is already charged with a known type of charge, it can also be used to determine whether an unknown charge is positive or negative.

**5. Answer: A**

**Explanation:** An electric cell converts chemical energy into electrical energy and maintains a nearly constant potential difference (voltage) across its terminals.

**6. Answer: B**

**Explanation:** Primary cells undergo irreversible chemical reactions. Once discharged, they cannot be recharged and must be replaced.

**7. Answer: C**

**Explanation:** Electromotive force (EMF) is the energy supplied by a source per unit charge and is measured in volts (V).

**8. Answer: D**

**Explanation:** MCB stands for Miniature Circuit Breaker. It automatically disconnects the circuit during an overload or short circuit, protecting electrical appliances and wiring.

**9. Answer: B**

**Explanation:** Like charges repel each other, whereas unlike charges attract each other.

**10. Answer: B**

**Explanation:** An electric cell converts chemical energy into electrical energy while supplying current to a circuit.

**11. Answer: D**

**Explanation:** Electrolysis is used in electroplating, extraction of metals, and purification (refining) of metals.

**12. Answer: D**

**Explanation:** An electrolyte is a substance that dissociates into ions when molten or dissolved in water, allowing it to conduct electricity.

**13. Answer: D**

**Explanation:** The chemical effects produced during electrolysis depend on the nature of the electrolyte, the electrodes used, and the amount of current passing through the solution.

**14. Answer: A**

**Explanation:** Drinking water contains dissolved salts and minerals, which provide free ions that conduct electricity. Pure distilled water is a poor conductor because it contains very few ions.

**15. Answer: B**

**Explanation:** Electroplating is mainly done to protect the metal from corrosion and improve its appearance by coating it with a thin layer of another metal.

**16. Answer: C**

**Explanation:** Common salt is added to distilled water because it dissociates into ions, increasing the water's electrical conductivity.

**17. Answer: C**

**Explanation:** A greenish-blue spot is observed around the wire connected to the positive terminal because copper from the wire reacts during electrolysis, forming copper compounds.

**18. Answer: B**

**Explanation:** The passage of electric current through a conducting solution produces chemical effects, such as the formation of gases, deposition of metals, or colour changes in the solution**.**

**19. Answer: C**

**Explanation:** Electroplating is the process of depositing a thin layer of one metal on the surface of another material by passing an electric current through an electrolyte. It is used to improve appearance, prevent corrosion, and enhance the durability of objects.

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## LEVEL – II

### Single Correct Answer Type

**1. Answer: A) Fuse**

**Explanation:**

A fuse is a safety device containing a thin wire with a low melting point. When excessive current flows through a circuit, the wire heats up and melts, breaking the circuit. This protects electrical appliances from damage due to overloading or short circuits.

**2. Answer: B) The rod is positively charged**

**Explanation:**

The electroscope is initially negatively charged, so its leaves diverge due to the repulsion between like charges. When the rod touches the metal cap, charge is transferred. Since the leaves move closer together, the negative charge on the electroscope decreases. This occurs when a positively charged rod attracts electrons from the electroscope, reducing the repulsive force between the leaves.

**3. Answer: D) Flow of electrons (negative charge) from lower potential to higher potential**

**Explanation:**

In metallic conductors, electric current is produced by the movement of free electrons. Electrons drift from the negative terminal (lower potential) to the positive terminal (higher potential).

Note: Conventional current is defined as the flow of positive charge from higher potential to lower potential, but the actual charge carriers in metals are electrons.

**ONE OR MORE THAN ONE CORRECT ANSWER TYPE**

**4. Correct Answers: A, B and C**

**Explanation:**

A) Prevent rusting of metals

Electroplating deposits a thin layer of a less reactive metal (such as chromium or zinc) on iron. This protective coating prevents iron from coming into contact with air and moisture, thereby preventing rusting.

B) Repairing worn-out parts of machinery

Electroplating is used in industries to deposit a controlled layer of metal on worn-out machine parts. This restores their original dimensions and extends their service life.

C) Ornamentation and decoration purposes

Cheap metals are often electroplated with a thin layer of precious metals like gold or silver to improve their appearance and provide a decorative finish at a lower cost.

**5. Correct Answer: C) Oxygen**

**Explanation:**

When an electric current is passed through acidified water, water decomposes into hydrogen and oxygen gases.

$$2H_2O \rightarrow 2H_2 + O_2$$

- Hydrogen gas ($H_2$) is produced at the cathode (negative electrode).
- Oxygen gas ($O_2$) is produced at the anode (positive electrode).

Since hydrogen is not given in the options, Oxygen (C) is the correct answer.

**6. Correct Answers: B, C and D**

**Explanation:**

Pure distilled water contains almost no dissolved ions and is therefore a poor conductor of electricity.

- B) Common salt solution: Dissociates into $Na^+$ and $Cl^-$ ions, making the solution conductive.
- C) Copper sulfate solution ($CuSO_4$): Dissociates into $Cu^{2+}$ and $SO_4^{2-}$ ions, allowing electric current to pass.
- D) Lemon juice: Contains citric acid, which ionizes in water and provides free ions that conduct electricity.

Thus, B, C and D are good conductors.

**7. Correct Answers: A and B**

**Explanation:**

Insulators are materials that do not allow electric current to pass through them because they lack free-moving electrons.

- A) Rubber tyres: Rubber is an excellent electrical insulator.
- B) Plastic bangles: Plastic has very high electrical resistance and acts as an insulator.
- C) Steel rods: Steel is a metal and conducts electricity.
- D) Iron rods: Iron is also a good conductor because it contains free electrons.

Therefore, A and B are the correct answers.

**8. Correct Answers: A, B and C**

**Explanation:**

Liquids conduct electricity only if they contain free-moving ions.

- A) Acids: Produce $H^+$ ions, making the solution conductive.
- B) Bases: Produce $OH^-$ ions, allowing current to flow.
- C) Salt solutions: Dissociate into positive and negative ions that conduct electricity.
- D) Distilled water: Contains almost no ions and is therefore a poor conductor.

Hence, A, B and C conduct electricity.

**9. Correct Answers: A and C**

**Explanation:**

A) Charge remains at the point where it is supplied on an insulator.

Insulators do not have free-moving electrons. Therefore, any charge placed on an insulator remains localized and does not spread throughout the material.

C) Excess charge on a conductor resides only on its outer surface.

Conductors contain free electrons. When excess charge is given to a conductor, the charges repel one another and move freely until they spread over the outer surface, where they remain in electrostatic equilibrium**.**

**Assertion–Reason Type**

**10. Answer: D) Assertion is false, but Reason is true**

**Explanation:**

- Assertion (A): False

  Charging by induction occurs without physical contact between the charged body and the neutral object.
- Reason (R): True

  Induction is the process of charging an object by redistributing its charges without touching it.

  Therefore, the Assertion is false, while the Reason is true.

**11. Answer: B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion**

**Explanation:**

- Assertion (A): True

  According to the Law of Conservation of Charge, electric charge can neither be created nor destroyed.
- Reason (R): True

  During charging by friction, electrons are transferred from one object to another. Although both statements are correct, the transfer of electrons during rubbing is only an example of conservation of charge and does not explain why charge is conserved.

**12. Answer: B) Rubber**

**Explanation:**

Insulators do not allow electric current to pass through them. Rubber is an insulator and is commonly used for covering electrical wires because it prevents electric shock.

**13. Answer: C) Conductors**

**Explanation:**

Materials that allow electric current to pass through them easily are called conductors. Metals such as copper and aluminium are common examples.

**14. Answer: A) Free electrons**

**Explanation:**

Metals contain a large number of free electrons. These electrons move freely when a potential difference is applied, making metals excellent conductors of electricity.

**15. Answer: C) To protect the objects from rusting**

**Explanation:**

Electroplating coats an object with a thin layer of another metal. This protective coating prevents rusting and corrosion while also improving appearance.

**16. Answer: A) Electroplating**

**Explanation:**

The process of depositing a thin layer of one metal over another using electric current is called electroplating.

**17. Answer: A) Chemical effect of current**

**Explanation:**

Electroplating works because electric current causes chemical reactions in an electrolyte, resulting in the deposition of metal on the object. Hence, it is based on the chemical effect of electric current.

**Matrix Match Type**

**18. Correct Matching: a → R, b → T, c → P, d → Q**

**19. Correct Matching: a → R, b → T, c → Q (or S, depending on the given options) d → P**

**Integer Type**

**20. Answer: 2.5**

**Explanation:**

Given: Charge, Q = 600 C

- Time, t = 4 minutes = 240 s

Using the formula:

$$I = \frac{Q}{t}$$

$$I = \frac{600}{240} = 2.5 \text{ A}$$

Answer = 2.5 A

**21. Answer: 600**

**Explanation:**

Given:

- Current, I = 5 A
- Time, t = 2 minutes = 120 s

Using the formula:

$$Q = I \times t$$

$$Q = 5 \times 120 = 600 \text{ C}$$

Answer = 600 C

**22. Answer: 3**

**Explanation:**

Given: Charge, Q = 900 C

- Time, t = 5 minutes = 300 s

Using the formula:

$$I = \frac{Q}{t}$$

$$I = \frac{900}{300} = 3 \text{ A}$$

# TOPIC-2
# Electrical Energy and Magnetic Effects of Electric Current

1. **Answer: D) Ampere**
   **Explanation:** The Ampere (A) is the standard SI unit used to measure electric current, which is the rate of flow of electric charge.
2. **Answer: A) 1 coulomb/second**
   **Explanation:** Electric current is the amount of charge flowing per unit time. 1 ampere = 1 coulomb/second.
3. **Answer: B) Galvanometer**
   **Explanation:** A galvanometer is a highly sensitive instrument used to detect and measure very small electric currents**.**
4. **Answer: B) Positive to negative**
   **Explanation:** By convention, electric current flows from the positive terminal to the negative terminal of a source, although electrons move in the opposite direction.
5. **Answer: D) V = W/Q**
   **Explanation:** Potential difference is defined as the work done per unit charge. V = W/Q.
6. **Answer: B) Electric potential**
   **Explanation:** Electric potential (or potential difference) is the work done in moving a unit positive charge between two points.
7. **Answer: C) Volt**
   **Explanation:** The SI unit of electric potential (potential difference) is the volt (V).
8. **Answer: A) 1 joule/1 coulomb**
   **Explanation:** One volt is the potential difference when one joule of work is done in moving one coulomb of charge.
9. **Answer: D) Voltmeter**
   **Explanation:** A voltmeter measures the potential difference between two points in an electric circuit and is connected in parallel.
10. **Answer: B) Switch or key**
    **Explanation:** A switch or key is used to open or close an electric circuit, thereby controlling the flow of current.
11. **Answer: C) Open circuit**
    **Explanation:** An open circuit has a break in the conducting path, preventing the flow of electric current.
12. **Answer: A) Circuit diagram**
    **Explanation:** A circuit diagram uses standard electrical symbols to represent components and their connections**.**
13. **Answer: D) Electric circuit**
    **Explanation:** An electric circuit is a complete conducting path through which electric current flows.
14. **Answer: C) Energy**
    **Explanation:** Chemical reactions inside an electric cell convert chemical energy into electrical energy.
15. **Answer: A) Resistance**
    **Explanation:** Resistance is the property of a conductor that opposes the flow of electric current.

**16. Answer: C) [The zigzag line]**
**Explanation:** A resistor is represented by a zigzag line in a standard circuit diagram.

**17. Answer: B) Ohm**
**Explanation:** The SI unit of electrical resistance is the ohm (Ω).

**18. Answer: A) $10^6$ Ω**
**Explanation:** The prefix "Mega" means one million, so 1 megaohm equals $10^6$ ohms.

**19. Answer: B) Statement A is true, Statement B is false.**
**Explanation:** Statement A correctly defines potential difference, whereas Statement B is incorrect because the SI unit of potential difference is the volt, not the ampere.

**20. Answer: B) Both statements are correct.**
**Explanation:** Closing a key completes the circuit, and a voltmeter is always connected in parallel across the component.

**21. Answer: B) A simple circuit with one bulb and two cells.**
**Explanation:** Two cells provide a higher potential difference, allowing more current to flow through the bulb and making it glow brighter.

**22. Answer: B) Dilute sulphuric acid**
**Explanation:** A simple voltaic cell uses dilute sulphuric acid as the electrolyte to enable chemical reactions.

**23. Answer: C) 1.0 V**
**Explanation:** A simple copper-zinc voltaic cell produces an emf of approximately 1.0 volt.

**24. Answer: A) Leclanché cell**
**Explanation:** The dry cell is an improved form of the Leclanché cell in which the liquid electrolyte is replaced by a moist paste.

**25. Answer: B) Two**
**Explanation:** Every electric cell has two terminals: a positive terminal and a negative terminal**.**

**26. Answer: D) Chemical**
**Explanation:** An electric cell stores energy in the form of chemical energy.

**27. Answer: D) Electrical**
**Explanation:** During operation, an electric cell converts stored chemical energy into electrical energy.

**28. Answer: C) Series**
**Explanation:** When cells or electrical components are connected end-to-end in a single continuous line, the current has only one path to flow through. This arrangement is called a series connection.

**29. Answer: B) Parallel**
**Explanation:** In household circuits, appliances are connected in parallel so that each appliance receives the same supply voltage. This arrangement also allows each appliance to operate independently. Switching off one appliance does not affect the operation of the others.

**30. Answer: B) Kilowatt-hour (kWh)**
**Explanation:** The incorrectly mentions the unit of electrical power. The intended answer is the commercial unit of electrical energy, which is the Kilowatt-hour (kWh). Electricity boards use kWh to measure and bill the electrical energy consumed by consumers.

**31. Answer: C) 1 kWh**

**Explanation:**

Electrical energy is calculated using the formula:

$$\text{Electrical Energy} = \text{Power} \times \text{Time}$$

For a 1 kW appliance used for 1 hour:

$$1\,\text{kW} \times 1\,\text{hour} = 1\,\text{kWh}$$

Therefore, the electrical energy consumed is 1 kWh.

**32. Answer: D) 1 unit**

**Explanation:**

The commercial unit of electrical energy used for billing purposes is the kilowatt-hour (kWh). By definition:

$$1\ \text{unit} = 1\ \text{kWh}$$

Thus, 1 commercial unit of electricity is equal to 1 kWh.

**33. Answer: C) Low resistance**

**Explanation:**

Copper is widely used for household wiring because it has low electrical resistance, allowing electric current to flow easily with minimal energy loss as heat. This makes copper an efficient and safe conductor for electrical wiring.

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## LEVEL – II

### Single Correct Option Type

**1. Correct Answer: (B)**

**Explanation:** A fuse wire is intentionally made of a material with a low melting point so that it melts when excessive current flows, thereby breaking the circuit and protecting electrical appliances from damage.

**2. Correct Answer: (A)**

**Explanation:** A battery consists of two or more electrochemical cells connected together (usually in series) to provide a higher voltage than a single cell**.**

**3. Correct Answer: (B)**

**Explanation**: Traditional decorative light strings are connected in series, meaning current flows through one bulb after another in a single path.

**4. Correct Answer: (C)**

**Explanation:** According to Ohm's Law,

$$I = \frac{V}{R}$$

If the resistance remains constant, adding more cells increases the total voltage, which increases the current flowing through the circuit.

**5. Correct Answer: (B)**

**Explanation:** An electromagnet produces magnetism only when electric current flows through its coil. The magnetic field disappears immediately when the current is switched off.

**6. Correct Answer: (A)**

**Explanation:** The magnetic field around a current-carrying conductor is directly proportional to the current flowing through it.

$$B \propto I$$

Hence, increasing the current increases the strength of the magnetic field.

**7. Correct Answer: (C)**

**Explanation:** Watt (W) is the SI unit of electrical power, which represents the rate at which electrical energy is consumed or converted.

**8. Correct Answer: (D)**

**Explanation:** Among the given materials, tungsten has the highest electrical resistivity. Its high resistivity and very high melting point make it suitable for incandescent lamp filaments**.**

**9. Correct Answer: (B)**

**Explanation:** The thin coiled wire inside an incandescent electric bulb that glows due to heating is called the filament.

**ONE OR MORE THAN ONE CORRECT ANSWER TYPE**

**10. Answer: A, B**

**Explanation:**

Electric cells and batteries are sources of electrical energy because they convert chemical energy into electrical energy. Switches and keys do not produce electricity; they only control the flow of current by opening or closing the circuit.

**11. Answer: A, C**

**Explanation:**

An ammeter is used to measure the magnitude of electric current in a circuit. A galvanometer detects and measures small currents and can also indicate their direction. A voltmeter measures potential difference, whereas an odometer measures the distance travelled by a vehicle.

**12. Answer: A, D**

**Explanation:** Electric potential is defined as the work done per unit charge.

$$V = \frac{W}{Q}$$

Rearranging the equation,

$$W = VQ$$

Hence, both equations are correct.

**13. Answer: B, C**

**Explanation:** Electric current is the rate of flow of electric charge.

$$I = \frac{Q}{t}$$

Its SI unit is the ampere (A), which is equivalent to coulomb per second (C/s).

$$1\ \text{A} = 1\ \frac{\text{C}}{\text{s}}$$

Joule per second (J/s) is the unit of power (watt), not current.

**14. Answer: A, D**

**Explanation:** A fuse protects electrical circuits by melting when excessive current flows, thereby breaking the circuit. An MCB (Miniature Circuit Breaker) automatically switches off the circuit during overloads or short circuits. Both are important electrical safety devices.

**15. Answer: C**

**Explanation:** The electrolyte used in a Leclanché cell is an aqueous solution (or paste) of ammonium chloride ($NH_4Cl$).

**16. Answer: C**

**Explanation:** In a Leclanché cell, the positive terminal is a carbon rod surrounded by manganese dioxide ($MnO_2$), while the negative terminal is a zinc container or zinc rod.

**17. Answer: A, B, C**

**Explanation:** The following statements about a bichromate cell are correct:

- It has two carbon plates as the positive electrodes and a zinc plate as the negative electrode.
- It uses a mixture of potassium bichromate and dilute sulphuric acid as the electrolyte.
- It produces an electromotive force (emf) of approximately 2 V.

Therefore, all the scientifically correct statements should be selected.

**18. Answer: A, C**

**Explanation:** The wattage of a bulb represents the rate at which it converts electrical energy into light and heat.

For the same operating voltage:

- A higher wattage bulb consumes more electrical power.
- It generally produces more light (greater brightness).

Hence, both statements are correct.

**19. Answer: B, C**

**Explanation:** Electricity supplied to homes is billed based on the electrical energy consumed, which is measured in kilowatt-hour (kWh), commonly called a unit of electricity.

$$1 \text{ Unit} = 1 \text{ kWh}$$

Note: If the question were asking about the unit of power, the correct unit would be watt (W). However, household electricity bills measure energy consumed, not power.

**20. Answer: C, D**

**Explanation:**

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s}$$
$$= 3{,}600{,}000 \text{ J}$$
$$= 3.6 \times 10^6 \text{ J}$$

This is also equal to

$$36 \times 10^5 \text{ J}$$

Therefore, both options are correct.

**21. Answer: B, D**

**Explanation:**

A power source supplies electrical energy to a circuit.

- A dry cell converts chemical energy into electrical energy.
- A solar cell converts light energy into electrical energy.

A bulb is an electrical appliance that consumes electrical energy, while a candle produces light through combustion rather than supplying electrical energy. Hence, the correct examples of power sources are the dry cell and the solar cell.

**Assertion and Reason Type**

**22. Correct Answer: (A)**

Assertion (A): True

Reason (R): True

Explanation: An electromagnet works because electric current flowing through a coil produces a magnetic field. Thus, the reason correctly explains why an electromagnet becomes magnetic.

**23. Correct Answer: (A)**

Assertion (A): True

Reason (R): True

**Explanation:** Connecting more cells in series increases the total voltage. According to Ohm's Law,

$$I = \frac{V}{R}$$

an increase in voltage increases the current (for constant resistance). Since

$$P = VI,$$

the electrical power increases, making the bulb glow brighter. Therefore, the reason correctly explains the assertion.

**Passage – I**

**24. Correct Answer: (A)**

**Explanation:** By convention, electric current is considered to flow from the positive terminal of a battery to its negative terminal, although electrons actually move in the opposite direction**.**

**25. Correct Answer: (C)**

**Explanation:** An ammeter is an instrument used to measure electric current. It has very low resistance so that it does not significantly affect the current in the circuit.

**26. Correct Answer: (B)**

**Explanation:** An ammeter is always connected in series with the circuit so that the entire current passes through it, allowing an accurate measurement.

**Passage – II**

**27. Correct Answer: (C)**

Explanation: Electrical power is defined as the rate at which electrical energy is consumed or converted.

$$P = \frac{E}{t}$$

where

- $P$= Power (W)
- $E$= Energy (J)
- $t$= Time (s)

**28. Correct Answer: (B)**

**Explanation:**

Given:

$$E = 10000 \text{ J}, t = 20 \text{ s}$$

Using

$$P = \frac{E}{t}$$

$$P = \frac{10000}{20} = 500 \text{ W}$$

Therefore, the correct answer is 500 W.

**29. Correct Answer: (D)**

**Explanation:** Daily energy consumption: 100 W × 10 h = 1000 Wh = 1 kWh

For 31 days: 31 × 1 = 31 kWh

Electricity bill: 31 × 5 = $\boxed{\text{Rs. 155}}$

**Matrix Matching Type (Q30–Q31)**

**30. Correct Matching**

| Column I | Column II |
|---|---|
| Ammeter | S (Connected in series) |
| Resistor | T (Zig-zag symbol) |
| Voltmeter | P (Connected in parallel) |
| Open switch | Q (Circuit broken/open symbol) |

**31. Correct Matching**

| Column I | Column II |
|---|---|
| Series circuit | S (Current remains the same throughout) |
| Household wiring | T (Connected in parallel) |
| Household wiring | Q (Same voltage across appliances) |
| Safety devices | P, R (Fuse and MCB/RCB) |

**Integer Type**

**32. Correct Answer: 5**

**Explanation:**

Given:

- Charge $Q = 20$ C
- Time $t = 4$ s

Using

$$I = \frac{Q}{t}$$

$$I = \frac{20}{4} = 5 \text{ A}$$

Answer: 5

**33. Correct Answer: 60**

**Explanation:**

Given:

- Energy $E = 2400$ J
- Time $t = 40$ s

Using

$$P = \frac{E}{t}$$

$$P = \frac{2400}{40} = 60 \text{ W}$$

Answer: 60

**34. Correct Answer: 24**

**Explanation:**

Given:

- Voltage $V = 12$ V
- Current $I = 2$ A

Using

$$P = VI$$

$$P = 12 \times 2 = 24 \text{ W}$$

Answer: 24

# Previous Year Questions

**1. Correct Answer: B (F, F, F, T)**

**Explanation**

Circuit Arrangement:

- Bulbs $L_1$ and $L_3$ are connected in parallel.
- This parallel combination is connected in series with bulb $L_2$.

Current through the Bulbs:

Let the resistance of each bulb be R.

- Equivalent resistance of the parallel combination:

$$R_{L_1 \| L_3} = \frac{R}{2}$$

- Since $L_2$ is in series with the parallel combination, it carries the total current $I$.
- The current divides equally in the parallel branch:

$$I_{L_1} = I_{L_3} = \frac{I}{2}$$

Brightness Comparison:

Brightness depends on electrical power.

$$P = I^2R$$

Since $L_2$ carries the largest current, it dissipates the greatest power and is therefore the brightest.

Hence:

- (i) False
- (ii) False
- (iii) False because $I_{L_2} = 2I_{L_1}$

Potential Difference:

Using Ohm's law,

$$V = IR$$

For bulb $L_2$, $V_{L_2} = IR$

For bulb $L_3$, $V_{L_3} = \frac{I}{2}R$

Therefore, $V_{L_2} = 2V_{L_3}$

So,

- (iv) True

Final Answer: B (F, F, F, T)

**2. Correct Answer: D ((i), (iii) and (iv) only)**

**Explanation**

Diagram I

The four identical bulbs share the 100 V supply equally because of the symmetry of the circuit.

Each bulb gets

$$\frac{100}{4} = 25 \text{ V}$$

Diagram II

The circuit consists of two parallel branches, each containing two identical bulbs in series.

Each branch receives the full supply voltage: 100 V

Since the bulbs in each branch are identical, the voltage divides equally:

$$\frac{100}{2} = 50 \text{ V per bulb}$$

Brightness Comparison

Brightness depends on power.

Since each bulb in Diagram II receives 50 V, while each bulb in Diagram I receives only 25 V, the bulbs in Diagram II glow brighter.

Therefore, the incorrect statements are:

- (i)
- (iii)
- (iv)

Final Answer: D ((i), (iii) and (iv) only)

**3. Correct Answer: C**

**Explanation**

Earthing is a safety measure that provides a low-resistance path for electric current to flow safely into the Earth.

If the insulation inside an electrical appliance is damaged and the live wire touches the metallic body, the current flows directly to the ground through the earth wire.

As a result:

- The appliance casing remains at earth potential.
- Excess current flows safely into the ground.
- The fuse or MCB trips, disconnecting the supply.
- The risk of electric shock is greatly reduced.

**4. Correct Answer: A**

**Explanation**

The strength of an electromagnet depends mainly on:

- Number of turns in the coil ($N$)
- Current flowing through the coil ($I$)

Mathematically, $B \propto NI$

A rheostat can change the current flowing through the coil.

- Increasing current makes the electromagnet stronger.
- Decreasing current makes it weaker.

Therefore, the magnetic strength can be varied by changing the current.

**5. Correct Answer: A (1: Dimmer, 2: Dimmer, 3: Brighter)**

**Explanation**

Initial Circuit

- Bulbs 1 and 2 are connected in parallel.

Equivalent resistance:

$$R_{12} = \frac{R}{2}$$

This combination is in series with bulb 3.

Total resistance:

$$R_{\text{total}} = \frac{R}{2} + R = \frac{3R}{2}$$

After Adding One More Bulb

A new bulb is connected parallel to bulb 1, so now bulbs 1, 2, and the new bulb are all in parallel.

Equivalent resistance becomes

$$R_{\text{parallel}} = \frac{R}{3}$$

Total resistance:

$$R_{\text{total}} = \frac{R}{3} + R = \frac{4R}{3}$$

Since the total resistance decreases, the battery supplies more current.

Effect on Bulb 3

Bulb 3 carries the total current.

Since the total current increases, its power increases.

Hence, Bulb 3 becomes brighter.

Effect on Bulbs 1 and 2

Although the total current increases, it is now divided among three parallel branches instead of two.

Therefore, the current through each original bulb decreases.

Hence,

- Bulb 1 → Dimmer
- Bulb 2 → Dimmer

**Final Answer: A (1: Dimmer, 2: Dimmer, 3: Brighter)**

**6. Correct Answer: A (P, R and T only)**

**Explanation**

To light only bulb Y, the current must travel through the branch containing Y alone.

Required switch positions:

- P: Closed (main switch connected to the battery)
- R: Closed (allows current to pass through bulb Y)
- T: Closed (completes the return path)

The remaining switches must stay open:

- Q: Open (prevents bulb X from glowing)
- S: Open (prevents bulb Z from glowing)

Thus, only bulb Y receives current.

**7. Correct Answer: C**

**Explanation**

In domestic wiring, the switch must always be connected in the live wire.

If the switch is connected in the neutral wire:

- Switching OFF stops the appliance from operating.
- However, the appliance remains connected to the live wire.
- Internal metallic parts may still be at high potential.
- Touching them can cause an electric shock.

Therefore, the switch should always interrupt the live wire.

**8. Correct Answer: B**

**Explanation**

Whenever electric current flows through a conductor, it produces a magnetic field around it.

This magnetic field exists only while current is flowing.

As soon as the current is switched OFF:

- Movement of charges stops.
- The magnetic field disappears immediately.

Therefore, the magnetic field lasts only as long as current flows through the conductor.

**9. Correct Answer: D**

**Explanation**

The circuit contains a wire directly connected across bulb S, forming a short circuit.

A short circuit provides a path of negligible resistance.

Therefore:

- Almost all the current passes through the wire.
- Practically no current flows through bulb S.
- Bulb S does not glow even initially.

If bulb S burns out, it simply becomes an open circuit.

Since the bypass wire is still present, the current continues to flow through it.

Therefore:

- Bulbs P, Q, and R continue to glow normally.
- Bulb S remains off.

Final Answer: D

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